

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\116-CPVElite.pvdb

Date of Analysis : Apr 10,2026 3:22pm

PV Elite 28, January 2026



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FileName : 116-C_PVElite

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Vessel Design Summary:

Step: 32 3:22pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1130.000 mm ID
Vessel Design Length, Tangent to Tangent 14.82 m
Specified Datum Line Distance 0.00 m
Shell Side Design Temperature 150 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 10.501 N/mm²
Channel Side Design Pressure 1.000 N/mm²
Shop Shell Side Test Pressure 14.772 N/mm²
Shop Channel Side Test Pressure 1.680 N/mm²
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Flange	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Jacket	SA-516 70	...	...	K02700	
Jkt Closure	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel cover lt	1.000	0.10	1.000	3.0000	N/A	No
Channel flng. left	1.000	0.10	2.000	3.0000	N/A	No
Channel left	1.011	0.10	4.000	3.0000	N/A	No
Shell-1	10.501	0.00	13.000	3.0000	N/A	No
Shell	10.501	0.00	11.000	3.0000	N/A	No
Shell-2	10.501	0.00	13.000	3.0000	N/A	No
Channel right	1.011	0.10	4.000	3.0000	N/A	No
Channel flng. rht	1.000	0.10	2.000	3.0000	N/A	No
Channel cover rht	1.000	0.10	1.000	3.0000	N/A	No

Liquid Level: 1.13 m Dens.: 992.545 kg/m³ Sp. Gr.: 0.993



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Vessel Design Summary:

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Element Types and Properties:

Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	0.1	0.1	18.0	70.0	64.3	...	1.00	1.00
Body Flg	0.2	0.1	18.0	102.0	96.7	...	1.00	1.00
Cylinder	1.4	1.2	18.0	18.0	7.2	...	1.00	1.00
Cylinder	2.7	1.2	60.0	60.0	48.3	...	1.00	1.00
Cylinder	12.1	9.4	50.0	50.0	48.3	...	1.00	1.00
Cylinder	13.3	1.2	60.0	60.0	48.3	...	1.00	1.00
Cylinder	14.6	1.2	18.0	18.0	7.2	...	1.00	1.00
Body Flg	14.7	0.1	18.0	102.0	96.7	...	1.00	1.00
Body Flg	14.8	0.1	18.0	70.0	64.3	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	70	44.1198	0.10131	No Calc
20	30	102	96.7486	0.10131	No Calc
30	Tsht	18	6.75622	0.10131	1.76383
Tsht	40	18	No Calc	0.10131	No Calc
40	Jckt	60	No Calc	...	7.45873
Jckt	50	60	No Calc	...	7.45873
50	Jckt	50	No Calc	...	6.25019
Jckt	60	50	No Calc	...	5.08242
60	Jckt	60	No Calc	...	6.54547
Jckt	Jckt	60	No Calc	...	7.45873
Jckt	Tsht	60	No Calc	...	7.45873
Tsht	70	60	No Calc	...	No Calc
70	80	18	6.75631	0.10131	1.7638
80	90	102	96.7486	0.10131	No Calc
90	100	70	44.1198	0.10131	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1.19997	24.0096	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Jckt	0.20003	No Calc	No Calc	No Calc
Jckt	50	1.15	No Calc	No Calc	No Calc
50	Jckt	1.15	No Calc	No Calc	No Calc
Jckt	60	9.23	No Calc	No Calc	No Calc
60	Jckt	9.23	No Calc	No Calc	No Calc
Jckt	Jckt	1.15	No Calc	No Calc	No Calc
Jckt	Tsht	0.01997	No Calc	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	1.20003	24.01	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

	Analysis	Max Stress	High Stress	Pass
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Vessel Design Summary:

Step: 32 3:22pm Apr 10,2026

Description	Type	Ratio	Location	Fail
Cooling water	WRC-297	0.929	Shell	Passed
N4- Syn gas ou	WRC-297	0.967	Shell	Passed
N3-Syn Gas inl	WRC-297	0.967	Shell	Passed
Cooling water	WRC-297	0.930	Shell	Passed

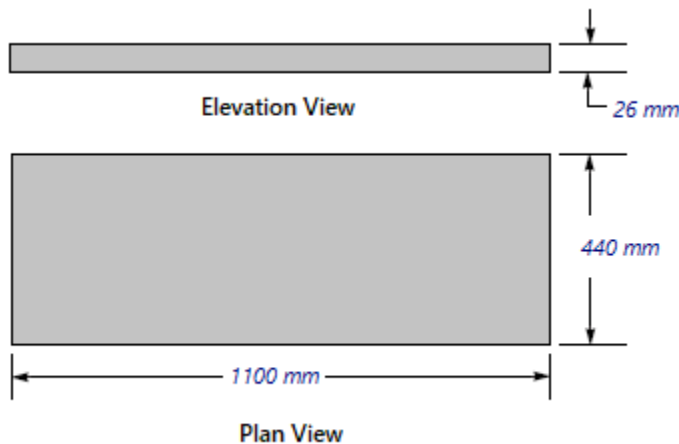
Foundation/Support Summary

Saddle Parameters:

Saddle Width	204.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1000.000 mm
Wear Pad Width	306.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1200.000 mm

Baseplate Length	1100.000 mm
Baseplate Thickness	26.000 mm
Baseplate Width	440.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	300.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	302081.53 N
Maximum Transverse Saddle Shear Load	13136.90 N
Maximum Longitudinal Saddle Shear Load	26273.79 N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	299542.53 N
Maximum Transverse Saddle Shear Load	2443.70 N
Maximum Longitudinal Saddle Shear Load	720.91 N

Weights:

Fabricated - Bare w/o removable internals	42989.7 kgm
Shop Test - Fabricated + water (full)	56410.3 kgm
Shipping - Fab. + rem. intls.+ shipping app.	42989.7 kgm



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Vessel Design Summary: Step: 32 3:22pm Apr 10,2026

Erected	- Fab. + rem. intls.+ insul. (etc)	42989.7	kgm
Empty	- Fab. + intls. + details + wghts.	43123.5	kgm
Operating	- Empty + operating liquid (no ca)	46025.2	kgm

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Nozzle Summary: Step: 30 3:22pm Apr 10,2026

Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Cooling water i	3	OK	...	OK	7.13	OK	Passed
N4- Syn gas out	11	...	...	OK	21.47	OK	Passed
N3-Syn Gas inle	11	...	...	OK	21.47	OK	Passed
Cooling water o	3	OK	...	OK	7.18	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 11 Nozzle : N3-Syn Gas inle [Shellside]
 Minimum MAWP Nozzles : 3 Nozzle : Cooling water o [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	Cooling water i	808.350	270.000	831.201
40	N4- Syn gas out	2308.350	270.000	926.400
60	N3-Syn Gas inlet	12483.351	90.000	926.400
70	Cooling water o	14008.351	90.000	831.201

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 29 3:22pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Cooling water i	18.000 in	80	WNF	18.000	23.800	697.20	10.00	316.31	150
N4- Syn gas out	18.000 in Actual		None	19.969	25.000	...	...	422.75	...
N3-Syn Gas inlet	18.000 in Actual		None	19.969	25.000	...	...	422.75	...
Cooling water o	18.000 in	80	WNF	18.000	23.800	697.20	10.00	316.31	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Cooling wat	SA-106 B	8.000	10.000	10.000	10.000	...
N4- Syn gas	SA-266 2	50.000	20.000	...	...	...
N3-Syn Gas	SA-266 2	50.000	10.000	...	...	...
Cooling wat	SA-106 B	8.000	10.000	10.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Cooling water i	0.808	270.0	250.00	0.00	Channel left
N4- Syn gas out	2.433	270.0	250.00	0.00	Shell-1
N3-Syn Gas inlet	12.483	90.0	250.00	0.00	Shell-2
Cooling water o	14.008	90.0	250.00	0.00	Channel right

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Bill of Material:

Step: 28 3:22pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 70.0mm THK X 1350.0mm OD	SA-266 2
1	WELD NECK FLANGE: 102.0mm THK X 1350.0mm OD	SA-266 2
1	WELD NECK FLANGE: 102.0mm THK X 1350.0mm OD	SA-516 70
1	BLIND FLANGE: 70.0mm THK X 1350.0mm OD	SA-516 70
2	INSULATION: 1200mm X 100mm THK	
17	BAFFLES: 1120.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 1250.0mm X 125.0mm THK	SA-266 2
808	TUBES: 12000.0mm X 19.0mm DIA X 2.1mm THK	SA-179
2	JACKET CYL: 1150.0mm LONG X 1384.0mm DIA X 60.0mm THK	SA-516 70
2	JACKET SEALING RING: 115.0mm THK	SA-516 70
2	SADDLE: 204.0mm X 120 DEG	SA-516 70
2	GASKET: 1199.0mm OD X 1159.0mm ID	...
80	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
160	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
2	CYLINDRICAL SEGMENT 18.0mm THK X 1412.5mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 725.0mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 60.0mm THK X 3047.0mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 60.0mm THK X 502.9mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 50.0mm THK X 502.0mm X 9.4m	SA-516 70
1	CYLINDRICAL SEGMENT 50.0mm THK X 3048.0mm X 9.4m	SA-516 70
1	CYLINDRICAL SEGMENT 60.0mm THK X 502.0mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 60.0mm THK X 3048.0mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 502.9mm X 1.2m	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 3047.0mm X 1.2m	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 31 3:22pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell-1	[8]	B	21	10		0.811	60.000	1.00	Yes
Shell	[8]	B	17	16		0.984	50.000	1.00	Yes
Shell-2	[8]	B	21	10		0.811	60.000	1.00	Yes
Left Saddle	[24]	B	-46	-46		0.984	10.000	1.00	No
Right saddle	[24]	B	-46	-46		0.984	10.000	1.00	No
N4- Syn gas ou	[1]	B	21	19		0.972	60.000	1.00	No
N3-Syn Gas inl	[1]	B	21	19		0.972	60.000	1.00	No
Tubesheet: SS	[13]	B	4	-12		0.710	60.000	1.00	No

Warmest MDMT: 21 19

Channel cover	[11]	B	-11	-16	-29	0.916	17.500	1.00	No
Channel flng.	[11]	B	-11	-104	-29	0.333	18.000	1.00	No
Channel flng.	[11]	B	-11	-104	-29	0.333	18.000	1.00	No
Channel cover	[11]	B	-11	-16	-29	0.916	17.500	1.00	No
Channel left	[8]	B	-11	-104	-29	0.329	18.000	1.00	No
Channel right	[8]	B	-11	-104	-29	0.329	18.000	1.00	No
Cooling water	[1]	B	13	-104		0.099	20.825	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Cooling water	[1]	B	13	-104		0.100	20.825	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: CS	[14]	B	5	-104		0.092	31.250	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: 13 -16

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	19.0	-12.0	Fail
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



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MDMT Summary:

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UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

Note:

For nozzles that are welded to both the Core and Jacket of jacketed vessels, the 2021 ASME Code requires that the opening in the jacket also be checked. This is accomplished by modelling an additional nozzle in the same location as the nozzle that penetrates through to the vessel core. The flange specification if any can be removed if needed. Other items such as weld sizes, pad specification etc. should also be addressed.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 3:22pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	10.5	N/mm ²
Channel Side Design Pressure	1	N/mm ²
Shell Side Design Temperature	150.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	Sour	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	N/mm ²
User defined MAWP, Channel Side	0	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 3:22pm Apr 10,2026

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year



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Input Echo:

Step: 1 3:22pm Apr 10,2026

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Channel cover lt
Distance "FROM" to "TO"	0.07 m
Flange Outside Diameter	1350 mm
Element Thickness	70 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	137990 kPa
Allowable Stress, Operating	137990 kPa
Allowable Stress, Hydrotest	224983 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230070 kPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Channel flng. left
Distance "FROM" to "TO"	0.132 m
Flange Inside Diameter	1130 mm
Element Thickness	102 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²



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Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Channel left
Distance "FROM" to "TO"	1.2 m
Inside Diameter	1130 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	137990 kPa
Allowable Stress, Operating	137990 kPa
Allowable Stress, Hydrotest	233982 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242583 kPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Cooling water
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.13 m
Liquid Density	992.54 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	1.2 m
Thickness of Insulation	100 mm
Density	140.16 kg/m ³

Element From Node	30
Detail Type	Nozzle
Detail ID	Cooling water i
Dist. from "FROM" Node / Offset dist	0.6 m
Nozzle Diameter	18 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1582 N
Grade of Attached Flange	GR 1.1

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Input Echo:

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Nozzle Mat1

SA-106 B

Element#	4/9	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	1.2	m
Inside Diameter	1130	mm
Element Thickness	60	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.501	N/mm ²
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N4- Syn gas out	
Dist. from "FROM" Node / Offset dist	0.9	m
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	3742	N
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	9.35 m
Inside Diameter	1130 mm
Element Thickness	50 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	50 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.501 N/mm ²
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1.2 m



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Input Echo:

Step: 1 3:22pm Apr 10,2026

Width of Saddle	204	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	306	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	8.4	m
Width of Saddle	204	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	306	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m

Element From Node	50	
Detail Type	Liquid	
Detail ID	Syn. Gas	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	1.13	m
Liquid Density	24.789	kg/m³

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	1.2 m
Inside Diameter	1130 mm
Element Thickness	60 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.501 N/mm²
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 N/mm²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Nozzle
Detail ID	N3-Syn Gas inlet



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Input Echo: Step: 1 3:22pm Apr 10,2026

Dist. from "FROM" Node / Offset dist	0.4	m
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	3836.6	N
Grade of Attached Flange	None	
Nozzle Matl	SA-266	2

Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Channel right
Distance "FROM" to "TO"	1.2 m
Inside Diameter	1130 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Cooling water
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.13 m
Liquid Density	992.54 kg/m ³

Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	1.2 m
Thickness of Insulation	100 mm
Density	140.16 kg/m ³

Element From Node	70
Detail Type	Nozzle
Detail ID	Cooling water o
Dist. from "FROM" Node / Offset dist	0.6 m
Nozzle Diameter	18 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1582 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	8/9
Element From Node	80



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Input Echo:

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Element To Node	90
Element Type	Flange
Description	Channel flng. rht
Distance "FROM" to "TO"	0.132 m
Flange Inside Diameter	1130 mm
Element Thickness	102 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Flange
Description	Channel cover rht
Distance "FROM" to "TO"	0.07 m
Flange Outside Diameter	1350 mm
Element Thickness	70 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations: Step: 2 3:22pm Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
Channel cover l		0.07635	...	0.07	...
Channel flng. l		0.20835	...	0.132	...
Channel left		1.40835	...	1.2	...
Shell-1		2.73335	...	1.2	...
Shell		12.0834	...	9.35	...
Shell-2		13.2834	...	1.2	...
Channel right		14.6084	...	1.2	...
Channel flng. r		14.7467	...	0.132	...
Channel cover r		14.8167	...	0.07	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: Chnl cover-L

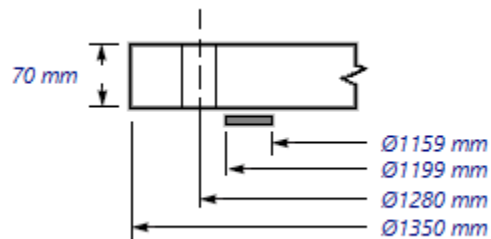
Flng: 9 3:22pm Apr 10,2026

Flange Input Data Values

Description: Chnl cover-L :

Channel cover It

Flange Type		Blind	
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1350.000	mm
Flange Thickness	t	70.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Perimeter along the Center of the Bolts	L	4021.239	mm
Diameter of Bolt Circle	C	1280.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		UNC Thread Series	
Number of Bolts		40	
Flange Face Outside Diameter	Fod	1210.000	mm
Flange Face Inside Diameter	Fid	1130.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1199.000	mm
Gasket Inside Diameter	Gi	1159.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68945.99	kPa
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	20.000 mm
Basic Gasket Width,	$bo = N/2$	10.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.969 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1183.062 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: Chnl cover-L Flng: 9 3:22pm Apr 10,2026

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1183.0625^2 \cdot 1.0$$

$$= 1099242.750 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 1183.0625 \cdot 3.0 \cdot 1.0$$

$$= 177698.531 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1099243 + 177699 + 0, 0)$$

$$= 1276941.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68945.99 \cdot 7.9687 \cdot \pi \cdot 1183.062 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2041933.625 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1276941 + 0 + 4 \cdot 0 / 1183.06) / 171987, 2041934 / 171987)$$

$$= 0.012 \text{ m}^2$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.012	0.023	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	100.428	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 0.023 \cdot 171986.91 / (68945.99 \cdot \pi (1199.0 + 1159.0))$$

$$= 7.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 171986.91(0.0119 + 0.023) / 2$$

$$= 2995984.75 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1276941.25 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1280.0 - 1183.0625) / 2$$

$$= 48.4687 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1276941 \cdot 48.4687 \cdot 1.0 / (67.0^2 \cdot 1183.0625) + 0.3 \cdot 1.0 \cdot 1.0 \cdot 1183.0625^2 / 67.0^2$$

$$= 115680.96 \text{ kPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2995985 \cdot 48.4687 \cdot 1.0 / (67.0^2 \cdot 1183.0625)$$



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Flg Calc [Int P]: Chnl cover-L

Flng: 9 3:22pm Apr 10,2026

= 51952.80 kPa

Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$

= 1276941/0.023

= 55598.78 kPa

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= (2041934/0.023)

= 88907.00 kPa

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	115680.96	137989.50	51952.80	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Reqd. Blind Flange Thickness at Center

64.346 mm

Reqd. Blind Flange Thickness at Gasket

44.111 mm

Estimated M.A.W.P. (Operating)

1 N/mm²

Estimated Finished Weight of Flange at given Thk.

776.6 kgm

Estimated Unfinished Weight of Forging at given Thk

776.6 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.916, Temperature Reduction per Fig. UCS 66.1 = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-11 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-16 °C

Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	105977.92	137989.50	47595.12	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 61.366 mm

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Flg Calc [Int P]: Chnl. flng L

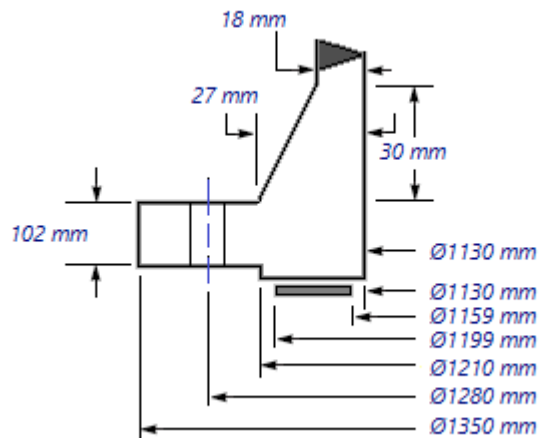
Flng: 10 3:22pm Apr 10,2026

Flange Input Data Values

Description: Chnl. flng L :

Channel flng. left

Flange Type	Integral Weld Neck		
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1130.000	mm
Flange Outside Diameter	A	1350.000	mm
Flange Thickness	t	102.0000	mm
Thickness of Hub at Small End	go	18.0000	mm
Thickness of Hub at Large End	gl	27.0000	mm
Length of Hub	h	30.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Diameter of Bolt Circle	C	1280.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	UNC Thread Series		
Number of Bolts		40	
Flange Face Outside Diameter	Fod	1210.000	mm
Flange Face Inside Diameter	Fid	1130.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1199.000	mm
Gasket Inside Diameter	Gi	1159.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68945.99	kPa
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: Chnl. flng L Flng: 10 3:22pm Apr 10,2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.0 \cdot (1130.0/2 + 3.0)) / (137989.5 \cdot 1.0 - 0.6 \cdot 1.0) + Ca$$

$$= 7.1342 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 15.0) / (568.0 + 0.6 \cdot 15.0)$$

$$= 3.587 \text{ N/mm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 18.0) / (565.0 + 0.6 \cdot 18.0)$$

$$= 4.314 \text{ N/mm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	99.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1136.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	24.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	15.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	48.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	20.000 mm
Basic Gasket Width, $b_o = N/2$	10.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.969 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1183.062 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1183.0625^2 \cdot 1.0$$

$$= 1099242.750 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 1183.0625 \cdot 3.0 \cdot 1.0$$

$$= 177698.531 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1136.0^2 \cdot 1.0$$

$$= 1013525.938 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1099243 - 1013526$$

$$= 85716.781 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(1099243 + 177699 + 0, 0)$$

$$= 1276941.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68945.99 \cdot 7.9687 \cdot \pi \cdot 1183.062 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2041933.625 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((1276941 + 0 + 4 \cdot 0/1183.06)/171987, 2041934/171987)$$

$$= 0.012 \text{ m}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 99.0/(3.0 + 0.5)$$

$$= 233.214 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1280.0 \cdot \sin(\pi/40)$$

$$= 100.428 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{100.428/(2 \cdot 31.75 + 99.0)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.012	0.023	
Radial Distance between Hub and Bolts:	44.450	48.000	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	100.428	233.214

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 0.023 \cdot 171986.91 / (68945.99 \cdot \pi(1199.0 + 1159.0))$$

$$= 7.734 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 171986.91(0.0119 + 0.023) / 2$$

$$= 2995984.75 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1276941 - 1099243$$

$$= 177698.58 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1280.0 - 1183.0625) / 2$$

$$= 48.4687 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1280.0 - 1136.0) / 2 + 48.4687)$$

$$= 60.2343 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1280.0 - 1136.0 - 24.0) / 2$$

$$= 60.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-m)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: Chnl. flng L Flng: 10 3:22pm Apr 10,2026

End Pressure, Md	1013526.	60.0000	1.0000	60836.
Face Pressure, Mt	85717.	60.2343	1.0000	5165.
Gasket Load, Mg	177699.	48.4687	1.0000	8616.
Gasket Seating, Matm	2995985.	48.4687	1.0000	145270.

Total Moment for Operation, Mo 74618. N-m
Total Moment for Gasket seating, Mg 145270. N-m

Effective Hub Length, ho = sqrt(Bcor*goCor) 130.537 mm
Hub Ratio, h/h0 = HL / H0 0.230
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.600

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.893
Factor V per 4.16.5 0.384
Factor f per 4.16.5 1.588

Stress factor from Table 4.16.4 [K]:

= A / B
= 1350.0/1136.0
= 1.188

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.188² • log10(1.188)) / (1.188² - 1)) / (1.188 - 1)
= 11.341

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.188²(1 + 8.55246 • log10(1.188)) - 1) / ((1.0472 + 1.9448 • 1.188²) • (1.188 - 1))
= 1.844

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.188²(1 + 8.55246 • log10(1.188)) - 1) / (1.36136(1.188² - 1) • (1.188 - 1))
= 12.463

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.188² + 1) / (1.188² - 1)
= 5.851

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 12.463 • 15.0² • 130.537 / 0.384
= 953521.625 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.893 / 130.537
= 0.007 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (99.0 • 0.007 + 1) / 1.844 + 99.0³/953522
= 1.927

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.5884 • 74618 / 1136.0) / (1.9274 • 24.0²)
= 93942.36 kPa

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.5884 \cdot 145270 / 1136.0) / (1.9274 \cdot 24.0^2)$$

$$= 182892.86 \text{ kPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.903 \cdot 74618 / 1136.0) / (1.9274 \cdot 99.0^2)$$

$$= 6614.67 \text{ kPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.903 \cdot 145270 / 1136.0) / (1.9274 \cdot 99.0^2)$$

$$= 12877.85 \text{ kPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (11.341 \cdot 74618 / (99.0^2 \cdot 1136.0)) - 5.8515 \cdot 6615$$

$$= 37271.09 \text{ kPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (11.341 \cdot 145270 / (99.0^2 \cdot 1136.0)) - 5.8515 \cdot 12878$$

$$= 72561.70 \text{ kPa}$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (93942 + \max(6615, 37271)) / 2$$

$$= 65606.73 \text{ kPa}$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (182893 + \max(12878, 72562)) / 2$$

$$= 127727.28 \text{ kPa}$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 1276941 / 0.023$$

$$= 55598.78 \text{ kPa}$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (2041934 / 0.023)$$

$$= 88907.00 \text{ kPa}$$

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	93942.36	206984.23	182892.86	206984.23
Radial Flange	6614.67	137989.50	12877.85	137989.50
Tangential Flange	37271.09	137989.50	72561.70	137989.50
Maximum Average	65606.73	137989.50	127727.28	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Minimum Required Flange Thickness

96.749 mm

Estimated M.A.W.P. (Operating)

2 N/mm²

Estimated Finished Weight of Flange at given Thk.

357.7 kgm

Estimated Unfinished Weight of Forging at given Thk

438.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (kPa):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499961.97	499961.97	1.000	499961.97	499961.97	1.000
Tangential Flange	St	199469.95	374971.50	0.532	199469.95	374971.50	0.532
Tangential Flange	Sto	107105.72	249980.98	0.428	107105.72	249980.98	0.428
Combined	Sh/f + St	516003.28	749943.00	0.688	516003.28	749943.00	0.688
Combined	Sr + St	234870.80	749943.00	0.313	234870.80	749943.00	0.313
Combined	Sh + Sto	607067.69	749943.00	0.809	607067.69	749943.00	0.809
Radial Flange	Sr	35400.83	499961.97	0.071	35400.83	499961.97	0.071
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.7625/1.1159)

= 1.5795

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3378/0.9983) if $h/h_0 < 0.35$, else max (0.25; 0.3011/0.7818)

= 0.3384

In this case, the hub ratio h/h_0 , 0.2298 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Flg Calc [Int P]: Chnl. flng L

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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507470.6 kPa
Minimum permissible bolt stress	Sbmin	289983.2 kPa
Max. permissible bolt stress prior to flange damage	Sfmax	358594.8 kPa
Target assembly gasket stress	SgT	206988.0 kPa
Maximum permissible gasket stress	Sgmax	275984.0 kPa
Minimum gasket operating stress	Sgmin-O	96994.4 kPa
Minimum gasket seating stress	Sgmin-S	139991.9 kPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	0.0006 m ²
Single flange rotation at Sfmax	Thetafmax	0.320 deg
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	249981.0 kPa
Flange Material Operating Yield Stress	Syo	230069.6 kPa
Single flange rotation at Sfmax	Thetafmax	0.443 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1199.0^2 - 1159.0^2)$$

$$= 0.1 \text{ m}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 206987.984 \cdot 0.074 / (40 \cdot 0.001)$$

$$= 667607.8 \text{ kPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(667608; 507471)$$

$$= 507470.6 \text{ kPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507471; 289983.156)$$

$$= 507470.6 \text{ kPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507471; 358594.844)$$

$$= 358594.8 \text{ kPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 358594.844 \geq 139991.875 (0.074 / (40 \cdot 0.001))$$

$$:: 358594.8 \geq 451522.2 \text{ kPa Check Fails}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 358594.844 \geq (96994.367 \cdot 0.074 + \pi/4 \cdot 1.0 \cdot 1159.0^2) / (0.7 \cdot 0.001 \cdot 40)$$

$$:: 358594.8 \geq 512535.6 \text{ kPa Check Fails}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 358594.844 \leq 275983.969 (0.074 / (40 \cdot 0.001))$$

$$:: 358594.8 \leq 890143.8 \text{ kPa Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$



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Flg Calc [Int P]: Chnl. flng L Flng: 10 3:22pm Apr 10,2026

:: 358594.844 ≤ 358594.844(1.0/0.32)
:: 358594.8 ≤ 1120609.0 kPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
= Sbsel • K • Abs • φb
= 358594.844 • 0.2 • 0.001 • 31.75
= 1308.0 N-m

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118084 kPa
Bolt Allowable Stress for: SA-193 B7 Sb 171986 kPa

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14 • 145270.4 / 1.0 • 999622 • 0.384 / (1.754 • 0.20270E+09 • 15.0² • 130.537 • 0.3)
= 0.927 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14 • 74617.7 / 1.0 • 999622 • 0.384 / (1.754 • 0.19921E+09 • 15.0² • 130.537 • 0.3)
= 0.485 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
= 52.14 • 145270.4 / 1.0 • 999622 • 0.384 / (1.927 • 0.20270E+09 • 15.0² • 130.537 • 0.3)
= 0.844 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
= 52.14 • 74617.7 / 1.0 • 999622 • 0.384 / (1.927 • 0.19921E+09 • 15.0² • 130.537 • 0.3)
= 0.441 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.333, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	81405.26	206984.23	155051.98	206984.23
Radial Flange	7226.68	137989.50	13764.61	137989.50
Tangential Flange	30354.14	137989.50	57815.31	137989.50
Maximum Average	55879.70	137989.50	106433.65	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 83.236 mm

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Jacket/VB Calcs: Jacket

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Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

Design per: Figure 4.11.1 Type 1

Attachment per: Table 4.11.1, Detail 5, Figure: (a)

Input Values:

Jacket Design Internal Pressure	Pj	10.50	N/mm ²
Jacket Design Temperature (Internal)		150	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.00	N/mm ²

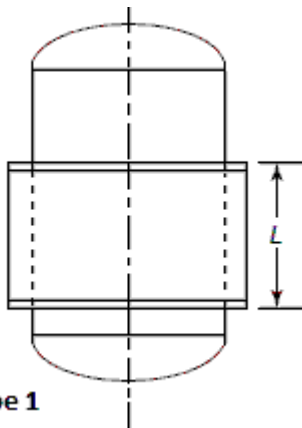
Closure Bar Material		SA-516 70	
Closure Bar Design Stress	[S or f]	137989.50	kPa
Closure Bar Thickness New	tc	115.0000	mm
Closure Bar Corrosion Allowance	cc	3.0000	mm

Inner Shell Outside Diameter	Dso	1250.000	mm
Inner Shell Thickness New	ts	60.0000	mm
Corroded Local Shell Thickness at Closure	ts,c	47.0000	mm
Inner Shell Corrosion Allow. Inside	csi	3.0000	mm
Inner Shell Corrosion Allow. Outside	cso	0.0000	mm
Inner Shell Vacuum Pressure		0.0000	N/mm ²

Jacket Material		SA-516 70	
Jacket Design Stress	Sj	137989.50	kPa
Jacket Inside Diameter New	Dji	1384.000	mm
Jacket Thickness New	tj	60.0000	mm
Jacket Corrosion Allowance Inside	cji	3.0000	mm
Jacket Length	L	1150.00	mm
Jacket Design Length for External Pressure	SL	960.00	mm
Specific Gravity of contents in Jacket	sg	1.000	
Jacket Long. Weld Joint Efficiency	E	1.00	

Weld Dimension a	a	45.000	mm
Weld Dimension b	b	45.000	mm

This has been specified as a vapor/distribution belt.



Type 1

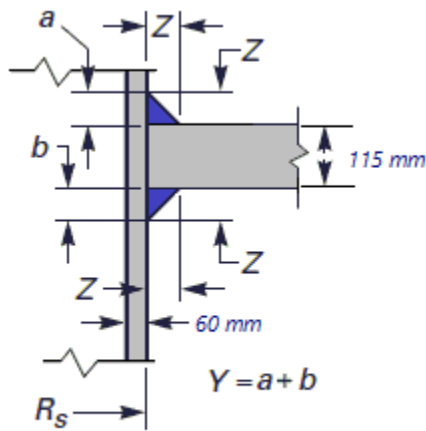
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Jacket/VB Calcs: Jacket

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Div. 2 Detail 5 Type (a)

Results:

Jacket Inside Radius [Rj]:

$$\begin{aligned} &= (D_{ji} + 2 \cdot c_i) / 2 \\ &= (1384.0 + 2 \cdot 3.0) / 2 \\ &= 695.000 \text{ mm} \end{aligned}$$

Jacket Corroded Thickness [tjc]:

$$\begin{aligned} &= t_j - c_{ji} \\ &= 60.0 - 3.0 \\ &= 57.000 \text{ mm} \end{aligned}$$

Shell Outside Radius [Rs]:

$$\begin{aligned} &= (D_{so} - 2 \cdot c_{so}) / 2 \\ &= (1250.0 - 2 \cdot 0.0) / 2 \\ &= 625.000 \text{ mm} \end{aligned}$$

Shell Corroded Thickness [tsc]:

$$\begin{aligned} &= ts - csi - cso \\ &= 60.0 - 3.0 - 0.0 \\ &= 57.000 \text{ mm} \end{aligned}$$

Jacket Space Width [j]:

$$\begin{aligned} &= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2 \\ &= (1384.0 + 2 \cdot 3.0 - 1250.0 + 2 \cdot 0.0) / 2 \\ &= 70.000 \text{ mm} \end{aligned}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (10.502 \cdot 695.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 10.502) \\ &= 55.4213 + 3.0000 = 58.4213 \text{ mm} \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (137989.5 \cdot 1.0 \cdot 57.0) / (695.0 + 0.6 \cdot 57.0) \\ &= 11 - 0 = 11 \text{ N/mm}^2 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$



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Jacket/VB Calcs: Jacket

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$$= (137989.5 \cdot 1.0 \cdot 60.0) / (692.0 + 0.6 \cdot 60.0)$$

$$= 11 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.502 \cdot (695.0 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0)$$

$$= 134342.234 \text{ kPa}$$

Jacket Minimum Design Metal Temperature Calculations:

Govrn. thk, tg = 60.0, tr = 57.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 C

Min Metal Temp. w/o impact per UCS-66, Curve B 21 C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit 21 C

Required Closure Thickness [trc]:

$$= 0.707 \cdot j \cdot \sqrt{(P_j / S)} + c$$

$$= 0.707 \cdot 70.0 \cdot \sqrt{(10.502 / 137.997)} + 3.0$$

$$= 13.653 + 3.0 = 16.653 \text{ mm}$$

$$\text{but trc is max(trc, } 2 \cdot \text{trj) + c}$$

$$= \text{max(13.653, 55.421)} + 3.0$$

$$= 113.843 \text{ mm}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$= ((tc - c) / (0.707 \cdot j))^2 \cdot S - p_{\text{Static}}$$

$$= ((115.0 - 3.0) / (0.707 \cdot 70.0))^2 \cdot 137989.5 - 0.001$$

$$= 706.760 \text{ N/mm}^2$$

Minimum Value of Dimension a [aRequired]:

$$= \min(1/4 \text{ inch (6mm), } tc, ts)$$

$$= \min(6.0, 112.0, 57.0)$$

$$= 6.000 \text{ mm}$$

Minimum Value of Dimension b [bRequired]:

$$= \min(1/4 \text{ inch (6mm), } tc, ts)$$

$$= \min(6.0, 112.0, 57.0)$$

$$= 6.000 \text{ mm}$$

Required Value of Y [YRequired]:

$$= \min(1.5 \cdot tc, 1.5 \cdot ts)$$

$$= \min(1.5 \cdot 112.0, 1.5 \cdot 57.0)$$

$$= 85.500 \text{ mm}$$

Value of Y [Y]:

$$= a + b$$

$$= 45.0 + 45.0$$

$$= 90.000 \text{ mm}$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f(1 - R_f/R_o)$): 4.155 %

External Pressure results for the Outer Jacket Cylinder:

Elastic Modulus from Chart: CS-2 at 150 C: 199818 N/mm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
57.000	1504.00	960.00	26.39	0.6383	0.0172750	122669.40

$$\text{MAEP} = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 122669.3984) / (3 \cdot 26.386) = 6.1991 \text{ N/mm}^2$$



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Jacket/VB Calcs: Jacket

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Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
57.000	1504.00	960.00	26.39	0.6383	0.0172750	122669.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*122669.3984)/(3*26.386) = 6.1991 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	58.421	60.000	Passed
Closure Bar Thickness	113.843	115.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure 10.79 N/mm²

Overall MAWP of Jacket Components 10.79 N/mm²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure 6.20 N/mm²

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 10.501 \cdot 1.0$$

$$= 13.651 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot \min(689.493, 10.787) \cdot 1.0$$

$$= 14.023 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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Jacket/VB Calcs: Jacket

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Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

Design per: Figure 4.11.1 Type 1

Attachment per: Table 4.11.1, Detail 5, Figure: (a)

Input Values:

Jacket Design Internal Pressure	Pj	10.50	N/mm ²
Jacket Design Temperature (Internal)		150	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.00	N/mm ²

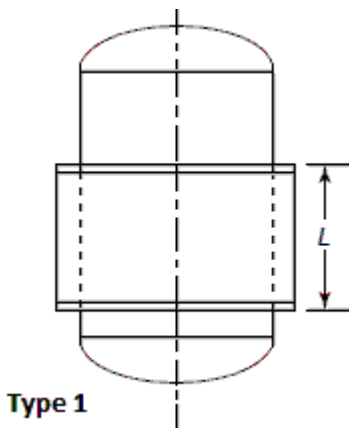
Closure Bar Material		SA-516 70
Closure Bar Design Stress	[S or f]	137989.50 kPa
Closure Bar Thickness New	tc	115.0000 mm
Closure Bar Corrosion Allowance	cc	3.0000 mm

Inner Shell Outside Diameter	Dso	1250.000 mm
Inner Shell Thickness New	ts	60.0000 mm
Corroded Local Shell Thickness at Closure	ts,c	57.0000 mm
Inner Shell Corrosion Allow. Inside	csi	3.0000 mm
Inner Shell Corrosion Allow. Outside	cso	0.0000 mm
Inner Shell Vacuum Pressure		0.0000 N/mm ²

Jacket Material		SA-516 70
Jacket Design Stress	Sj	137989.50 kPa
Jacket Inside Diameter New	Dji	1384.000 mm
Jacket Thickness New	tj	60.0000 mm
Jacket Corrosion Allowance Inside	cji	3.0000 mm
Jacket Length	L	1150.00 mm
Jacket Design Length for External Pressure	SL	960.00 mm
Specific Gravity of contents in Jacket	sg	0.000
Jacket Long. Weld Joint Efficiency	E	1.00

Weld Dimension a	a	45.000 mm
Weld Dimension b	b	45.000 mm

This has been specified as a vapor/distribution belt.





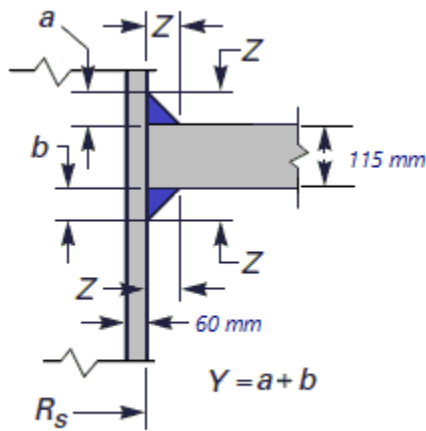
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Jacket/VB Calcs: Jacket

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Div. 2 Detail 5 Type (a)

Results:

Jacket Inside Radius [Rj]:

$$= (D_{ji} + 2 \cdot c_{ji}) / 2$$

$$= (1384.0 + 2 \cdot 3.0) / 2$$

$$= 695.000 \text{ mm}$$

Jacket Corroded Thickness [tjc]:

$$= t_j - c_{ji}$$

$$= 60.0 - 3.0$$

$$= 57.000 \text{ mm}$$

Shell Outside Radius [Rs]:

$$= (D_{so} - 2 \cdot c_{so}) / 2$$

$$= (1250.0 - 2 \cdot 0.0) / 2$$

$$= 625.000 \text{ mm}$$

Shell Corroded Thickness [tsc]:

$$= t_s - c_{si} - c_{so}$$

$$= 60.0 - 3.0 - 0.0$$

$$= 57.000 \text{ mm}$$

Jacket Space Width [j]:

$$= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2$$

$$= (1384.0 + 2 \cdot 3.0 - 1250.0 + 2 \cdot 0.0) / 2$$

$$= 70.000 \text{ mm}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.501 \cdot 695.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 10.501)$$

$$= 55.4144 + 3.0000 = 58.4144 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 57.0) / (695.0 + 0.6 \cdot 57.0)$$

$$= 11 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 60.0) / (692.0 + 0.6 \cdot 60.0)$$



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Jacket/VB Calcs: Jacket

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$$= 11 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\ &= (10.501 \cdot (695.0 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0) \\ &= 134326.312 \text{ kPa} \end{aligned}$$

Jacket Minimum Design Metal Temperature Calculations:

Govrn. thk, tg = 60.0, tr = 57.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 1.0, Temp. Reduction = 0 C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	21 C

Required Closure Thickness [trc]:

$$\begin{aligned} &= 0.707 \cdot j \cdot \sqrt{(P_j / S)} + c \\ &= 0.707 \cdot 70.0 \cdot \sqrt{(10.501 / 137.997)} + 3.0 \\ &= 13.652 + 3.0 = 16.652 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{but trc is } \max(\text{trc}, 2 \cdot \text{trj}) + c \\ &= \max(13.652, 55.414) + 3.0 \\ &= 113.829 \text{ mm} \end{aligned}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$\begin{aligned} &= ((tc - c) / (0.707 \cdot j))^2 \cdot S - p_{\text{Static}} \\ &= ((115.0 - 3.0) / (0.707 \cdot 70.0))^2 \cdot 137989.5 - 0.0 \\ &= 706.761 \text{ N/mm}^2 \end{aligned}$$

Minimum Value of Dimension a [aRequired]:

$$\begin{aligned} &= \min(1/4 \text{ inch (6mm)}, tc, ts) \\ &= \min(6.0, 112.0, 57.0) \\ &= 6.000 \text{ mm} \end{aligned}$$

Minimum Value of Dimension b [bRequired]:

$$\begin{aligned} &= \min(1/4 \text{ inch (6mm)}, tc, ts) \\ &= \min(6.0, 112.0, 57.0) \\ &= 6.000 \text{ mm} \end{aligned}$$

Required Value of Y [YRequired]:

$$\begin{aligned} &= \min(1.5 \cdot tc, 1.5 \cdot ts) \\ &= \min(1.5 \cdot 112.0, 1.5 \cdot 57.0) \\ &= 85.500 \text{ mm} \end{aligned}$$

Value of Y [Y]:

$$\begin{aligned} &= a + b \\ &= 45.0 + 45.0 \\ &= 90.000 \text{ mm} \end{aligned}$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f(1 - R_f/R_o)$): 4.155 %

External Pressure results for the Outer Jacket Cylinder:

Elastic Modulus from Chart: CS-2 at 150 C: 199818 N/mm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
57.000	1504.00	960.00	26.39	0.6383	0.0172750	122669.40

$$\text{MAEP} = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 122669.3984) / (3 \cdot 26.386) = 6.1991 \text{ N/mm}^2$$



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Jacket/VB Calcs: Jacket

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Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
57.000	1504.00	960.00	26.39	0.6383	0.0172750	122669.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*122669.3984)/(3*26.386) = 6.1991 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	58.414	60.000	Passed
Closure Bar Thickness	113.829	115.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure 10.79 N/mm²

Overall MAWP of Jacket Components 10.79 N/mm²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure 6.20 N/mm²

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 10.501 \cdot 1.0$$

$$= 13.651 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot \min(689.493, 10.787) \cdot 1.0$$

$$= 14.023 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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Flg Calc [Int P]: Chn. flng- R

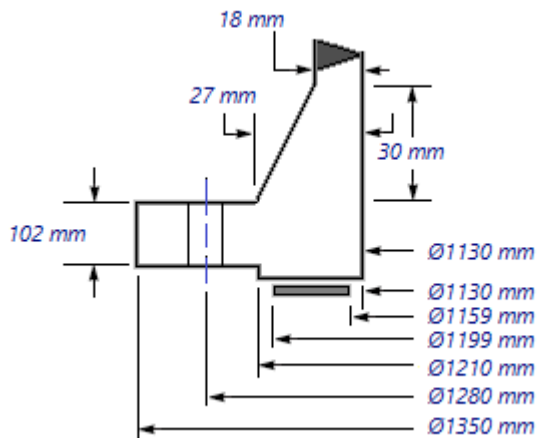
Flng: 11 3:22pm Apr 10,2026

Flange Input Data Values

Description: Chn. flng- R :

Channel flng. rht

Flange Type	Integral Weld Neck		
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1130.000	mm
Flange Outside Diameter	A	1350.000	mm
Flange Thickness	t	102.0000	mm
Thickness of Hub at Small End	go	18.0000	mm
Thickness of Hub at Large End	gl	27.0000	mm
Length of Hub	h	30.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Diameter of Bolt Circle	C	1280.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	UNC Thread Series		
Number of Bolts		40	
Flange Face Outside Diameter	Fod	1210.000	mm
Flange Face Inside Diameter	Fid	1130.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1199.000	mm
Gasket Inside Diameter	Gi	1159.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68945.99	kPa
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: Chn. flng- R Flng: 11 3:22pm Apr 10,2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.0 \cdot (1130.0/2 + 3.0)) / (137989.5 \cdot 1.0 - 0.6 \cdot 1.0) + Ca$$

$$= 7.1342 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 15.0) / (568.0 + 0.6 \cdot 15.0)$$

$$= 3.587 \text{ N/mm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 18.0) / (565.0 + 0.6 \cdot 18.0)$$

$$= 4.314 \text{ N/mm}^2$$

Corroded Flange Thickness, tc = T-ci	99.000 mm
Corroded Flange ID, Bcor = B+2*ci	1136.000 mm
Corroded Large Hub, g1Cor = g1-ci	24.000 mm
Corroded Small Hub, g0Cor = go-ci	15.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	48.000 mm

Gasket Contact Width, N = (Go - Gi)/2	20.000 mm
Basic Gasket Width, bo = N/2	10.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.969 mm
Gasket Reaction Diameter, G = Go - 2 * b	1183.062 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1183.0625^2 \cdot 1.0$$

$$= 1099242.750 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 1183.0625 \cdot 3.0 \cdot 1.0$$

$$= 177698.531 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1136.0^2 \cdot 1.0$$

$$= 1013525.938 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1099243 - 1013526$$

$$= 85716.781 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1099243 + 177699 + 0, 0)$$

$$= 1276941.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68945.99 \cdot 7.9687 \cdot \pi \cdot 1183.062 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2041933.625 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1276941 + 0 + 4 \cdot 0 / 1183.06) / 171987, 2041934 / 171987)$$

$$= 0.012 \text{ m}^2$$



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Flg Calc [Int P]: Chn. flng- R Flng: 11 3:22pm Apr 10,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 99.0/(3.0 + 0.5)$$

$$= 233.214 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1280.0 \cdot \sin(\pi/40)$$

$$= 100.428 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{100.428/(2 \cdot 31.75 + 99.0)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.012	0.023	
Radial Distance between Hub and Bolts:	44.450	48.000	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	100.428	233.214

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 0.023 \cdot 171986.91 / (68945.99 \cdot \pi(1199.0 + 1159.0))$$

$$= 7.734 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 171986.91(0.0119 + 0.023) / 2$$

$$= 2995984.75 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1276941 - 1099243$$

$$= 177698.58 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1280.0 - 1183.0625) / 2$$

$$= 48.4687 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1280.0 - 1136.0) / 2 + 48.4687)$$

$$= 60.2343 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1280.0 - 1136.0 - 24.0) / 2$$

$$= 60.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-m)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: Chn. flng- R Flng: 11 3:22pm Apr 10,2026

End Pressure, Md	1013526.	60.0000	1.0000	60836.
Face Pressure, Mt	85717.	60.2343	1.0000	5165.
Gasket Load, Mg	177699.	48.4687	1.0000	8616.
Gasket Seating, Matm	2995985.	48.4687	1.0000	145270.

Total Moment for Operation, Mo 74618. N-m
Total Moment for Gasket seating, Mg 145270. N-m

Effective Hub Length, ho = sqrt(Bcor*goCor) 130.537 mm
Hub Ratio, h/h0 = HL / H0 0.230
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.600

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.893
Factor V per 4.16.5 0.384
Factor f per 4.16.5 1.588

Stress factor from Table 4.16.4 [K]:

= A / B
= 1350.0/1136.0
= 1.188

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.188² • log10(1.188))/(1.188² - 1))/(1.188 - 1)
= 11.341

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.188²(1 + 8.55246 • log10(1.188)) - 1)/((1.0472 + 1.9448 • 1.188²) • (1.188 - 1))
= 1.844

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.188²(1 + 8.55246 • log10(1.188)) - 1)/(1.36136(1.188² - 1) • (1.188 - 1))
= 12.463

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.188² + 1)/(1.188² - 1)
= 5.851

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 12.463 • 15.0² • 130.537/0.384
= 953521.625 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.893/130.537
= 0.007 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (99.0 • 0.007 + 1)/1.844 + 99.0³/953522
= 1.927

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.5884•74618/1136.0)/(1.9274•24.0²)
= 93942.36 kPa

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.5884 \cdot 145270 / 1136.0) / (1.9274 \cdot 24.0^2)$$

$$= 182892.86 \text{ kPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.903 \cdot 74618 / 1136.0) / (1.9274 \cdot 99.0^2)$$

$$= 6614.67 \text{ kPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.903 \cdot 145270 / 1136.0) / (1.9274 \cdot 99.0^2)$$

$$= 12877.85 \text{ kPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (11.341 \cdot 74618 / (99.0^2 \cdot 1136.0)) - 5.8515 \cdot 6615$$

$$= 37271.09 \text{ kPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (11.341 \cdot 145270 / (99.0^2 \cdot 1136.0)) - 5.8515 \cdot 12878$$

$$= 72561.70 \text{ kPa}$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (93942 + \max(6615, 37271)) / 2$$

$$= 65606.73 \text{ kPa}$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (182893 + \max(12878, 72562)) / 2$$

$$= 127727.28 \text{ kPa}$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 1276941 / 0.023$$

$$= 55598.78 \text{ kPa}$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (2041934 / 0.023)$$

$$= 88907.00 \text{ kPa}$$

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	93942.36	206984.23	182892.86 206984.23
Radial Flange	6614.67	137989.50	12877.85 137989.50
Tangential Flange	37271.09	137989.50	72561.70 137989.50
Maximum Average	65606.73	137989.50	127727.28 137989.50
Bolting	55598.78	171986.91	88907.00 171986.91

Minimum Required Flange Thickness 96.749 mm

Estimated M.A.W.P. (Operating) 2 N/mm²

Estimated Finished Weight of Flange at given Thk. 357.7 kgm

Estimated Unfinished Weight of Forging at given Thk 438.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (kPa):



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		Actual	Operating Allowed	stress ratio	Gasket Seating Actual Allowed		stress ratio
Longitudinal Hub	Sh	519960.41	519960.41	1.000	519960.41	519960.41	1.000
Tangential Flange	St	207448.77	389970.31	0.532	207448.77	389970.31	0.532
Tangential Flange	Sto	111389.94	259980.20	0.428	111389.94	259980.20	0.428
Combined	Sh/f + St	536643.44	779940.62	0.688	536643.44	779940.62	0.688
Combined	Sr + St	244265.62	779940.62	0.313	244265.62	779940.62	0.313
Combined	Sh + Sto	631350.38	779940.62	0.809	631350.38	779940.62	0.809
Radial Flange	Sr	36816.86	519960.41	0.071	36816.86	519960.41	0.071
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.7625/1.1159)

= 1.5795

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.3378/0.9983) if $h/h_0 < 0.35$, else max (0.25; 0.3011/0.7818)

= 0.3384

In this case, the hub ratio h/h_0 , 0.2298 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507470.6 kPa
Minimum permissible bolt stress	Sbmin	289983.2 kPa
Max. permissible bolt stress prior to flange damage	Sfmax	372938.6 kPa
Target assembly gasket stress	SgT	206988.0 kPa
Maximum permissible gasket stress	Sgmax	275984.0 kPa
Minimum gasket operating stress	Sgmin-O	96994.4 kPa
Minimum gasket seating stress	Sgmin-S	139991.9 kPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	0.0006 m ²
Single flange rotation at Sfmax	Thetafmax	0.320 deg
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	259980.2 kPa
Flange Material Operating Yield Stress	Syo	242583.1 kPa
Single flange rotation at Sfmax	Thetafmax	0.460 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1199.0^2 - 1159.0^2)$$

$$= 0.1 \text{ m}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 206987.984 \cdot 0.074 / (40 \cdot 0.001)$$

$$= 667607.8 \text{ kPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(667608; 507471)$$

$$= 507470.6 \text{ kPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507471; 289983.156)$$

$$= 507470.6 \text{ kPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507471; 372938.594)$$

$$= 372938.6 \text{ kPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 372938.594 \geq 139991.875 (0.074 / (40 \cdot 0.001))$$

$$:: 372938.6 \geq 451522.2 \text{ kPa Check Fails}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 372938.594 \geq (96994.367 \cdot 0.074 + \pi/4 \cdot 1.0 \cdot 1159.0^2) / (0.7 \cdot 0.001 \cdot 40)$$

$$:: 372938.6 \geq 512535.6 \text{ kPa Check Fails}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 372938.594 \leq 275983.969 (0.074 / (40 \cdot 0.001))$$

$$:: 372938.6 \leq 890143.8 \text{ kPa Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$



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:: 372938.594 ≤ 372938.594(1.0/0.32)
 :: 372938.6 ≤ 1165433.1 kPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 = Sbsel • K • Abs • φb
 = 372938.594 • 0.2 • 0.001 • 31.75
 = 1360.3 N-m

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118084 kPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171986 kPa

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14 • 145270.4 / 1.0 • 999622 • 0.384 / (1.754 • 0.20270E+09 • 15.0² • 130.537 • 0.3)
 = 0.927 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14 • 74617.7 / 1.0 • 999622 • 0.384 / (1.754 • 0.19921E+09 • 15.0² • 130.537 • 0.3)
 = 0.485 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14 • 145270.4 / 1.0 • 999622 • 0.384 / (1.927 • 0.20270E+09 • 15.0² • 130.537 • 0.3)
 = 0.844 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14 • 74617.7 / 1.0 • 999622 • 0.384 / (1.927 • 0.19921E+09 • 15.0² • 130.537 • 0.3)
 = 0.441 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.333, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	81405.26	206984.23	155051.98	206984.23
Radial Flange	7226.68	137989.50	13764.61	137989.50
Tangential Flange	30354.14	137989.50	57815.31	137989.50
Maximum Average	55879.70	137989.50	106433.65	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 83.236 mm

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Flg Calc [Int P]: Chn. Cover-R

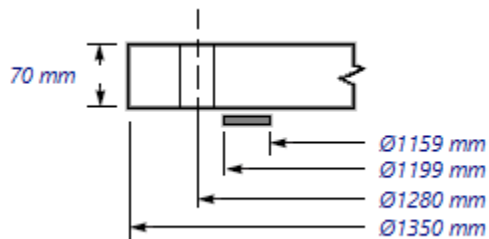
Flng: 12 3:22pm Apr 10,2026

Flange Input Data Values

Description: Chn. Cover-R :

Channel cover rht

Flange Type		Blind	
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1350.000	mm
Flange Thickness	t	70.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Perimeter along the Center of the Bolts	L	4021.239	mm
Diameter of Bolt Circle	C	1280.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		UNC Thread Series	
Number of Bolts		40	
Flange Face Outside Diameter	Fod	1210.000	mm
Flange Face Inside Diameter	Fid	1130.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1199.000	mm
Gasket Inside Diameter	Gi	1159.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68945.99	kPa
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	20.000 mm
Basic Gasket Width,	$bo = N/2$	10.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.969 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1183.062 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: Chn. Cover-R Flng: 12 3:22pm Apr 10,2026

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1183.0625^2 \cdot 1.0$$

$$= 1099242.750 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 1183.0625 \cdot 3.0 \cdot 1.0$$

$$= 177698.531 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1099243 + 177699 + 0, 0)$$

$$= 1276941.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68945.99 \cdot 7.9687 \cdot \pi \cdot 1183.062 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 2041933.625 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1276941 + 0 + 4 \cdot 0 / 1183.06) / 171987, 2041934 / 171987)$$

$$= 0.012 \text{ m}^2$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.012	0.023	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	100.428	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 0.023 \cdot 171986.91 / (68945.99 \cdot \pi (1199.0 + 1159.0))$$

$$= 7.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 171986.91(0.0119 + 0.023) / 2$$

$$= 2995984.75 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1276941.25 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1280.0 - 1183.0625) / 2$$

$$= 48.4687 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1276941 \cdot 48.4687 \cdot 1.0 / (67.0^2 \cdot 1183.0625) + 0.3 \cdot 1.0 \cdot 1.0 \cdot 1183.0625^2 / 67.0^2$$

$$= 115680.96 \text{ kPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2995985 \cdot 48.4687 \cdot 1.0 / (67.0^2 \cdot 1183.0625)$$



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= 51952.80 kPa

Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$

= 1276941/0.023

= 55598.78 kPa

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= (2041934/0.023)

= 88907.00 kPa

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	115680.96	137989.50	51952.80	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Reqd. Blind Flange Thickness at Center

64.346 mm

Reqd. Blind Flange Thickness at Gasket

44.111 mm

Estimated M.A.W.P. (Operating)

1 N/mm²

Estimated Finished Weight of Flange at given Thk.

776.6 kgm

Estimated Unfinished Weight of Forging at given Thk

776.6 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.916, Temperature Reduction per Fig. UCS 66.1 = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-11 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-16 °C

Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	105977.92	137989.50	47595.12	137989.50
Bolting	55598.78	171986.91	88907.00	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 61.366 mm

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Internal Pressure Calculations: Step: 9 3:22pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
Channel cover lt		1	18	3	1350	137990	1
Channel flng. le		1	18	3	1130	137990	2
Channel left		1.0111	18	3	1130	137990	3
Shell-1		10.501	60	3	1130	137990	4
Shell		10.501	50	3	1130	137990	5
Shell-2		10.501	60	3	1130	137990	6
Channel right		1.0111	18	3	1130	137990	7
Channel flng. rh		1	18	3	1130	137990	8
Channel cover rh		1	18	3	1350	137990	9

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.W.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
Channel cover lt		1.00006	1.1929	1.30213	70	64.3455	1
Channel flng. le		1.00006	1.97653	2.46955	102	96.7486	2
Channel left		1.00006	3.57643	4.31392	18	7.18003	3
Shell-1		10.5006	13.0619	13.7768	60	48.2884	4
Shell		10.5006	10.8784	11.5964	50	48.2896	5
Shell-2		10.5006	13.0619	13.7768	60	48.2884	6
Channel right		1.00006	3.57643	4.31392	18	7.18003	7
Channel flng. rh		1.00006	1.97653	2.46955	102	96.7486	8
Channel cover rh		1.00006	1.1929	1.30213	70	64.3455	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 11 N/mm²
 Shell Side MAPnc = 11 N/mm²
 Channel Side MAWP = 1 N/mm²
 Channel Side MAPnc = 1 N/mm²

Elements Suitable for Design Internal Pressure.

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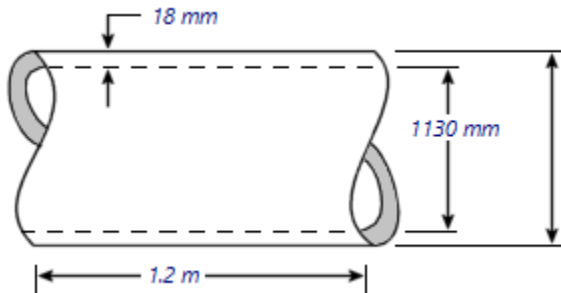
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Element: Channel left

Step: 10 3:22pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel left

Cylindrical Shell Design Information:			
Design Pressure P	1.011 N/mm ²	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1.200 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1130.000 mm	Actual Stress	38890.9 kPa
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	4.4 m ²	SG of Contents	0.993
Element Volume	1.2 m ³	Weight of Contents	11713.4 N
Empty Weight	5920.6 N	Operating Weight	17634.0 N
		Comp. All. Ope. Stress	135721.4 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.011 \cdot 568.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 1.011) \\
 &= 4.1800 + 3.0000 = 7.1800 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 15.0) / (568.0 + 0.6 \cdot 15.0) \\
 &= 4 - 0 = 4 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 18.0) / (565.0 + 0.6 \cdot 18.0) \\
 &= 4 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.011 \cdot (568.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0) \\
 &= 38890.871 \text{ kPa}
 \end{aligned}$$



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Element: Channel left

Step: 10 3:22pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.568 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 18.0$, $t_r = 4.936$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.329$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.193 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.329	Temperature Difference	93 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Channel left

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 N/mm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1166.00	1199.97	77.73	1.0291	0.0019203	102825.07

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 102825.0703) / (3 \cdot 77.7333) = 1.7638 \text{ N/mm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.756	1166.00	1199.97	310.42	1.0291	0.0002359	23584.88

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 23584.877) / (3 \cdot 310.4182) = 0.1013 \text{ N/mm}^2$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1166.00	24009.61	77.73	20.5914	0.0001867	18666.03

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 18666.0293) / (3 \cdot 77.7333) = 0.3202 \text{ N/mm}^2$$



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FileName : 116-C_PVElite

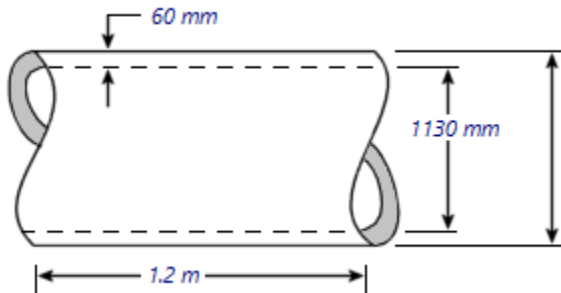
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Element: Shell-1

Step: 11 3:22pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	10.501 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	1.200 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1130.000 mm	Actual Stress	110931.6 kPa
Spec. Min. Thk t,e, or sa	60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area	4.7 m ²	SG of Contents	0.000
Element Volume	0.9 m ³	Weight of Contents	0.0 N
Empty Weight	20457.4 N	Operating Weight	20457.4 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (10.501 \cdot 568.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 10.501) \\
 &= 45.2884 + 3.0000 = 48.2884 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 57.0) / (568.0 + 0.6 \cdot 57.0) \\
 &= 13 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 60.0) / (565.0 + 0.6 \cdot 60.0) \\
 &= 14 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (10.501 \cdot (568.0 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0) \\
 &= 110931.578 \text{ kPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 5.042 %



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Element: Shell-1

Step: 11 3:22pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 60.0$, $t_r = 46.227$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.811$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.708 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	60.000 mm	Basic MDMT	21 °C
Stress/Thk. Ratio	0.811	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	10 °C
		Joint Efficiency E*	100 %

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FileName : 116-C_PVElite

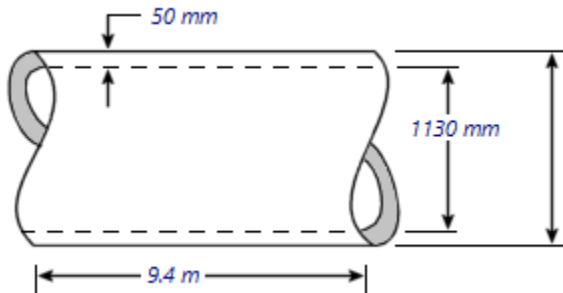
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Element: Shell

Step: 12 3:22pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	10.501 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	9.350 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1130.000 mm	Actual Stress	133197.1 kPa
Spec. Min. Thk t,e, or sa	50.0000 mm	Spec. Nominal Thk.	50.000 mm
Surface Area	36.1 m ²	SG of Contents	0.025
Element Volume	7.1 m ³	Weight of Contents	1755.9 N
Empty Weight	131715.1 N	Operating Weight	133471.0 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (10.501 \cdot 568.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 10.501) \\
 &= 45.2896 + 3.0000 = 48.2896 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 47.0) / (568.0 + 0.6 \cdot 47.0) \\
 &= 11 - 0 = 11 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 50.0) / (565.0 + 0.6 \cdot 50.0) \\
 &= 12 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (10.501 \cdot (568.0 + 0.6 \cdot 47.0)) / (1.0 \cdot 47.0) \\
 &= 133197.109 \text{ kPa}
 \end{aligned}$$



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Element: Shell

Step: 12 3:22pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 4.237 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 50.0$, $t_r = 46.227$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.984$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 17 °C

Min Metal Temp. at Required thickness (UCS 66.1) 16 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.708 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	50.000 mm	Basic MDMT	17 °C
Stress/Thk. Ratio	0.984	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	16 °C
		Joint Efficiency E^*	100 %

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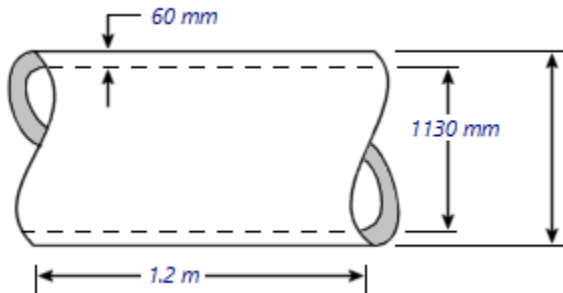
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Element: Shell-2

Step: 13 3:22pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	10.501 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	1.200 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1130.000 mm	Actual Stress	110931.6 kPa
Spec. Min. Thk t,e, or sa	60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area	4.7 m ²	SG of Contents	0.000
Element Volume	0.9 m ³	Weight of Contents	0.0 N
Empty Weight	20457.4 N	Operating Weight	20457.4 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.501 \cdot 568.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 10.501)$$

$$= 45.2884 + 3.0000 = 48.2884 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 57.0) / (568.0 + 0.6 \cdot 57.0)$$

$$= 13 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137989.5 \cdot 1.0 \cdot 60.0) / (565.0 + 0.6 \cdot 60.0)$$

$$= 14 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.501 \cdot (568.0 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0)$$

$$= 110931.578 \text{ kPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 5.042 %



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Element: Shell-2

Step: 13 3:22pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 60.0$, $t_r = 46.227$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.811$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	10.708 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	60.000 mm	Basic MDMT	21 °C
Stress/Thk. Ratio	0.811	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	10 °C
		Joint Efficiency E*	100 %

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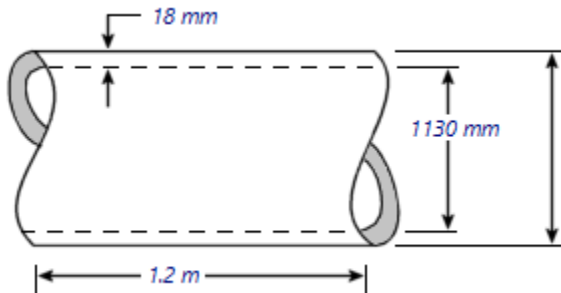
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Element: Channel right

Step: 14 3:22pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel right

Cylindrical Shell Design Information:			
Design Pressure P	1.011 N/mm ²	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1.200 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1130.000 mm	Actual Stress	38890.9 kPa
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	4.4 m ²	SG of Contents	0.993
Element Volume	1.2 m ³	Weight of Contents	11713.4 N
Empty Weight	5920.6 N	Operating Weight	17634.0 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.011 \cdot 568.0) / (137989.5 \cdot 1.0 - 0.6 \cdot 1.011) \\
 &= 4.1800 + 3.0000 = 7.1800 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.011 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 15.0) / (568.0 + 0.6 \cdot 15.0) \\
 &= 4 - 0 = 4 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137989.5 \cdot 1.0 \cdot 18.0) / (565.0 + 0.6 \cdot 18.0) \\
 &= 4 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.011 \cdot (568.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0) \\
 &= 38890.871 \text{ kPa}
 \end{aligned}$$



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Element: Channel right

Step: 14 3:22pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.568 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18.0$, $t_r = 4.936$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.329$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.193 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.329	Temperature Difference	93 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Channel right

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 N/mm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1166.00	1200.03	77.73	1.0292	0.0019202	102823.83

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 102823.8281) / (3 \cdot 77.7333) = 1.7638 \text{ N/mm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.756	1166.00	1200.03	310.41	1.0292	0.0002359	23584.34

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 23584.3438) / (3 \cdot 310.4113) = 0.1013 \text{ N/mm}^2$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1166.00	24010.03	77.73	20.5918	0.0001867	18666.02

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 18666.0176) / (3 \cdot 77.7333) = 0.3202 \text{ N/mm}^2$$



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Shop Pressure Test Results: Step: 15 3:22pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	13.921 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.651 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14.772 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	11.779 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	15.016 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	15.016 N/mm ²
App 27-4	= M.A.W.P.	10.708 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.551 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.680 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	1.312 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 N/mm ²
App 27-4	= M.A.W.P.	1.193 N/mm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel left	65028.7	233982.2	0.278	1.69
Shell-1	156175.1	233982.2	0.667	14.78
Shell	187516.8	233982.2	0.801	14.78
Shell-2	156175.1	233982.2	0.667	14.78
Channel right	65028.7	233982.2	0.278	1.69

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Cooling water i	Nozzle	117991.02	117991.02	1.000
Cooling water i	Pad	137989.50	137989.50	1.000
N4- Syn gas out	Nozzle	137989.50	137989.50	1.000
N3-Syn Gas inlet	Nozzle	137989.50	137989.50	1.000
Cooling water o	Nozzle	117991.02	117991.02	1.000
Cooling water o	Pad	137989.50	137989.50	1.000

Minimum 1.000

Stress ratios for Jacket/Vapor Belt Materials:

Description	Cylinder/Closure	Ambient	Operating	Ratio
Jacket	Cylinder	137989.50	137989.50	1.000
Jacket	Closure	137989.50	137989.50	1.000
Jacket	Cylinder	137989.50	137989.50	1.000
Jacket	Closure	137989.50	137989.50	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kPa):



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Shop Pressure Test Results:

Step: 15 3:22pm Apr 10,2026

Description	Ambient	Operating	Ratio
Channel cover lt	137989.50	137989.50	1.000
Channel flng. left	137989.50	137989.50	1.000
Channel left	137989.50	137989.50	1.000
Shell-1	137989.50	137989.50	1.000
Shell	137989.50	137989.50	1.000
Shell-2	137989.50	137989.50	1.000
Channel right	137989.50	137989.50	1.000
Channel flng. rht	137989.50	137989.50	1.000
Channel cover rht	137989.50	137989.50	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	92392.97	92392.97	1.000
Tubesheet Material	137989.50	137989.50	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Cooling water i	17903.44	215983.55	0.083
N4- Syn gas out	156029.34	224982.88	0.694
N3-Syn Gas inlet	156029.34	224982.88	0.694
Cooling water o	17903.44	215983.55	0.083

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 3:22p Apr 10,2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		14.77 N/mm ²
Shell Thickness	ts	60.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1130.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel left

Channel Type:		Cylinder
Channel Test Pressure		1.68 N/mm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1130.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	808
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	12000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11750.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	179986.31 kPa
Tube Pitch (Center to Center Spacing)	P	30.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	457.685 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm



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ASME TS Calc: Pressure Test

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Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 137989.50 kPa
Tubesheet Allowable Stress at Ambient Tt 137989.50 kPa
Thickness of Tubesheet h 125.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1250.000 mm

Area of the Untubed Lanes AL 0.000 m²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 850.000 mm
Tube End condition corresponding to Span (l) k 1.00

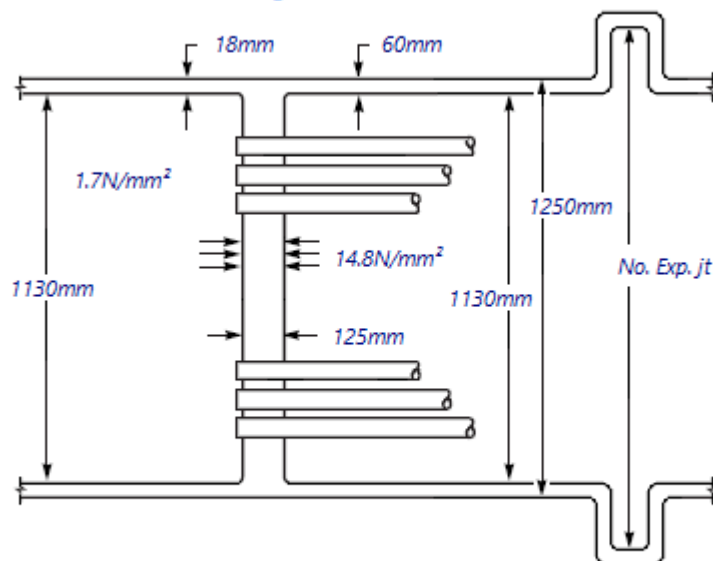
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell and Channel



Configuration a:



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ASME TS Calc: Pressure Test

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Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(1.68 \cdot (19.05/2 - 0)) / (92392.97 \cdot 1.0 + 0.4 \cdot 1.68)$
= $0.1719 + 0.0000 = 0.1719$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199955.00 N/mm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	11750.00	9.03	50.0000	0.0134948	95834.94

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15.0201$ N/mm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0874	19.05	11750.00	9.13	50.0000	0.0132070	95834.94

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 14.7734$ N/mm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.0874 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm ²
Required Internal Design Pressure	1.68 N/mm ²



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Allowable External Pressure at Corroded thickness	15.02 N/mm ²
Required External Design Pressure	14.77 N/mm ²
Required Thickness due to Shell Side pressure	2.0874 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

= l_{tx} / h (modified for corrosion if present)

= 0/125.0

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

= $\max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(19.05 - 2 \cdot 2.11 (202713/202713) \cdot$

$(92392/137989) \cdot (0); 19.05 - 2 \cdot 2.11)$

= 19.0500 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$

= $2 \cdot 457.685 + 19.05$

= 934.420 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(30.0 - 19.05) / 30.0$

= 0.365

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= $934.42 / 2$

= 467.2100 mm

Effective tube pitch [p^*]:

= $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

= $30.0 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 1000248 ; 4 \cdot 934.42 \cdot 30.0) / (\pi \cdot 934.42^2)) }$

= 30.0000 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(30.0 - 19.05) / 30.0$

= 0.3650

Ratio [ps]:

= as / ao

= $565.0 / 467.21$

= 1.2093

Ratio [pc]:

= ac / ao

= $565.0 / 467.21$

= 1.2093

Compute parameter [xt]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

= $1 - 808 ((19.05 - 2 \cdot 2.11) / (2 \cdot 467.21))^2$

= 0.7965



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ASME TS Calc: Pressure Test

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Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 808 (19.05 / (2 \cdot 467.21))^2$$

$$= 0.6642$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 60.0 (1130.0 + 60.0)202713/11750.0$$

$$= 3869500.7500 \text{ N/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (19.05 - 2.11)202713/11750.0$$

$$= 1937.1023 \text{ N/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 3869500 / (808 \cdot 1937.102)$$

$$= 2.47224$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 3869500/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1130.0 + 60.0) 60.0)$$

$$= 0.0068 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.007 \cdot 202713 \cdot 60.0^3 / (6(1 - 0.3^2))$$

$$= 54556844.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1130.0 + 60.0) \cdot 54556844 / 125.0^3 \cdot (1 + 125.0 \cdot 0.007 + (125.0 \cdot 0.007)^2 / 2)$$

$$= 441159.6875 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1130.0^2 / (4 \cdot 202713 \cdot 60.0) \cdot (1 - 1130.0 \cdot 0.3 / (2(1130.0 + 60.0)))$$

$$= 0.22509\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc) tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1130.0 + 18.0) 18.0)$$

$$= 0.0126 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.013 \cdot 202713 \cdot 18.0^3 / (6(1 - 0.3^2))$$

$$= 2738135.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$



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ASME TS Calc: Pressure Test

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Channel coefficient [λ_c]:

$$= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta_c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(1130.0 + 18.0/\cos(0)) \cdot 2738135/125.0^3 \cdot (1.0 + 125.0 \cdot 0.013 \cdot (1.0 + 1.0)/2 + (125.0 \cdot 0.013)^2/2 \cdot 1.0)$$

$$= 36985.1 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= Dc^2/(4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc)))$$

$$= 1130.0^2/(4 \cdot 202713 \cdot 18.0) \cdot (1 - 1130.0 \cdot 0.3/(2(1130.0 + 18.0)))$$

$$= 0.74574\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [ζ_{etac}]:

$$\zeta_{etac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 4.166667 ; \mu^* = 0.365000 ; E^* = 74408 \text{ N/mm}^2$$

$$E^*/E = 0.367063 ; \nu^* = 0.320752$$

Note: As h/p (4.167) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2)/(E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.321^2)808 \cdot 202713 \cdot 2.11 (19.05 - 2.11)467.21^2)/(74408 \cdot 11750.0 \cdot 125.0^3))^{1/4}$$

$$= 2.0035$$

Values from Table 4.18.3

$$Z_d = 0.172927 ; Z_v = 0.162518 ; Z_m = 0.676541$$

$$Z_a = 0.803015\text{E}+00 ; Z_w = 0.162518$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1250.0/934.42$$

$$= 1.3377$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32)/74408 \cdot (441159.69 + 36985.12 + 202713 \cdot \ln(1.34))$$

$$= 4.9033$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)F$$

$$= (1 + 0.3208)4.9033$$

$$= 6.4760$$

Compute parameter [Q_1]:

$$= (ps - 1 - \phi \cdot Z_v)/(1 + \phi \cdot Z_m)$$

$$= (1.2093 - 1 - 6.476 \cdot 0.1625)/(1 + 6.476 \cdot 0.6765)$$

$$= -0.156684$$

Compute Parameter [Q_{z1}]:

$$= (Z_d + Q_1 \cdot Z_w)/2 \cdot X_a^4$$

$$= (0.17293 + -0.15668 \cdot 0.16252)/2 \cdot 2.00353^4$$

$$= 1.1881$$

Compute parameter [Q_{z2}]:

$$= (Z_v + Q_1 \cdot Z_m)/2 \cdot X_a^4$$

$$= (0.16252 + -0.15668 \cdot 0.67654)/2 \cdot 2.00353^4$$

$$= 0.4553$$



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ASME TS Calc: Pressure Test

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Compute parameter [U]:

$$= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.1625 + (1.2093 - 1)0.6765)2.00353^4 / (1 + 6.476 \cdot 0.6765)$$

$$= 0.9106$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.2093 \cdot 54556844 \cdot 0.0068 \cdot 0.225E-04 (1 + 125.0 \cdot 0.0068)$$

$$= 18693.2441 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$

$$= 467.21^2 (1.209^2 - 1) \cdot (1.209 - 1) / 4 - 18693.244$$

$$= -13411.4062 \text{ mm}^2$$

Parameter [wc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.209 \cdot 2738135 [0.746E-04 (1.0 + 0.013 \cdot 1.0 \cdot 125.0) -$$

$$0 (1.0/125.0 + 1.0 \cdot 0.013/2)]$$

$$= 8058.3193 \text{ mm}^2$$

Parameter [ω*c]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega c$$

$$= 467.20999^2 [(1.20931^2 + 1) \cdot (1.20931 - 1) / 4 - (1.20931 - 1) / 2] - 8058.31934$$

$$= -2776.4800 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs -$$

$$[(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj)^2 - (Ds)^2 / Do^2] \}$$

$$= 14.772 \{ 0.664 + 2(1 - 0.664)0.3 + [2/2.472 \cdot$$

$$(1130.0/934.42)^2] 0.3 - [(1.209^2 - 1)/(1.0 \cdot 2.472)] -$$

$$[(1 - 1.0)/(2 \cdot 1.0 \cdot 2.472)] [(0^2 - (1130.0)^2 / 934.42^2] \}$$

$$= 15.2678 \text{ N/mm}^2$$

Pressure [P't]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt$$

$$= [0.796 + 2(1 - 0.796)0.3 + 1/(1 \cdot 2.472)] 1.68$$

$$= 2.2222 \text{ N/mm}^2$$

Pressure [Pv]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 808 \cdot 1937.102 \cdot 0 / (3.142 \cdot 467.21^2)$$

$$= 0.0000 \text{ N/mm}^2$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 0.911 \cdot 0 / (2 \cdot \pi \cdot 467.21^2)$$



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ASME TS Calc: Pressure Test

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$$= -0.0000 \text{ N/mm}^2$$

Calculate the pressure [Prim]:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega * s \cdot Ps - \omega * c \cdot Pt) \\ &= -(0.911/467.21^2) \cdot (-13411.406 \cdot 14.772 - -2776.48 \cdot 1.68) \\ &= 0.8070 \text{ N/mm}^2 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega s \cdot P * s - \omega c \cdot P * c) \\ &= (0.911/467.21^2) \cdot (18693.2441 \cdot 0 - 8058.3193 \cdot 0) \\ &= 0.0000 \text{ N/mm}^2 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1) Qz2)) \cdot \\ &\quad (P * s - P * t + P * y + P * w + P * prim) \\ &= 0.1000E+01 \cdot 2.472 / (1 + 1.0 \cdot 2.472 (1.188 + (1.209 - \\ &\quad 1) 0.455)) \cdot (15.268 - 2.222 + 0 + 0 + 0 + 0.807) \\ &= 8.2073 \text{ N/mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega * s \cdot Ps - \omega * c \cdot Pt) - (\omega s \cdot P * s - \omega c \cdot P * c)) + \\ &\quad W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ &= [((-13411.406 \cdot 14.772 - -2776.48 \cdot 1.68) - \\ &\quad (18693.244 \cdot 0 - 8058.319 \cdot 0)) \cdot 1. + \\ &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.47599 \cdot 0.67654) \\ &= -35946.2266 \text{ N} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ &= -0.157 + 2 \cdot -35946.227 / (8.207 \cdot 467.21^2) \\ &= -0.19682 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.098408

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.0984 / 0.365) \cdot (2 \cdot 467.21 / (125.0 - 0))^2 \cdot 8.21 \\ &= 185467.8906 \text{ kPa} \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 249980.98 = 356222.88 \text{ kPa}, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.0991 / 0.365) \cdot (2 \cdot 467.21 / (85.673 - 0))^2 \cdot 7.35 \\ &= 356214.0938 \text{ kPa} \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 85.6728 + 0 + 0 \\ &= 85.6728 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy) \\ &= (2 \cdot 0.365 \cdot 125.0 / 467.21) \cdot 189985.547 \\ &= 37105.7617 \text{ kPa} \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taujlimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (1 / (2 \cdot \mu)) \cdot (ao / h) \cdot Pe$$



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$$= (1 / (2 \cdot 0.365)) \cdot (467.21 / 7.245) \cdot 2.063$$

$$= 182254.1719 \text{ kPa}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + \text{Cats} + \text{Cats}$$

$$= 7.245 + 0 + 0$$

$$= 7.2450 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(\text{HreqB}; \text{HreqS})$$

$$= \max(85.6728; 7.245)$$

$$= 85.6728 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$\text{Ftmin} = 0.7912, \text{ Ftmax} = 1.1355$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((\text{Ps} \cdot \text{xs} - \text{Pt} \cdot \text{xt}) - \text{Pe} \cdot \text{Ftmin}) / (\text{Xt} - \text{Xs})$$

$$= ((14.77 \cdot 0.6642 - 1.68 \cdot 0.7965) - (8.207) \cdot 0.7912) /$$

$$(0.7965 - 0.6642)$$

$$= 14966.0098 \text{ kPa}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((\text{Ps} \cdot \text{xs} - \text{Pt} \cdot \text{xt}) - \text{Pe} \cdot \text{Ftmax}) / (\text{Xt} - \text{Xs})$$

$$= ((14.77 \cdot 0.6642 - 1.68 \cdot 0.7965) - (8.207) \cdot 1.136) /$$

$$(0.7965 - 0.6642)$$

$$= -6392.6987 \text{ kPa}$$

Maximum Tube Axial Stress [$\sigma_{t,\max}$]:

$$= \max(|\sigma_{t1}|; |\sigma_{t2}|) = 14966.01 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot \text{Sy},t = 170987.0000 \text{ kPa}$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(\text{Zd} + \text{Q3} \cdot \text{Zw})\text{Xa}^4); 1.25)$$

$$= \max((3.25 - 0.25(0.173 + -0.197 \cdot 0.163)2.004^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{(\text{dt}^2 + (\text{dt} - 2 \cdot \text{tt})^2)}) / 4$$

$$= (\sqrt{(19.05^2 + (19.05 - 2 \cdot 2.11)^2)}) / 4$$

$$= 6.0355 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Et} / \text{Sy},t)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713 / 179986)}$$

$$= 149.1573$$

Determine the factor [Fi]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 850.0 / 6.035$$

$$= 140.8340$$

Buckling Allowable Stress [Stb]:

$$= (\text{Sy},t / \text{Fs}) \cdot (1 - \text{Fi} / (2 \cdot \text{Ct}))$$

$$= (179986 / 2.0) \cdot (1 - 140.834 / (2 \cdot 149.157))$$

$$= 170987.000 \text{ kPa (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.



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Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 14966.01 \cdot 0.0001$$

$$= 1680.51 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.05 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 170987.0$$

$$= 19199.818 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 170987.0$$

$$= 19199.8184 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 170987.0 / 170987.0$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 170987.0$$

$$= 12756.4717 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 19199.82 is $\geq$ tube strength 19199.82 N

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 259980.2 = 370471.75 \text{ kPa}$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 467.21^2 / ((1130.0 + 60.0) 60.0) \cdot [8.21 + (1.209^2 - 1)(14.77 - 1.68)] + 565.0^2 \cdot 1.68 / ((1130.0 + 60.0) 60.0)$$

$$= 51106.9727 \text{ kPa}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 60.0 mm, the minimum Shell length adjacent to the tubesheet should be 468.692 mm



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Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	185467.9	<= 356222.9 kPa	Pass
Tubesheet Shear Stress	42019.7	<= 189985.5 kPa	Pass
Maximum Tube Stress	14966.0	<= 170987.0 kPa	Pass
Minimum Tube Stress (Buckling)	-6392.7	<= -170987.0 kPa	Pass
Maximum Force on any one Tube	1680.5	<= 19199.8 N	Pass
Axial Membrane Stress in Shell	51107.0	<= 370471.8 kPa	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	85.673	125.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	11.730 ...	31814	356223 6800 189986	Ps+Pt-Th D1	Ok
D2uc	93.210 ...	210272	356223 48819 189986	Ps+Pt-Th D2	Ok
D3uc	85.673 ...	185468	356223 42020 189986	Ps+Pt-Th D3	Ok
Max:	93.2101 ... mm	0.590	0.257 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+/-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses Ten Allwd Cmp Allwd	----	Shell Band Stress Ten Allwd Cmp Allwd	---- : Pass : Fail
D1uc	1074 370471 ...	...	...	Ok
D2uc	50033 370471 ...	...	...	Ok
D3uc	51107 370471 ...	...	...	Ok



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Max RATIO 0.138 ...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	2593	170987	-2593	-170987	291	19199	...	370471	...	370471	Ok
D2uc	17560	170987	-8129	-170987	1972	19199	...	370471	...	370471	Ok
D3uc	14966	170987	-6393	-170987	1681	19199	...	370471	...	370471	Ok

Max RATIO 0.103 0.048 0.103 ...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	93.210	125.000	Ok
Tube thickness :	2.087	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (60.000) adj. to tubesheet: 468.692 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -1.014 mm
 Due to Pressure + Thermal Shell Compresses by : -1.014 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0325	...	...	...	...	...
2	-1.0138	...	...	...	...	...
3	-0.9813	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 16 3:22pm Apr 10,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	...	67	No Calc	No Calc
20	30	No Calc	...	99	No Calc	No Calc
30	Tsht	1.19997	1166	15	0.0019203	102825
Tsht	40	No Calc	...	15	No Calc	No Calc
40	Jckt	0.20003	1250	57	0.093934	122669
Jckt	50	1.15	1250	57	0.015107	122669
50	Jckt	1.15	1230	47	0.011229	122669
Jckt	60	9.23	1230	47	0.0017031	99750.1
60	Jckt	9.23	1250	57	0.0024298	107649
Jckt	Jckt	1.15	1250	57	0.015107	122669
Jckt	Tsht	0.01997	1250	57	0.093934	122669
Tsht	70	No Calc	...	57	No Calc	No Calc
70	80	1.20003	1166	15	0.0019202	102824
80	90	No Calc	...	99	No Calc	No Calc
90	100	No Calc	...	67	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	70	44.1198	0.10131	No Calc
20	30	102	96.7486	0.10131	No Calc
30	Tsht	18	6.75622	0.10131	1.76383
Tsht	40	18	No Calc	0.10131	No Calc
40	Jckt	60	No Calc	...	7.45873
Jckt	50	60	No Calc	...	7.45873
50	Jckt	50	No Calc	...	6.25019
Jckt	60	50	No Calc	...	5.08242
60	Jckt	60	No Calc	...	6.54547
Jckt	Jckt	60	No Calc	...	7.45873
Jckt	Tsht	60	No Calc	...	7.45873
Tsht	70	60	No Calc	...	No Calc
70	80	18	6.75631	0.10131	1.7638
80	90	102	96.7486	0.10131	No Calc
90	100	70	44.1198	0.10131	No Calc

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1.19997	24.0096	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Jckt	0.20003	No Calc	No Calc	No Calc
Jckt	50	1.15	No Calc	No Calc	No Calc
50	Jckt	1.15	No Calc	No Calc	No Calc



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Jckt	60	9.23	No Calc	No Calc	No Calc
60	Jckt	9.23	No Calc	No Calc	No Calc
Jckt	Jckt	1.15	No Calc	No Calc	No Calc
Jckt	Tsht	0.01997	No Calc	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	1.20003	24.01	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

For vessels with Jackets, the External Design Pressure includes the core vacuum pressure. The External MAWP is the total allowable external pressure for that segment minus the core vacuum pressure. The External MAWP can be less than the External Design pressure and this is acceptable given that the actual thickness is greater than the required thickness.

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Element and Detail Weights:

Step: 17

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	776.579	...	776.579	...	77.6579
20	30	357.704	0.13032	346.779	0.13063	35.7704
30	40	603.776	1.17502	504.461	1.18754	60.3776
40	50	2086.22	0.90519	1986.9	0.91771	208.622
50	60	13432.1	7.05297	12658.3	7.15045	1343.21
60	70	2086.22	0.90519	1986.9	0.91771	208.622
70	80	603.776	1.17502	504.461	1.18754	60.3776
80	90	357.704	0.13032	346.779	0.13063	35.7704
90	100	776.579	...	776.579	...	77.6579
Total		21080	11.47	19887	11.62	2108



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset Dtl. Cent. m	Description
30	Liqd	1194.51	0.6	...	...	Cooling water
30	Ins1	66.8962	0.6	...	...	Insulation
30	Noz1	177.459	0.6	0.7698	...	Cooling water i
40	Noz1	419.764	0.9	0.7936	...	N4- Syn gas out
40	Jckt	3084.04	0.775	...	...	Jacket
50	Sad1	228.947	1.2	0.7525	...	Left Saddle
50	Sad1	228.947	8.4	0.7525	...	Right saddle
50	Liqd	179.068	4.675	...	...	Syn. Gas
60	Noz1	430.377	0.4	-0.7936	...	N3-Syn Gas inlet
60	Jckt	3084.04	0.605	...	...	Jacket
70	Liqd	1194.51	0.6	...	...	Cooling water
70	Ins1	66.8962	0.6	...	...	Insulation
70	Noz1	177.459	0.6	-0.7698	...	Cooling water o
30	FTsh	1062.38	1.2625	...	...	
30	Tube	9282.41	7.2	...	...	Tubes
30	RTsh	1062.38	13.1375	...	...	Tubesheet
40	Baf1	33.1036	0.76	...	...	Baffles
40	Baf1	33.1036	1.185	...	...	Baffles
40	Baf1	33.1036	1.61	...	...	Baffles
40	Baf1	33.1036	2.035	...	...	Baffles
40	Baf1	33.1036	2.46	...	...	Baffles
40	Baf1	33.1036	2.885	...	...	Baffles
40	Baf1	33.1036	3.31	...	...	Baffles
40	Baf1	33.1036	3.735	...	...	Baffles
40	Baf1	33.1036	4.16	...	...	Baffles
40	Baf1	33.1036	4.585	...	...	Baffles
40	Baf1	33.1036	5.01	...	...	Baffles
40	Baf1	33.1036	5.435	...	...	Baffles
40	Baf1	33.1036	5.86	...	...	Baffles
40	Baf1	33.1036	6.285	...	...	Baffles
40	Baf1	33.1036	6.71	...	...	Baffles
40	Baf1	33.1036	7.135	...	...	Baffles
40	Baf1	33.1036	7.56	...	...	Baffles



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Element and Detail Weights: Step: 17 3:22pm Apr 10,2026

Total Weight of Each Detail Type:

Saddles	457.9
Liquid	2568.1
Insulation	133.8
Nozzles	1205.1
Exchanger Components	11969.9
Jacket, Closure, Insul	6168.1

Sum of the Detail Weights 22836.5 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	23188.7	23188.7	23188.7	23188.7	23188.7	23188.7
Saddles	457.9	457.9	457.9	457.9	457.9	457.9
Nozzles	1205.1	1205.1	1205.1	1205.1	1205.1	1205.1
Exchanger	11969.9	11969.9	11969.9	11969.9	11969.9	11969.9
Jacket	6168.1	6168.1	6168.1	6168.1	6168.1	6168.1
Insulation	...	...	...	...	133.8	133.8 *
Ope. Liquid	...	...	...	...	...	2568.1
Jacket Liquid	...	...	...	...	...	333.7
Test Liquid	...	11746.5	...	...	...	...
Tube Tst Lqd	...	1674.1	...	...	...	...
Totals	42989.7	56410.3	42989.7	42989.7	43123.5	46025.2

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	42989.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	56410.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	42989.7 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	42989.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	43123.5 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	46025.2 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	44713.0 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	1.6 m ³
Weight of Ope Liq in tubes :	0.0 kgm
Weight of Water in tubes :	1674.1 kgm

Field Test weight is computed in the corroded condition.

Field Test weight includes test liquid in the jacket only for Type 1 jackets specified as vapor belts.

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
10	20	1.72784	...
20	30	0.97246	0.46848



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Element and Detail Weights:

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30	40	4.39463	4.25894
40	50	4.71122	4.25894
50	60	36.1209	33.1843
60	70	4.71122	4.25894
70	80	4.39463	4.25894
80	90	0.97246	0.46848
90	100	1.72784	...

Total	59.7	51.2 m ²
Total	643.1	550.8 ft ²

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Nozzle Flange MAWP:

Step: 18 3:22pm Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Cooling water	1.82	1.96	82	150	GR 1.1	1.796	1.821	1.821	0.935
Cooling water	1.82	1.96	82	150	GR 1.1	1.796	1.821	1.821	0.935

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1.821 N/mm²

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1.960 N/mm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-m	Axial Force FE N	MAWP PD N/mm ²	Maximum Pressure Pmax N/mm ²	Pass Fail
Cooling water i	500.888	1.20	40093	33750	1.193	1.821	Pass
Cooling water o	500.888	1.20	40093	33750	1.193	1.821	Pass

Evaluating Flange on Nozzle [Cooling water i]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(40093) + 4 \cdot \text{abs}(33750) \cdot 500.888 \leq \pi \cdot 500.888^3 [(1.821 - 1.193) + 1.200 \cdot 1.821]$$

$$= 0.70914E+06 \leq 0.11106E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Cooling water i [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.821, 1.200 \cdot 1.821 + 1.821 - 709135.9 / (\pi \cdot 500.888^3))$$

$$= \min(1.821, 2.210)$$

$$= 1.821 \text{ N/mm}^2$$

Evaluating Flange on Nozzle [Cooling water o]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(40093) + 4 \cdot \text{abs}(33750) \cdot 500.888 \leq \pi \cdot 500.888^3 [(1.821 - 1.193) + 1.200 \cdot 1.821]$$

$$= 0.70914E+06 \leq 0.11106E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Cooling water o [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.821, 1.200 \cdot 1.821 + 1.821 - 709135.9 / (\pi \cdot 500.888^3))$$

$$= \min(1.821, 2.210)$$

$$= 1.821 \text{ N/mm}^2$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 19 3:22pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.982)/(1 + 1.7 \cdot 3.4 \cdot 0.228))) \\
 &= \min(0.85, 0.916) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.629
Structure Height to Diameter ratio	12.133

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274.32$ m

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1.0 + 0.3 = 1.3 \text{ m} \\
 &= 4.265 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (4.265 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900.0)^{2/9.5} \\
 &= 0.849
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 19 3:22pm Apr 10, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 60.398²)

= 16.0 psf [0.766] kPa

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.629 • 1.221

= 10.4 lbs. [46.4] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
Channel cover l	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Channel flng. l	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Channel left	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Shell	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Shell-2	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Channel right	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Channel flng. r	1.3	0.000	0.000	0.000	0.849	1.000	0.766
Channel cover r	1.3	0.000	0.000	0.000	0.849	1.000	0.766

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	1.3	1.62	0.11337	0.76606	51.0499
20	30	1.3	1.356	0.17895	0.76606	80.5779
30	40	1.3	1.3992	1.67862	0.76606	755.863
40	50	1.3	1.634	1.96031	0.76606	882.705
50	60	1.3	1.47858	13.8213	0.76606	6223.56
60	70	1.3	1.6541	1.98443	0.76606	893.563
70	80	1.3	1.3992	1.67862	0.76606	755.863
80	90	1.3	1.356	0.17895	0.76606	80.5779
90	100	1.3	1.62	0.11337	0.76606	51.0499

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 20 3:22pm Apr 10,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(5.49540.75)$
 $= 0.072$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.072, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.157 \cdot 446833 \cdot 1.25$
 $= 26273.791$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 446833$
 $= 14012.7$ N

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	0.675	40621.2	2755.99



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Earthquake Load Calculation: Step: 20 3:22pm Apr 10,2026

20	30	0.565	40621.2	2306.87
30	40	0.565	40621.2	2306.87
40	50	0.565	40621.2	2306.87
50	Sad1	0.565	40621.2	2306.87
Sad1	60	0.565	40621.2	2306.87
50	60	0.565	40621.2	2306.87
60	70	0.565	40621.2	2306.87
70	80	0.565	40621.2	2306.87
80	90	0.565	40621.2	2306.87
90	100	0.675	40621.2	2755.99

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 21 3:22pm Apr 10,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	7.533 m
Center of Gravity of the Liquid	7.408 m
Center of Gravity of the Insulation	7.408 m
Center of Gravity of the Nozzles	7.488 m
Center of Gravity of the Tubesheet(s)	7.408 m
Center of Gravity of the Tubes	7.408 m
Center of Gravity of the Jacket/Vapor Belt	7.498 m
Center of Gravity of the Baffles	5.693 m
Center of Gravity of Bare Shell New and Cold	7.409 m
Center of Gravity of Bare Shell Corroded	7.409 m
Vessel CG in the Operating Condition	7.448 m
Vessel CG in the Fabricated (Shop/Empty) Condition	7.403 m
Vessel CG in the Test Condition	7.404 m

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Saddle Calcs: Operating Case: Step: 22 3:22pm Apr 10, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

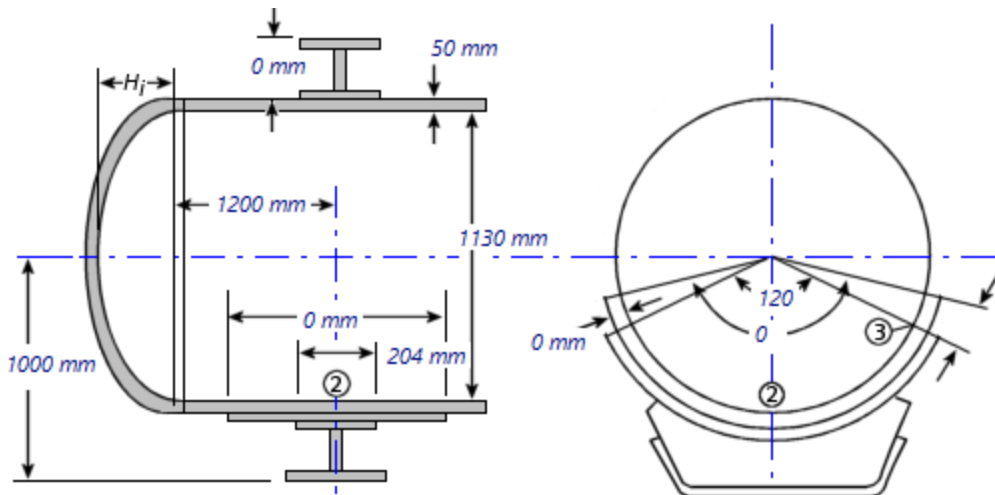
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{591.5 \cdot 47.0}, 2 \cdot 1200.0)$$

$$= 464.1064 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	591.50	mm
Shell Thickness used in this Case	t	50.000	mm
Stiffened Vessel Length per 4.15.6	L	11.75	m
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		137989.50	kPa
Head Allowable Stress used in Calculation		137989.50	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	299836.47	N
Pressure used in Analysis	P	10.501 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	55901.54	137989.50
Long. Stress at Bottom of Midspan	76243.80	137989.50
Long. Stress at Top of Saddles	71934.20	137989.50
Long. Stress at Bottom of Saddles	62823.85	137989.50



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Saddle Calcs: Operating Case: Step: 22 3:22pm Apr 10, 2026

Tangential Shear in Shell	10047.57	110391.59
Circ. Stress at Horn of Saddle	14197.49	172486.86
Circ. Compressive Stress in Shell	1045.06	137989.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(9774.8/2 + 0) \cdot 1000.0/1082.5319$
 $= 13544.4\ N$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1441.82, 0) \cdot 1000.0/7200.0$
 $= 200.3\ N$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(26273.79, 0) \cdot 1000.0/7200.0$
 $= 3649.1\ N$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(26274/2 + 0) \cdot 1000.0/1082.5319$
 $= 36406.0\ N$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 263430 + \max(200, 13544, 3649, 36406)$
 $= 299836.5\ N$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(1.23 \cdot 1.2)^2)$
 $= 1441.891\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	302081.53	N
Transverse Shear Load	13136.90	N
Longitudinal Shear Load	26273.79	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1130.0/2 + 50.0 + 10.0) \sin(120.0/2), 2 \cdot 591.5)$
 $= 1082.532\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -299836 \cdot 1.2 [1 - (1 - 1.2/11.75 + (0.592^2 - 0.0^2) / (2 \cdot 1.2 \cdot 11.75)) / (1 + (4 \cdot 0.0) / (3 \cdot 11.75))]$
 $= -32295.0\ N \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$



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Saddle Calcs: Operating Case: Step: 22 3:22pm Apr 10, 2026

$$= 299836 \cdot 12 / 4 \cdot ((1 + 2(0.6^2 - 0^2) / (12^2)) / (1 + (4 \cdot 0) / (3 \cdot 12))) - 4 \cdot 1 / 12$$

$$= 525643.0 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.501 \cdot 591.5 / (2 \cdot 47.0) - 525643 / (\pi \cdot 591.5^2 \cdot 47.0)$$

$$= 55901.54 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.501 \cdot 591.5 / (2 \cdot 47.0) + 525643 / (\pi \cdot 591.5^2 \cdot 47.0)$$

$$= 76243.80 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 10.501 \cdot 591.5 / (2 \cdot 47.0) - 32295.0 / (0.1066 \cdot \pi \cdot 591.5^2 \cdot 47.0)$$

$$= 71934.20 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 10.501 \cdot 591.5 / (2 \cdot 47.0) + 32295.0 / (0.1923 \cdot \pi \cdot 591.5^2 \cdot 47.0)$$

$$= 62823.85 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 299836(11.75 - 2 \cdot 1.2) / (11.75 + (4 \cdot 0.0 / 3))$$

$$= 238593.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 238593.28 / (591.5 \cdot 47.0)$$

$$= 10047.57 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(591.5 \cdot 47.0)}$$

$$= 130.053 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 299836 \cdot 0.1 / (47.0(204.0 + 130.05 + 130.05))$$

$$= -1045.06 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -299836 / (4 \cdot 47.0(204.0 + 130.053 + 130.053)) - 3 \cdot 0.0529 \cdot 299836 / (2 \cdot 47.0^2)$$

$$= -14197.49 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(591.5 \cdot 47.0)}, 2 \cdot 1.2)$$

$$= 464.11 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 7200.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 7200.0(150.0 - 21.1)$$

$$= 11.531 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length BpLen 1100.0000 mm



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Saddle Calcs: Operating Case: Step: 22 3:22pm Apr 10,2026

Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	231984.0	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(568.0 \cdot 47.0) + 306.0})^3) / 12$$

$$= 4852743.500 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 4852744 + 0.026 (143.848 - 23.5)^2$$

$$= 386667328.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	560.9	47.0	23.5	0.0	619499.6	0.387E+09
Wearplate	306.0	10.0	52.0	0.0	159120.0	0.258E+08
Web	10.0	349.0	231.5	0.0	807934.8	0.622E+08
BasePlate	440.0	26.0	419.0	0.0	4793359.0	0.867E+09
Totals	...	...	...	...	6379913.5	0.134E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6379914 / 0.044$$

$$= 143.848 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 299836.469$$

$$= 61023.2227 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 3392.1550 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139190.3906 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000.0 - 568.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 530.268 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 32371.7656 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3469.9102 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 154655.9844 \text{ kPa}$$



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Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(299836 + 2245) \cdot 440.0}{4 \cdot 1100.0 \cdot 154655.984}}$$

$$= 24.207 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (565.0 + 50.0 + 10.0)$$

$$= 1082.532 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1082.5317 / (3 - 1)$$

$$= 541.266 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 541.2659 \cdot 440.0 / 2$$

$$= 0.119 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 302081.531 / (1100.0 \cdot 440.0)$$

$$= 624290.250 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.1 \cdot 624290$$

$$= 74321.109 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.002 + 0.003$$

$$= 0.005 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 74321.1 / 0.0046$$

$$= 15996.093 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	...	...	0.707E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204.0 - 10.0$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 0.005$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0 / 2$$

$$= 102.000 \text{ mm}$$



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Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7074713 / 0.005}$$

$$= 39.021 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000.0 - \cos(120.0/2) (565.0 + 50.0 + 10.0) - 26.0$$

$$= 661.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 231984.0}$$

$$= 130.433$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 661.5 / 39.021$$

$$= 16.952$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 26273.8 / (2 \cdot 1100.0) \cdot 541.266 \cdot 661.5 / 2$$

$$= 2138.881 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($16.9524 < 130.4335$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (16.95)^2 / (2 \cdot 130.43^2)) 231984 /$$

$$(5/3 + 3 \cdot (16.95) / (8 \cdot 130.43) - (16.95^3) / (8 \cdot 130.43^3))$$

$$= 134115 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 15996.09 / 134114.94 + (2138.88 \cdot 102.0 / 7074720) / 154655.98$$

$$= 0.319$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	0.0	0.0	0.707E+07
Web	541.3	10.0	0.0	0.0	0.0	0.451E+05
Totals	...	...	...	...	...	0.712E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 0.007$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000.0 - \cos(625.0/2) (26.0 + 0.0 + 0.0) - 0.0$$

$$= 349.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7118985 / 0.007}$$

$$= 31.116 \text{ mm}$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 349.0/31.116$$

$$= 11.216$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 26273.793 / (2 \cdot 1100.0)$$

$$= 11.943 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 11.943 \cdot 541.266 \cdot 349.0$$

$$= 2256.899 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2256.899/69794.039$$

$$= 32324.332 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (11.216 < 130.4335) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.22)^2 / (2 \cdot 130.43^2)) 231984 /$$

$$(5/3 + 3 \cdot (11.22) / (8 \cdot 130.43) - (11.22)^3 / (8 \cdot 130.43^3))$$

$$= 136050 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 15715.36/136050.02 + (2256.9 \cdot 102.0/7118985) / 154655.98$$

$$= 0.325$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0006 m²
BasePlate Allowable Stress	S	137989.20 kPa
Area Available in a Single Bolt	BltArea	0.0005 m²
Saddle Load Q0 (Weight)	Q0	265675.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	3649.1 N
Maximum Transverse Force	Ft	13136.9 N
Maximum Longitudinal Force	F1	26273.8 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 26273.8 / (0. \cdot 4)$$

$$= 13495.1 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 13136.9 / (0. \cdot 4)$$

$$= 6747.6 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:



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= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 = $26273.79 / (82795.2 \cdot 4.0)$
 = 0.0001 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1000.0 \cdot 13136.9 + 0.0$
 = 13142.22 N-m

Eccentricity (e):
 = R_{mom} / QO
 = $13142.22 / 265675.53$
 = 49.45 mm < $B_{\text{plen}} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.0001, 0.0]$
 = 0.0001 m²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.984$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 = $\min(204.0 + 1.56 \cdot \sqrt{ 591.5 \cdot 47.0 }, 2 \cdot 2400.0)$
 = 464.1064 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	591.50	mm
Shell Thickness used in this Case	t	50.000	mm
Stiffened Vessel Length per 4.15.6	L	11.75	m
Distance from Saddle to Vessel tangent	a1 or a	2400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		137989.50	kPa
Head Allowable Stress used in Calculation		137989.50	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 287761.75 N
Pressure used in Analysis P 10.501 N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	62995.64	137989.50
Long. Stress at Bottom of Midspan	69149.70	137989.50
Long. Stress at Top of Saddles	90908.30	137989.50
Long. Stress at Bottom of Saddles	52307.23	137989.50
Tangential Shear in Shell	7167.75	110391.59
Circ. Stress at Horn of Saddle	13625.74	172486.86
Circ. Compressive Stress in Shell	1002.98	137989.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (9774.8 / 2 + 0) \cdot 1000.0 / 1082.5319$$

$$= 13544.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(1441.82, 0) \cdot 1000.0 / 7200.0$$

$$= 200.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(26273.79, 0) \cdot 1000.0 / 7200.0$$

$$= 3649.1 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (26274 / 2 + 0) \cdot 1000.0 / 1082.5319$$

$$= 36406.0 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 251356 + \max(200, 13544, 3649, 36406)$$

$$= 287761.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi / 4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi / 4 (1.23 \cdot 1.2)^2)$$

$$= 1441.891 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 290006.81 N

Transverse Shear Load 13136.90 N

Longitudinal Shear Load 26273.79 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2 (\text{ShellID} / 2 + t + \text{WearPadThickness}) \sin(\theta / 2), 2 \cdot R_m)$$

$$= \min(2 (1130.0 / 2 + 50.0 + 10.0) \sin(120.0 / 2), 2 \cdot 591.5)$$

$$= 1082.532 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923



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Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -287762 \cdot 2.4 [1 - (1 - 2.4/11.75 + (0.592^2 - 0.0^2)/(2 \cdot 2.4 \cdot 11.75))/(1 + (4 \cdot 0.0)/(3 \cdot 11.75))] \\
 &= -136835.7 \text{ N-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L] \\
 &= 287762 \cdot 12/4 [(1 + 2(0.6^2 - 0^2)/(12^2))/(1 + (4 \cdot 0)/(3 \cdot 12)) - 4 \cdot 2/12] \\
 &= 159020.7 \text{ N-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= 10.501 \cdot 591.5/(2 \cdot 47.0) - 159020.7/(\pi \cdot 591.5^2 \cdot 47.0) \\
 &= 62995.64 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 10.501 \cdot 591.5/(2 \cdot 47.0) + 159020.7/(\pi \cdot 591.5^2 \cdot 47.0) \\
 &= 69149.70 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\
 &= 10.501 \cdot 591.5/(2 \cdot 47.0) - 136835.7/(0.1066 \cdot \pi \cdot 591.5^2 \cdot 47.0) \\
 &= 90908.30 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 10.501 \cdot 591.5/(2 \cdot 47.0) + 136835.7/(0.1923 \cdot \pi \cdot 591.5^2 \cdot 47.0) \\
 &= 52307.23 \text{ kPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 287762(11.75 - 2 \cdot 2.4)/(11.75 + (4 \cdot 0.0/3)) \\
 &= 170208.0 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 170208.03/(591.5 \cdot 47.0) \\
 &= 7167.75 \text{ kPa}
 \end{aligned}$$

Decay Length (4.15.20) [x1, x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{Rm \cdot t} \\
 &= 0.78 \cdot \sqrt{591.5 \cdot 47.0} \\
 &= 130.053 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 287762 \cdot 0.1/(47.0(204.0 + 130.05 + 130.05)) \\
 &= -1002.98 \text{ kPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2) \\
 &= -287762/(4 \cdot 47.0(204.0 + 130.053 + 130.053)) - 3 \cdot 0.0529 \cdot 287762/(2 \cdot 47.0^2) \\
 &= -13625.74 \text{ kPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\
 &= \min(204.0 + 1.56 \cdot \sqrt{591.5 \cdot 47.0}, 2 \cdot 2.4) \\
 &= 464.11 \text{ mm}
 \end{aligned}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	231984.0	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(568.0 \cdot 47.0)} + 306.0)^3) / 12$$

$$= 4852743.500 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 4852744 + 0.026 (143.848 - 23.5)^2$$

$$= 386667328.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	560.9	47.0	23.5	0.0	619499.6	0.387E+09
Wearplate	306.0	10.0	52.0	0.0	159120.0	0.258E+08
Web	10.0	349.0	231.5	0.0	807934.8	0.622E+08
BasePlate	440.0	26.0	419.0	0.0	4793359.0	0.867E+09
Totals	...	...	...	...	6379913.5	0.134E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6379914 / 0.044$$

$$= 143.848 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 287761.75$$

$$= 58565.7539 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 3255.5496 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139190.3906 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000.0 - 568.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 530.268 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 31068.1211 \text{ N-m}$$



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Bending Stress, $S_b = (M \cdot C_1 / I) = 3330.1733 \text{ kPa}$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 154655.9844 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(287762 + 2245) \cdot 440.0}{4 \cdot 1100.0 \cdot 154655.9844}}$$

$$= 23.718 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (565.0 + 50.0 + 10.0)$$

$$= 1082.532 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1082.5317 / (3 - 1)$$

$$= 541.266 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 541.2659 \cdot 440.0 / 2$$

$$= 0.119 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 290006.812 / (1100.0 \cdot 440.0)$$

$$= 599336.250 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.1 \cdot 599336$$

$$= 71350.359 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.002 + 0.003$$

$$= 0.005 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 71350.4 / 0.0046$$

$$= 15356.699 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	...	...	0.707E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204.0 - 10.0$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 0.005$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 204.0/2
= 102.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7074713 / 0.005}$
= 39.021 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= $1000.0 - \cos(120.0/2) (565.0 + 50.0 + 10.0) - 26.0$
= 661.500 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199955 / 231984.0}$
= 130.433

Slenderness ratio [KL/r]:
= KL/r
= $1 \cdot 661.5 / 39.021$
= 16.952

Bending Moment [Rm]:
= $Fl / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= $26273.8 / (2 \cdot 1100.0) \cdot 541.266 \cdot 661.5 / 2$
= 2138.881 N-m

Compressive Allowable, $KL/r < Cc$ ($16.9524 < 130.4335$) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (16.95)^2 / (2 \cdot 130.43^2)) 231984 / (5/3 + 3 \cdot (16.95) / (8 \cdot 130.43) - (16.95)^3 / (8 \cdot 130.43^3))$
= 134115 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc / Sca + (Rm \cdot C1 / I) / Sba$
= $15356.7 / 134114.94 + (2138.88 \cdot 102.0 / 7074720) / 154655.98$
= 0.314

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	0.0	0.0	0.707E+07
Web	541.3	10.0	0.0	0.0	0.0	0.451E+05
Totals	...	...	...	...	...	0.712E+07

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/0.007
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 204.0/2
= 102.000 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= $1000.0 - \cos(625.0/2) (26.0 + 0.0 + 0.0) - 0.0$
= 349.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7118985 / 0.007}$$

$$= 31.116 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 349.0 / 31.116$$

$$= 11.216$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 26273.793 / (2 \cdot 1100.0)$$

$$= 11.943 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 11.943 \cdot 541.266 \cdot 349.0$$

$$= 2256.899 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2256.899 / 69794.039$$

$$= 32324.332 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (11.216 < 130.4335) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.22)^2 / (2 \cdot 130.43^2)) 231984 / (5/3 + 3 \cdot (11.22) / (8 \cdot 130.43) - (11.22)^3 / (8 \cdot 130.43^3))$$

$$= 136050 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1 / I) / S_{ba}$$

$$= 15087.19 / 136050.02 + (2256.9 \cdot 102.0 / 7118985) / 154655.98$$

$$= 0.320$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0006 m²
BasePlate Allowable Stress	S	137989.20 kPa
Area Available in a Single Bolt	BlArea	0.0005 m²
Saddle Load Q0 (Weight)	Q0	253600.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	3649.1 N
Maximum Transverse Force	Ft	13136.9 N
Maximum Longitudinal Force	F1	26273.8 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 26273.8 / (0. \cdot 4)$$

$$= 13495.1 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 13136.9 / (0. \cdot 4)$$

$$= 6747.6 \text{ kPa. Must be less than } 82795.2 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $26273.79 / (82795.2 \cdot 4.0)$
= 0.0001 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1000.0 \cdot 13136.9 + 0.0$
= 13142.22 N-m

Eccentricity (e):
= $Rmom / QO$
= $13142.22 / 253600.77$
= 51.80 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.0001, 0.0]$
= 0.0001 m²

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.984$, Temp. Reduction = 1 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

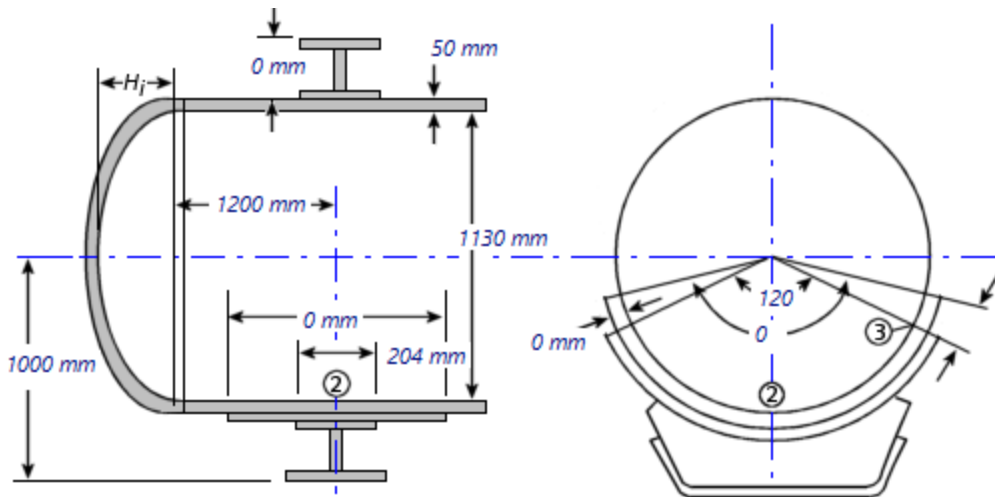
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{591.5 \cdot 47.0}, 2 \cdot 1200.0)$$

$$= 464.1064 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	591.50	mm
Shell Thickness used in this Case	t	50.000	mm
Stiffened Vessel Length per 4.15.6	L	11.75	m
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		233982.17	kPa
Head Allowable Stress used in Calculation		233982.17	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	297297.47	N
Pressure used in Analysis	P	14.778 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	82901.02	233982.17
Long. Stress at Bottom of Midspan	103071.03	233982.17
Long. Stress at Top of Saddles	98797.92	233982.17
Long. Stress at Bottom of Saddles	89764.72	233982.17



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Saddle Calcs: Test Case:

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Tangential Shear in Shell	9962.48	138569.45
Circ. Stress at Horn of Saddle	14077.27	350973.22
Circ. Compressive Stress in Shell	1036.21	233982.17

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (4887.4/2 + 0) \cdot 1000.0/1082.5319$$

$$= 6772.2 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(720.91, 0) \cdot 1000.0/7200.0$$

$$= 100.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 290525 + \max(100, 6772, 0, 0)$$

$$= 297297.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi/4 (1.23 \cdot 1.2)^2)$$

$$= 1441.891 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	299542.53	N
Transverse Shear Load	2443.70	N
Longitudinal Shear Load	720.91	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1130.0/2 + 50.0 + 10.0) \sin(120.0/2), 2 \cdot 591.5)$$

$$= 1082.532 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -297297 \cdot 1.2 [1 - (1 - 1.2/11.75 + (0.592^2 - 0.0^2)/(2 \cdot 1.2 \cdot 11.75)) / (1 + (4 \cdot 0.0)/(3 \cdot 11.75))]$$

$$= -32021.5 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 297297 \cdot 12/4 ((1 + 2(0.6^2 - 0^2)/(12^2)) / (1 + (4 \cdot 0)/(3 \cdot 12)) - 4 \cdot 1/12)$$

$$= 521191.9 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 14.778 \cdot 591.5/(2 \cdot 47.0) - 521192/(\pi \cdot 591.5^2 \cdot 47.0)$$

$$= 82901.02 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 14.778 \cdot 591.5 / (2 \cdot 47.0) + 521192 / (\pi \cdot 591.5^2 \cdot 47.0) \\
 &= 103071.03 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 14.778 \cdot 591.5 / (2 \cdot 47.0) - 32021.5 / (0.1066 \cdot \pi \cdot 591.5^2 \cdot 47.0) \\
 &= 98797.92 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 14.778 \cdot 591.5 / (2 \cdot 47.0) + 32021.5 / (0.1923 \cdot \pi \cdot 591.5^2 \cdot 47.0) \\
 &= 89764.72 \text{ kPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 297297(11.75 - 2 \cdot 1.2) / (11.75 + (4 \cdot 0.0 / 3)) \\
 &= 236572.9 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 1.1707 \cdot 236572.89 / (591.5 \cdot 47.0) \\
 &= 9962.48 \text{ kPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(591.5 \cdot 47.0)} \\
 &= 130.053 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 297297 \cdot 0.1 / (47.0(204.0 + 130.05 + 130.05)) \\
 &= -1036.21 \text{ kPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -297297 / (4 \cdot 47.0(204.0 + 130.053 + 130.053)) - \\
 &\quad 3 \cdot 0.0529 \cdot 297297 / (2 \cdot 47.0^2) \\
 &= -14077.27 \text{ kPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(204.0 + 1.56 \cdot \sqrt{(591.5 \cdot 47.0)}, 2 \cdot 1.2) \\
 &= 464.11 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw, c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(568.0 \cdot 47.0)} + 306.0)^3) / 12 \\
 &= 4852743.500 \text{ mm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I, shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 4852744 + 0.026(143.848 - 23.5)^2 \\
 &= 386667328.000 \text{ mm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	560.9	47.0	23.5	0.0	619499.6	0.387E+09
Wearplate	306.0	10.0	52.0	0.0	159120.0	0.258E+08
Web	10.0	349.0	231.5	0.0	807934.8	0.622E+08
BasePlate	440.0	26.0	419.0	0.0	4793359.0	0.867E+09
Totals	...	...	...	...	6379913.5	0.134E+10

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 6379914 / 0.044 \\
 &= 143.848 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 297297.469 \\
 &= 60506.4805 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 3363.4307 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1000.0 - 568.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 530.268 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 32097.6426 \text{ N-m}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 3440.5269 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\
 &= \sqrt{(3(297297 + 2245)440.0 / (4 \cdot 1100.0 \cdot 173320.141))} \\
 &= 22.770 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0 / 2) (565.0 + 50.0 + 10.0) \\
 &= 1082.532 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1082.5317 / (3 - 1) \\
 &= 541.266 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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$$= 541.2659 \cdot 440.0/2$$

$$= 0.119 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 299542.531 / (1100.0 \cdot 440.0)$$

$$= 619043.062 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.1 \cdot 619043$$

$$= 73696.438 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.002 + 0.003$$

$$= 0.005 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 73696.4 / 0.0046$$

$$= 15861.645 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	...	...	0.707E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204.0 - 10.0$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 0.005$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0 / 2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7074713 / 0.005)}$$

$$= 39.021 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000.0 - \cos(120.0/2) (565.0 + 50.0 + 10.0) - 26.0$$

$$= 661.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980.2)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 661.5 / 39.021$$

$$= 16.952$$



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Saddle Calcs: Test Case:

Step: 23 3:22pm Apr 10, 2026

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 720.9 / (2 \cdot 1100.0) \cdot 541.266 \cdot 661.5 / 2$$

$$= 58.687 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ (16.9524 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (16.95)^2 / (2 \cdot 123.21^2)) 259980 /$$

$$(5/3 + 3 \cdot (16.95) / (8 \cdot 123.21) - (16.95^3) / (8 \cdot 123.21^3))$$

$$= 149900 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 15861.64 / 149900.41 + (58.69 \cdot 102.0 / 7074720) / 173320.14$$

$$= 0.111$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	0.0	0.0	0.707E+07
Web	541.3	10.0	0.0	0.0	0.0	0.451E+05
Totals	...	...	...	...	...	0.712E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 0.007$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000.0 - \cos(625.0 / 2) (26.0 + 0.0 + 0.0) - 0.0$$

$$= 349.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (7118985 / 0.007) }$$

$$= 31.116 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 349.0 / 31.116$$

$$= 11.216$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 720.909 / (2 \cdot 1100.0)$$

$$= 0.328 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.328 \cdot 541.266 \cdot 349.0$$

$$= 61.926 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 61.926 / 69794.039$$

$$= 886.925 \text{ kPa}$$



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Saddle Calcs: Test Case:

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Compressive Allowable, $KL/r < C_c$ (11.216 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.22)^2 / (2 \cdot 123.21^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.22) / (8 \cdot 123.21) - (11.22^3) / (8 \cdot 123.21^3))$$

$$= 152232 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 15583.27 / 152232.39 + (61.93 \cdot 102.0 / 7118985) / 173320.14$$

$$= 0.107$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0006 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0005 m ²
Saddle Load Q0 (Weight)	Q0	292770.3 N
Saddle Load QL (Wind/Seismic contribution)	QL	100.1 N
Maximum Transverse Force	Ft	2443.7 N
Maximum Longitudinal Force	F1	720.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 720.9 / (0. \cdot 4)$$

$$= 370.3 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2443.7 / (0. \cdot 4)$$

$$= 1255.2 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 720.91 / (103192.15 \cdot 4.0)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000.0 \cdot 2443.7 + 0.0$$

$$= 2444.69 \text{ N-m}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 2444.69 / 292770.34$$

$$= 8.35 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



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Saddle Calcs: Test Case:

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= max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0., 0.0]
 = 0.0000 m²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min(b + $1.56 \cdot \sqrt{Rm \cdot t}$, 2a)
 = min(204.0 + $1.56 \cdot \sqrt{591.5 \cdot 47.0}$, 2 • 2400.0)
 = 464.1064 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	591.50	mm
Shell Thickness used in this Case	t	50.000	mm
Stiffened Vessel Length per 4.15.6	L	11.75	m
Distance from Saddle to Vessel tangent	a1 or a	2400.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		233982.17	kPa
Head Allowable Stress used in Calculation		233982.17	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		271165.28	N
Pressure used in Analysis	P	14.778	N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	90086.46	233982.17
Long. Stress at Bottom of Midspan	95885.60	233982.17
Long. Stress at Top of Saddles	116389.27	233982.17
Long. Stress at Bottom of Saddles	80014.51	233982.17
Tangential Shear in Shell	6754.35	138569.45
Circ. Stress at Horn of Saddle	12839.89	350973.22
Circ. Compressive Stress in Shell	945.13	233982.17

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(4887.4/2 + 0) • 1000.0/1082.5319
 = 6772.2 N

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls
 = max(720.91, 0) • 1000.0/7200.0
 = 100.1 N

Load Combination Results for Q + Wind or Seismic [Q]:



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Saddle Calcs: Test Case: Step: 23 3:22pm Apr 10, 2026

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 264393 + max(100, 6772, 0, 0)
 = 271165.3 N

Longitudinal Wind Force [F_L]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (1.23 • 1.2)²)
 = 1441.891 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	273410.34	N
Transverse Shear Load	2443.70	N
Longitudinal Shear Load	720.91	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1130.0/2 + 50.0 + 10.0)sin(120.0/2), 2•591.5)
 = 1082.532 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M₁]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -271165•2.4[1-(1-2.4/11.75+(0.592²-0.0²)/
 (2•2.4•11.75))/(1+(4•0.0)/(3•11.75))]
 = -128943.8 N-m

Moment per Equation 4.15.2 [M₂]:
 = Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 271165•12/4((1+2(0.6²-0²)/(12²))/(1+(4•0)/
 (3•12))-4•2/12)
 = 149849.3 N-m

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 14.778 • 591.5/(2•47.0) - 149849.3/(π •591.5²•47.0)
 = 90086.46 kPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 14.778 • 591.5/(2 • 47.0) + 149849.3/(π • 591.5² • 47.0)
 = 95885.60 kPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 14.778•591.5/(2•47.0)-128943.8/(0.1066• π •591.5²•47.0)
 = 116389.27 kPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 14.778•591.5/(2•47.0)+128943.8/(0.1923• π •591.5²•47.0)
 = 80014.51 kPa

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))
 = 271165(11.75 - 2 • 2.4)/(11.75 + (4 • 0.0/3))
 = 160391.4 N



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Saddle Calcs: Test Case:

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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 160391.41 / (591.5 \cdot 47.0)$$

$$= 6754.35 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(591.5 \cdot 47.0)}$$

$$= 130.053 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 271165 \cdot 0.1 / (47.0(204.0 + 130.05 + 130.05))$$

$$= -945.13 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -271165 / (4 \cdot 47.0(204.0 + 130.053 + 130.053)) -$$

$$3 \cdot 0.0529 \cdot 271165 / (2 \cdot 47.0^2)$$

$$= -12839.89 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(591.5 \cdot 47.0)}, 2 \cdot 2.4)$$

$$= 464.11 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw, c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(568.0 \cdot 47.0)} + 306.0)47.0^3) / 12$$

$$= 4852743.500 \text{ mm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 4852744 + 0.026(143.848 - 23.5)^2$$

$$= 386667328.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	560.9	47.0	23.5	0.0	619499.6	0.387E+09
Wearplate	306.0	10.0	52.0	0.0	159120.0	0.258E+08
Web	10.0	349.0	231.5	0.0	807934.8	0.622E+08
BasePlate	440.0	26.0	419.0	0.0	4793359.0	0.867E+09
Totals	...	...	...	...	6379913.5	0.134E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6379914 / 0.044$$

$$= 143.848 \text{ mm}$$



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Saddle Calcs: Test Case:

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Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 271165.281 \\ &= 55188.0156 \text{ N} \end{aligned}$$

$$\text{Tension Stress, } St = (Fh / As) = 3067.7881 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1000.0 - 568.0 \cdot \sin(120.0/2) / 1.0472 \\ &= 530.268 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 29276.2891 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3138.1079 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(271165 + 2245) \cdot 440.0) / (4 \cdot 1100.0 \cdot 173320.141)} \\ &= 21.755 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (565.0 + 50.0 + 10.0) \\ &= 1082.532 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1082.5317 / (3 - 1) \\ &= 541.266 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 541.2659 \cdot 440.0/2 \\ &= 0.119 \text{ m}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 273410.344 / (1100.0 \cdot 440.0) \\ &= 565037.500 \text{ N/m}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 0.1 \cdot 565038 \\ &= 67267.133 \text{ N} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 0.002 + 0.003 \\ &= 0.005 \text{ m}^2 \end{aligned}$$



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Saddle Calcs: Test Case:

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Compressive Stress [Sc]:

= P/Ar

= 67267.1/0.0046

= 14477.869 kPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	...	...	0.707E+07

Rib dimension [D]:

= Saddle Width - Web Thickness

= 204.0 - 10.0

= 194.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/0.005

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 204.0/2

= 102.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{7074713/0.005}$

= 39.021 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1000.0 - $\cos(120.0/2)$ (565.0 + 50.0 + 10.0) - 26.0

= 661.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/259980.2}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 661.5/39.021

= 16.952

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 720.9 / (2 • 1100.0) • 541.266 • 661.5/2

= 58.687 N-m

Compressive Allowable, $KL/r < Cc$ (16.9524 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (16.95)^2 / (2 \cdot 123.21^2)) 259980 /$

$(5/3 + 3 \cdot (16.95) / (8 \cdot 123.21) - (16.95)^3 / (8 \cdot 123.21^3))$

= 149900 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 14477.87/149900.41 + (58.69 • 102.0/7074720) / 173320.14

= 0.101

Check of Inside Ribs:



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Saddle Calcs: Test Case:

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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	0.0	0.0	0.707E+07
Web	541.3	10.0	0.0	0.0	0.0	0.451E+05
Totals	...	...	...	...	...	0.712E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/0.007$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000.0 - \cos(625.0/2) (26.0 + 0.0 + 0.0) - 0.0$$

$$= 349.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7118985/0.007)}$$

$$= 31.116 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 349.0/31.116$$

$$= 11.216$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 720.909 / (2 \cdot 1100.0)$$

$$= 0.328 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.328 \cdot 541.266 \cdot 349.0$$

$$= 61.926 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 61.926/69794.039$$

$$= 886.925 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (11.216 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.22)^2 / (2 \cdot 123.21^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.22) / (8 \cdot 123.21) - (11.22)^3 / (8 \cdot 123.21^3))$$

$$= 152232 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 14223.78/152232.39 + (61.93 \cdot 102.0/7118985) / 173320.14$$

$$= 0.099$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	171986.91 kPa



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Saddle Calcs: Test Case: Step: 23 3:22pm Apr 10, 2026

Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0006 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0005 m ²
Saddle Load Q0 (Weight)	Q0	266638.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	100.1 N
Maximum Transverse Force	Ft	2443.7 N
Maximum Longitudinal Force	F1	720.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $720.9 / (0. \cdot 4)$
 = 370.3 kPa. Must be less than 103192.1 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $2443.7 / (0. \cdot 4)$
 = 1255.2 kPa. Must be less than 103192.1 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $720.91 / (103192.15 \cdot 4.0)$
 = 0.0000 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$
 = $1000.0 \cdot 2443.7 + 0.0$
 = 2444.69 N-m

Eccentricity (e):

= $Rmom / Q0$
 = $2444.69 / 266638.16$
 = 9.16 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[0.0, 0., 0.0]$
 = 0.0000 m²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Input, Nozzle Desc: Cooling water i

From: 30

Pressure for Reinforcement Calculations	P	1.0001	N/mm ²
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	1130.00	mm
Design Length of Section	L	1199.9701	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		0.81	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117991.02	kPa
Allowable Stress At Ambient	Sna	117991.02	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		18.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137989.50	kPa
Pad Allowable Stress At Ambient	Spa	137989.50	kPa
Diameter of Pad along vessel surface	Dp	697.2000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		120.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

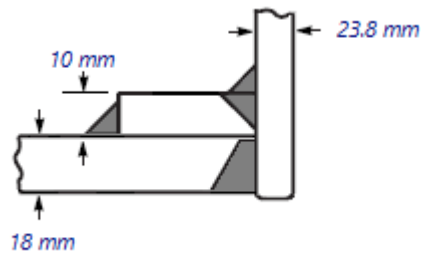


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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026



Reinforcement CALCULATION, Description: Cooling water i

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	16.126 in.
Actual Thickness Used in Calculation	0.937 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.0 \cdot 568.0 / (137990 \cdot 1.0 - 0.6 \cdot 1.0)$
 $= 4.1342 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.0 \cdot 207.8 / (117991 \cdot 1.0 - 0.6 \cdot 1.0)$
 $= 1.7702 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1355 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	831.2008 mm
Parallel to Vessel Wall, opening length	d	415.6004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991.0 / 137989.5)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991.0 / 137989.5)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137989.5 / 137989.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: Cooling water i

Nozl: 9 3:22pm Apr 10,2026

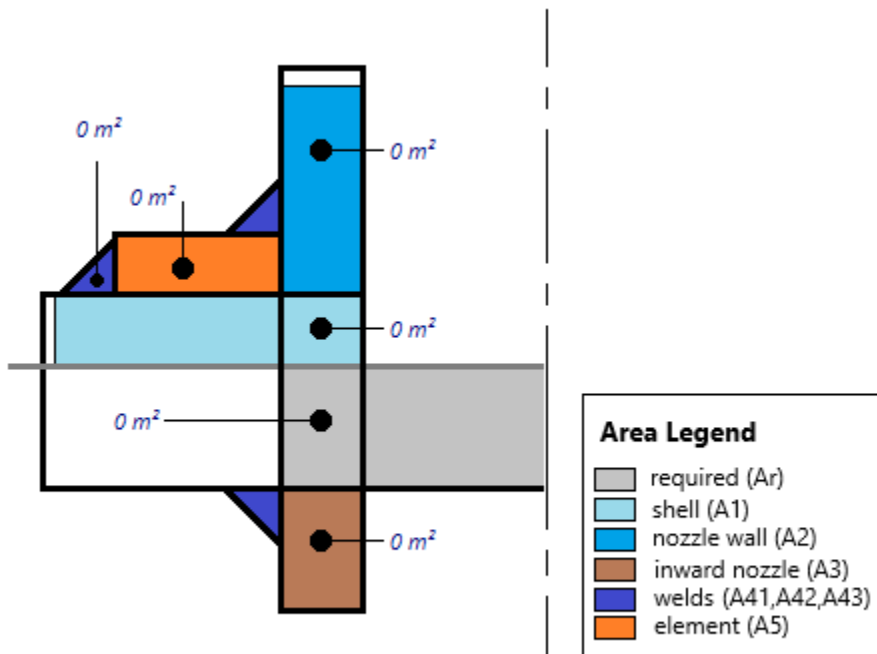


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.002	0.001	NA
Area in Shell	A1	0.004	0.005	NA
Area in Nozzle Wall	A2	0.001	0.001	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.002	0.002	NA
TOTAL AREA AVAILABLE	Atot	0.008	0.008	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (415.6004 \cdot 4.1342 \cdot 1.0 + 2 \cdot 20.7998 \cdot 4.1342 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 0.002 \text{ m}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 415.6(1.0 \cdot 15.0 - 1.0 \cdot 4.134) - 2 \cdot 20.8 \\
 &\quad (1.0 \cdot 15.0 - 1.0 \cdot 4.1342) \cdot (1 - 0.855) \\
 &= 0.004 \text{ m}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.5) \cdot (20.8 - 1.77) \cdot 0.8551
 \end{aligned}$$



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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

$$= 0.001 \text{ m}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (697.2 - 457.2) \cdot 10.0 \cdot 1.0$$

$$= 0.002 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.7702 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 7.1342 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.1342 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.4172 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(7.1342, 4.5)]$$

$$= 7.1342 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.7702, 7.1342)$$

$$= 7.1342 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 23.800 = 20.825 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 17039.1, Allowable : 117991.0 kPa Passed

Expansion : 0.0, Allowable : 277938.5 kPa Passed

Occasional : 4757.2, Allowable : 156928.1 kPa Passed

Shear : 9517.4, Allowable : 82593.7 kPa Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.825$, $tr = 1.77$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.099$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 13°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.276$, Temp. Reduction = 78°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Govrn. thk, $t_g = 10.0$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.276$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.276$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 4.134$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.276$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.00 / 1.96 = 0.510$

Weld Size Calculations, Description: Cooling water i

Intermediate Calc. for nozzle/shell Welds T_{min} 10.0000 mm

Intermediate Calc. for pad/shell Welds T_{minPad} 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.0017 - 0.0044 + 2 \cdot 20.7998 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 4.1342)) \cdot 137990)$$

$$= \max(0, -320222.56) \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.0012 + 0.0024 + 0.0002 - 0.0 \cdot 0.86) \cdot 137990$$

$$= 525158.25 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.0012 + 0.0 + 0.0001 + (0.0005)) \cdot 137990$$



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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

= 253817.47 N

Weld Load [W3]:

= (A2+A3+A4+A5+(2*tn*t*fr1))*S

= (0.0012 + 0.0 + 0.0002 + 0.0024 + (0.0005)) • 137990

= 598781.88 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw

= (3.1416/2.0) • 457.2 • 10.0 • 0.49 • 117991

= 415202. N

Shear, Pad Element Weld [Spew]:

= (π/2) • DP • WP • 0.49 • SEW

= (3.1416/2.0) • 697.2 • 10.0 • 0.49 • 137990

= 740470. N

Shear, Nozzle Wall [Snw]:

= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn

= (3.1416 • 218.2001) • (23.7998 - 3.0) • 0.7 • 117991

= 1177604. N

Tension, Pad Groove Weld [Tpgw]:

= (π/2) • Dlo • Wgpn • 0.74 • Seg

= (3.1416/2) • 457.2 • 10.0 • 0.74 • 137990

= 733318. N

Tension, Shell Groove Weld [Tngw]:

= (π/2) • Dlo • Wgnvi • 0.74 • Sng

= (3.1416/2.0) • 457.2 • (8.0 - 0.0) • 0.74 • 137990

= 586654. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (740470 + 1177604) = 1918074 N

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (415202 + 733318 + 586654 + 0) = 1735174 N

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (740470 + 586654 + 0) = 1327125 N

Summary of Failure Path Calculations:

Path 1-1 = 1918074 N , must exceed W = 0 N or W1 = 525158 N

Path 2-2 = 1735173 N , must exceed W = 0 N or W2 = 253817 N

Path 3-3 = 1327124 N , must exceed W = 0 N or W3 = 598781 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 N/mm²

Nozzle is O.K. for the External Pressure: 0 N/mm²

The Drop for this Nozzle is : 48.3115 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 316.3115 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 163.36 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 415.60 mm



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 Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: Cooling water :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1130.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	409.600	mm
Nozzle Thickness	23.7998	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 117991.02	kPa
Nozzle Hot S.I. Allowable	SNmh 117991.02	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	697.200	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.00	N/mm ²
Radial Load	P	-33750.00	N
Circumferential Shear	Vc	22499.85	N
Longitudinal Shear	Vl	22499.85	N
Circumferential Moment	Mc	28350.06	N-m
Longitudinal Moment	Ml	28350.06	N-m
Torsional Moment	Mt	40499.93	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{580.5(18.0 - 3.0)}$
 $= 765.136 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1161.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	25.000	mm
Nozzle Thickness used	(t) =	20.800	mm
T / t	=	1.202	
d / t	=	21.981	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.684$$



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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Nr/P = 0.054
 Mr/P = 0.097
 MØ/P = 0.035
 NØ/P = 0.120
 MrD/Mc = 0.231
 NrDL/Mc = 0.100
 MØd/Mc = 0.095
 NØDL/Mc = 0.119
 MrD/Ml = 0.106
 NrDL/Ml = 0.051
 MØD/Ml = 0.044
 NØDL/Ml = 0.170

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6467	6467	6467	6467
Outplane Bending (P)	11424	-11424	11424	-11424
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16822	-16822	16822	16822
Outplane Bending (ML)	-26192	26192	26192	-26192
Normal Pressure Stress	22228	23227	22228	23227
Outplane Stress Summary	-2895	27640	83135	8900

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2937	2937	2937	2937
Inplane Bending (P)	31370	-31370	31370	-31370
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5026	-5026	5026	5026
Inplane Bending (ML)	-62968	62968	62968	-62968
Inplane Pressure Stress	11114	11114	11114	11114
Inplane Stress Summary	-22572	40622	113416	-75261

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1247	1247	-1247	-1247
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4929	4929	4929	4929
Shear Stress Summary	6177	6177	3681	3681

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	24344	43091	113850	84479

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6467	6467	6467	6467
Outplane Bending (P)	11424	-11424	11424	-11424
Outplane Membrane (Mc)	-11762	-11762	11762	11762
Outplane Bending (Mc)	-56783	56783	56783	-56783
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11114	11114	11114	11114
Outplane Stress Summary	-39540	51178	97551	-38864

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2937	2937	2937	2937
Inplane Bending (P)	31370	-31370	31370	-31370
Inplane Membrane (Mc)	-9921	-9921	9921	9921
Inplane Bending (Mc)	-137374	137374	137374	-137374
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22228	23227	22228	23227
Inplane Stress Summary	-90760	122248	203831	-132659

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1247	-1247	1247	1247
Torsional Shear (Mt)	4929	4929	4929	4929
Shear Stress Summary	3681	3681	6177	6177

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	91022	122434	204183	133058

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6467	6467	6467	6467
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16822	-16822	16822	16822
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11410	12617	11410	12617
Hoop Stress Summary	1054	2261	34700	35907

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1178	1178	1178	1178
Axial Bending (P)	32576	-32576	32576	-32576
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9521	-9521	9521	9521
Axial Bending (ML)	-69166	69166	69166	-69166
Axial Pressure Stress	5701	5701	5701	5701

Axial Stress Summary	-39230	33949	118145	-85341
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1503	1503	-1503	-1503
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5922	5922	5922	5922

Shear Stress Summary	7425	7425	4419	4419
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	42932	35596	118373	121565
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6467	6467	6467	6467
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11762	-11762	11762	11762
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11410	12617	11410	12617

Hoop Stress Summary	6115	7322	29639	30846
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1178	1178	1178	1178
Axial Bending (P)	32576	-32576	32576	-32576
Axial Membrane (Mc)	-9521	-9521	9521	9521
Axial Bending (Mc)	-155438	155438	155438	-155438
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5701	5701	5701	5701

Axial Stress Summary	-125502	120221	204418	-171613
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1503	-1503	1503	1503
Shear due to Torsion	5922	5922	5922	5922

Shear Stress Summary	4419	4419	7425	7425
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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	131907	120393	204728	202997

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{575.5(18.0 - 3.0)}$$

$$= 763.807 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1151.000 mm
Nozzle Outside Diameter	(d) =	697.200 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	140.800 mm
T / t	=	0.107
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.306$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.031
MrD/Mc	=	0.240
NrDL/Mc	=	0.095
M0d/Mc	=	0.071
N0DL/Mc	=	0.058
MrD/Ml	=	0.049
NrDL/Ml	=	0.043
M0D/Ml	=	0.015
N0DL/Ml	=	0.029

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4715	4715	4715	4715
Outplane Bending (P)	17898	-17898	17898	-17898
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5170	-5170	5170	5170
Outplane Bending (ML)	-16512	16512	16512	-16512
Normal Pressure Stress	37368	38368	37368	38368
Outplane Stress Summary	38299	36527	81666	13844

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6611	6611	6611	6611
Inplane Bending (P)	63002	-63002	63002	-63002
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Nozzle Calcs.: Cooling water i Nozl: 9 3:22pm Apr 10,2026

Inplane Membrane (ML)	-7852	-7852	7852	7852
Inplane Bending (ML)	-53191	53191	53191	-53191
Inplane Pressure Stress	18684	18684	18684	18684

Inplane Stress Summary	27254	7632	149343	-83045
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1365	1365	-1365	-1365
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3530	3530	3530	3530

Shear Stress Summary	4895	4895	2164	2164
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40154	37334	149412	96986
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4715	4715	4715	4715
Outplane Bending (P)	17898	-17898	17898	-17898
Outplane Membrane (Mc)	-10466	-10466	10466	10466
Outplane Bending (Mc)	-77205	77205	77205	-77205
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18684	18684	18684	18684

Outplane Stress Summary	-46373	72241	128970	-61237
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6611	6611	6611	6611
Inplane Bending (P)	63002	-63002	63002	-63002
Inplane Membrane (Mc)	-17105	-17105	17105	17105
Inplane Bending (Mc)	-260602	260602	260602	-260602
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37368	38368	37368	38368

Inplane Stress Summary	-170724	225474	384691	-261519
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1365	-1365	1365	1365
Torsional Shear (Mt)	3530	3530	3530	3530

Shear Stress Summary	2164	2164	4895	4895
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	170758	225501	384780	261636

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22228	23227	22228	23227	22228	23227	22228	23227
Circ. Pl (SUS)	-10355	-10355	23289	23289	-6984	-6984	12858	12858
Circ. Q (SUS)	-14768	14768	37616	-37616	-106004	106004	168744	-168744
Long. Pm (SUS)	11114	11114	11114	11114	11114	11114	11114	11114
Long. Pl (SUS)	-2089	-2089	7963	7963	-5295	-5295	18229	18229
Long. Q (SUS)	-31598	31598	94338	-94338	-45359	45359	68207	-68207
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1247	1247	-1247	-1247	-1247	-1247	1247	1247
Shear Q (SUS)	4929	4929	4929	4929	4929	4929	4929	4929
Pm (SUS)	22228	23227	22228	23227	22228	23227	22228	23227
Pm+Pl (SUS)	12341	13240	45575	46572	15406	16390	35345	36308
Pm+Pl+Q (Total)	24350	43091	113856	84482	91023	122437	204187	133063

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23227	137989	Passed
Pm+Pl (SUS)	46572	206984	Passed
Pm+Pl+Q (TOTAL)	204187	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37368	38368	37368	38368	37368	38368	37368	38368
Circ. Pl (SUS)	-455	-455	9885	9885	-10494	-10494	23716	23716
Circ. Q (SUS)	1386	-1386	34410	-34410	-197600	197600	323604	-323604
Long. Pm (SUS)	18684	18684	18684	18684	18684	18684	18684	18684
Long. Pl (SUS)	-1241	-1241	14463	14463	-5751	-5751	15181	15181
Long. Q (SUS)	9811	-9811	116193	-116193	-59307	59307	95103	-95103
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1365	1365	-1365	-1365	-1365	-1365	1365	1365
Shear Q (SUS)	3530	3530	3530	3530	3530	3530	3530	3530
Pm (SUS)	37368	38368	37368	38368	37368	38368	37368	38368



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Pm+Pl (SUS)	37008	38003	47383	48375	27006	27997	61152	62149
Pm+Pl+Q (Total)	40156	37333	149409	96985	170763	225504	384781	261639

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38368	137989	Passed
Pm+Pl (SUS)	62149	206984	Passed
Pm+Pl+Q (TOTAL)	384781	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11410	12617	11410	12617	11410	12617	11410	12617
Circ. Pl (SUS)	-10355	-10355	23289	23289	-5295	-5295	18229	18229
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5701	5701	5701	5701	5701	5701	5701	5701
Long. Pl (SUS)	-8343	-8343	10699	10699	-8343	-8343	10699	10699
Long. Q (SUS)	-36590	36590	101742	-101742	-122862	122862	188014	-188014
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1503	1503	-1503	-1503	-1503	-1503	1503	1503
Shear Q (SUS)	5922	5922	5922	5922	5922	5922	5922	5922
Pm (SUS)	11410	12617	11410	12617	11410	12617	11410	12617
Pm+Pl (SUS)	4764	5751	34821	36021	9258	10407	29807	31000
Pm+Pl+Q (Total)	42936	35601	118375	121569	131915	120392	204728	203003

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12617	117991	Passed
Pm+Pl (SUS)	36021	176986	Passed
Pm+Pl+Q (TOTAL)	204728	353973	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4- Syn gas out

Nozl: 10 3:22pm Apr 10,2026

Input, Nozzle Desc: N4- Syn gas out

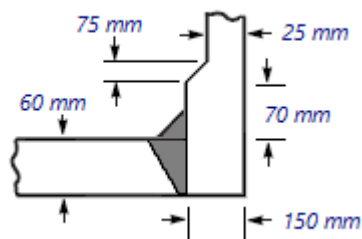
From: 40

Pressure for Reinforcement Calculations	P	10.5006	N/mm ²
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa
Inside Diameter of Cylindrical Shell	D	1384.00	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2.31	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137989.50	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	75.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4- Syn gas out



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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 18.000 in.
Actual Thickness Used in Calculation 0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.5 \cdot 695.0 / (137990 \cdot 1.0 - 0.6 \cdot 10.5)$
= 55.4144 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.5 \cdot 231.6 / (137990 \cdot 1.0 - 0.6 \cdot 10.5)$
= 18.4662 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.0/0.5773
= 216.5065 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	926.4000	mm
Parallel to Vessel Wall, opening length	d	463.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	142.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 137989.5/137989.5)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 137989.5/137989.5)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



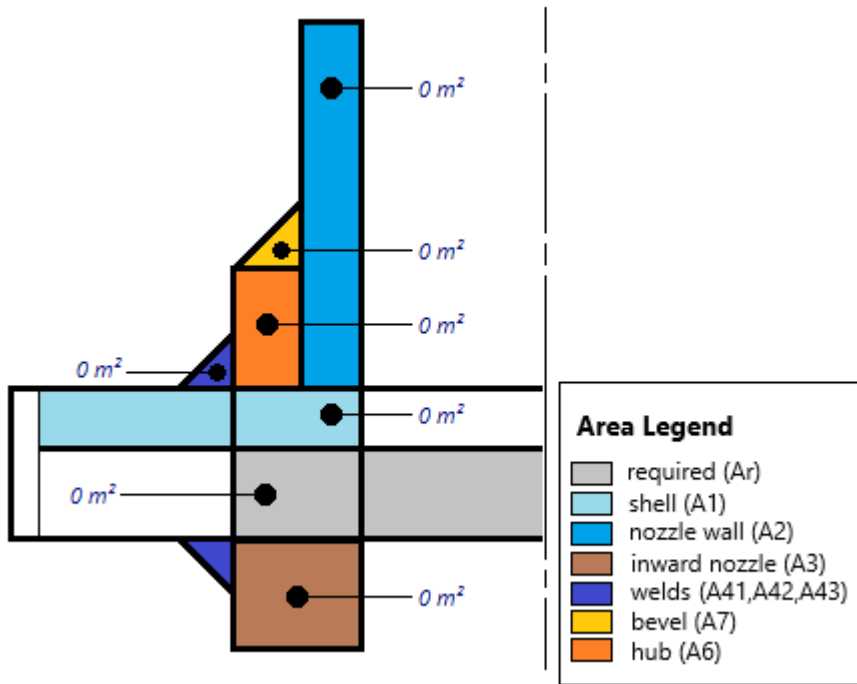
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Nozzle Calcs.: N4- Syn gas out

Nozl: 10 3:22pm Apr 10,2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.026	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.027	NA	NA
TOTAL AREA AVAILABLE	Atot	0.029	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (463.2 \cdot 55.4144 \cdot 1.0 + 2 \cdot 147.0 \cdot 55.4144 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.026 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 463.2(1.0 \cdot 57.0 - 1.0 \cdot 55.414) - 2 \cdot 147.0$$

$$(1.0 \cdot 57.0 - 1.0 \cdot 55.4144) \cdot (1 - 1.0)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 142.5)(22.0 - 18.47)1.0$$

$$= 0.001 \text{ m}^2$$



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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(142.5, 250.0, 70.0)) \cdot (150.0 - 25.0) \cdot 1.0$$

$$= 0.027 \text{ m}^2 \text{ Includes Bevel Area of } 0.009 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.4662 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 58.4144 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 58.4144 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(58.4144, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(21.4662, 11.3312)$$

$$= 21.4662 \text{ mm}$$

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 69707.1,	Allowable	: 137989.5 kPa	Passed
Expansion	: 0.0,	Allowable	: 275266.7 kPa	Passed
Occasional	: 52762.2,	Allowable	: 183526.0 kPa	Passed
Shear	: 11371.0,	Allowable	: 96592.6 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 60.0, tr = 55.414, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.972, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Weld Size Calculations, Description: N4- Syn gas out

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10, 2026

$$= \max(0, (0.0257 - 0.0007 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 57.0 - 55.4144)) \cdot 137990) \\ = 3504796.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv} \\ = (0.0279 + 0.0004 - 0.0 \cdot 1.0) \cdot 137990 \\ = 3901096.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{Sv} \\ = (0.0279 + 0.0 + 0.0004 + (0.0168)) \cdot 137990 \\ = 6213461.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{S} \\ = (0.0279 + 0.0 + 0.0004 + 0.0 + (0.0168)) \cdot 137990 \\ = 6213461.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw} \\ = (3.1416/2.0) \cdot 757.2 \cdot 20.0 \cdot 0.49 \cdot 137990 \\ = 1608388. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo})/4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn} \\ = (3.1416 \cdot 305.1) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 137990 \\ = 13609487. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgnvi} \cdot 0.74 \cdot \text{Sng} \\ = (3.1416/2.0) \cdot 757.2 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 137990 \\ = 6072486. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (1608388 + 13609487) = 15217874 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1608388 + 0 + 6072487 + 0) = 7680874 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (1608388 + 6072487 + 0) = 7680874 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 15217874N , must exceed W = 3504796 N or W1 = 3901096 N
 Path 2-2 = 7680874 N , must exceed W = 3504796 N or W2 = 6213461 N
 Path 3-3 = 7680874 N , must exceed W = 3504796 N or W3 = 6213461 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 N/mm²

The Drop for this Nozzle is : 112.7539 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 422.7539 mm



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 Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: N4- Syn gas ou:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1384.000	mm
Shell Thickness	60.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	10.50	N/mm ²
Radial Load	P	-47250.21	N
Circumferential Shear	Vc	35999.98	N
Longitudinal Shear	Vl	35999.98	N
Circumferential Moment	Mc	50624.98	N-m
Longitudinal Moment	Ml	50624.98	N-m
Torsional Moment	Mt	60750.03	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.503) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (5.048) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 757.2 + 2 \cdot 1.65 \cdot \sqrt{723.5(60.0 - 3.0)}$$

$$= 1427.348 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1447.000 mm
Nozzle Outside Diameter	(d) =	757.200 mm
Vessel Thickness used	(T) =	57.000 mm
Nozzle Thickness used	(t) =	147.000 mm
T / t	=	0.388
d / t	=	10.000



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Nozzle Calcs.: N4- Syn gas out

Nozl: 10 3:22pm Apr 10,2026

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.637$$

Nr/P	=	0.062
Mr/P	=	0.116
M0/P	=	0.036
N0/P	=	0.081
MrD/Mc	=	0.272
NrDL/Mc	=	0.097
M0d/Mc	=	0.082
N0DL/Mc	=	0.082
MrD/M1	=	0.097
NrDL/M1	=	0.060
M0D/M1	=	0.028
N0DL/M1	=	0.097

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1165	1165	1165	1165
Outplane Bending (P)	3130	-3130	3130	-3130
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1992	-1992	1992	1992
Outplane Bending (ML)	-3405	3405	3405	-3405
Normal Pressure Stress	122978	133479	122978	133479
Outplane Stress Summary	121875	132927	132672	130101

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	896	896	896	896
Inplane Bending (P)	10114	-10114	10114	-10114
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1227	-1227	1227	1227
Inplane Bending (ML)	-11934	11934	11934	-11934
Inplane Pressure Stress	61486	61486	61486	61486
Inplane Stress Summary	59334	62975	85658	41560

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	530	530	-530	-530
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1178	1178	1178	1178
Shear Stress Summary	1709	1709	648	648

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	121917	132969	132679	130101

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1165	1165	1165	1165
Outplane Bending (P)	3130	-3130	3130	-3130
Outplane Membrane (Mc)	-1689	-1689	1689	1689
Outplane Bending (Mc)	-10162	10162	10162	-10162
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61486	61486	61486	61486
Outplane Stress Summary	53929	67994	77633	51047

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	896	896	896	896
Inplane Bending (P)	10114	-10114	10114	-10114
Inplane Membrane (Mc)	-1999	-1999	1999	1999
Inplane Bending (Mc)	-33611	33611	33611	-33611
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	122978	133479	122978	133479
Inplane Stress Summary	98379	155873	169600	92649

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-530	-530	530	530
Torsional Shear (Mt)	1178	1178	1178	1178
Shear Stress Summary	648	648	1709	1709

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	98385	155873	169627	92718

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1165	1165	1165	1165
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1992	-1992	1992	1992
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15064	27668	15064	27668
Hoop Stress Summary	14237	26840	18222	30825

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

Axial Membrane (P)	165	165	165	165
Axial Bending (P)	1110	-1110	1110	-1110
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1378	-1378	1378	1378
Axial Bending (ML)	-1234	1234	1234	-1234
Axial Pressure Stress	7528	7528	7528	7528

Axial Stress Summary	6191	6439	11417	6729
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	199	199	-199	-199
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	455	455	455	455

Shear Stress Summary	654	654	255	255
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14285	26861	18229	30825
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1165	1165	1165	1165
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1689	-1689	1689	1689
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15064	27668	15064	27668

Hoop Stress Summary	14540	27144	17919	30522
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	165	165	165	165
Axial Bending (P)	1110	-1110	1110	-1110
Axial Membrane (Mc)	-1378	-1378	1378	1378
Axial Bending (Mc)	-4143	4143	4143	-4143
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7528	7528	7528	7528

Axial Stress Summary	3281	9349	14326	3819
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-199	-199	199	199
Shear due to Torsion	455	455	455	455

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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10, 2026

Shear Stress Summary	255	255	654	654
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14540	27144	18029	30536
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122978	133479	122978	133479	122978	133479	122978	133479
Circ. Pl (SUS)	-827	-827	3157	3157	-1103	-1103	2895	2895
Circ. Q (SUS)	-275	275	6535	-6535	-23497	23497	43725	-43725
Long. Pm (SUS)	61486	61486	61486	61486	61486	61486	61486	61486
Long. Pl (SUS)	-331	-331	2123	2123	-524	-524	2854	2854
Long. Q (SUS)	-1820	1820	22048	-22048	-7032	7032	13292	-13292
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	530	530	-530	-530	-530	-530	530	530
Shear Q (SUS)	1178	1178	1178	1178	1178	1178	1178	1178
Pm (SUS)	122978	133479	122978	133479	122978	133479	122978	133479
Pm+Pl (SUS)	122155	132655	126139	136639	121879	132379	125877	136377
Pm+Pl+Q (Total)	121922	132968	132678	130105	98387	155877	169629	92719

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133479	137989	Passed
Pm+P1 (SUS)	136639	206984	Passed
Pm+P1+Q (TOTAL)	169629	413968	Passed

*Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15064	27668	15064	27668	15064	27668	15064	27668
Circ. Pl (SUS)	-827	-827	3157	3157	-524	-524	2854	2854
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7528	7528	7528	7528	7528	7528	7528	7528
Long. Pl (SUS)	-1213	-1213	1543	1543	-1213	-1213	1543	1543
Long. Q (SUS)	-124	124	2344	-2344	-3033	3033	5253	-5253
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	199	199	-199	-199	-199	-199	199	199
Shear Q (SUS)	455	455	455	455	455	455	455	455



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Nozzle Calcs.: N4- Syn gas out Nozl: 10 3:22pm Apr 10,2026

Pm (SUS)	15064	27668	15064	27668	15064	27668	15064	27668
Pm+Pl (SUS)	14241	26842	18225	30826	14544	27145	17922	30523
Pm+Pl+Q (Total)	14289	26861	18230	30827	14545	27147	18033	30538

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27668	137989	Passed
Pm+Pl (SUS)	30826	206984	Passed
Pm+Pl+Q (TOTAL)	30827	413968	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N3-Syn Gas inle

Nozl: 11 3:22pm Apr 10,2026

Input, Nozzle Desc: N3-Syn Gas inlet

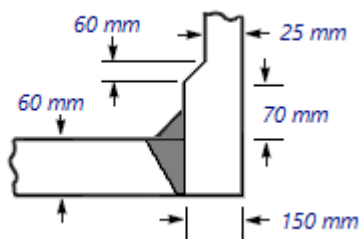
From: 60

Pressure for Reinforcement Calculations	P	10.5006	N/mm ²
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa
Inside Diameter of Cylindrical Shell	D	1384.00	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12.48	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137989.50	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3-Syn Gas inlet



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Nozzle Calcs.: N3-Syn Gas inle Nozl: 11 3:22pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 18.000 in.
Actual Thickness Used in Calculation 0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.5 \cdot 695.0 / (137990 \cdot 1.0 - 0.6 \cdot 10.5)$
= 55.4144 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $10.5 \cdot 231.6 / (137990 \cdot 1.0 - 0.6 \cdot 10.5)$
= 18.4662 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 125.0/0.5773
= 216.5065 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	926.4000	mm
Parallel to Vessel Wall, opening length	d	463.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	142.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 137989.5/137989.5)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 137989.5/137989.5)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



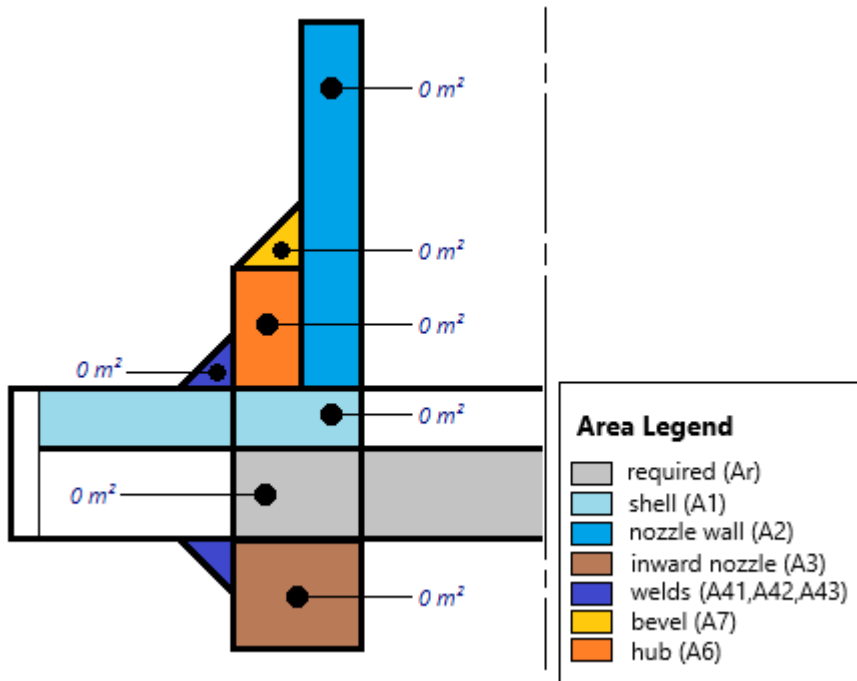
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Nozzle Calcs.: N3-Syn Gas inle

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.026	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.025	NA	NA
TOTAL AREA AVAILABLE	Atot	0.027	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (463.2 \cdot 55.4144 \cdot 1.0 + 2 \cdot 147.0 \cdot 55.4144 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.026 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 463.2(1.0 \cdot 57.0 - 1.0 \cdot 55.414) - 2 \cdot 147.0$$

$$(1.0 \cdot 57.0 - 1.0 \cdot 55.4144) \cdot (1 - 1.0)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 142.5)(22.0 - 18.47)1.0$$

$$= 0.001 \text{ m}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(142.5, 250.0, 70.0)) \cdot (150.0 - 25.0) \cdot 1.0$$

$$= 0.025 \text{ m}^2 \text{ Includes Bevel Area of } 0.007 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.4662 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 58.4144 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 58.4144 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(58.4144, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(21.4662, 11.3312)$$

$$= 21.4662 \text{ mm}$$

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 69707.1,	Allowable	: 137989.5 kPa	Passed
Expansion	: 0.0,	Allowable	: 275266.7 kPa	Passed
Occasional	: 52762.2,	Allowable	: 183526.0 kPa	Passed
Shear	: 11371.0,	Allowable	: 96592.6 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, tg = 60.0, tr = 55.414, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.972, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Weld Size Calculations, Description: N3-Syn Gas inle

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$



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Nozzle Calcs.: N3-Syn Gas inle Nozl: 11 3:22pm Apr 10, 2026

$$= \max(0, (0.0257 - 0.0007 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 57.0 - 55.4144)) \cdot 137990) \\ = 3504796.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv} \\ = (0.026 + 0.0001 - 0.0 \cdot 1.0) \cdot 137990 \\ = 3602414.25 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{Sv} \\ = (0.026 + 0.0 + 0.0001 + (0.0168)) \cdot 137990 \\ = 5914780.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{S} \\ = (0.026 + 0.0 + 0.0001 + 0.0 + (0.0168)) \cdot 137990 \\ = 5914780.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw} \\ = (3.1416/2.0) \cdot 757.2 \cdot 10.0 \cdot 0.49 \cdot 137990 \\ = 804194. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo})/4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn} \\ = (3.1416 \cdot 305.1) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 137990 \\ = 13609487. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgnvi} \cdot 0.74 \cdot \text{Sng} \\ = (3.1416/2.0) \cdot 757.2 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 137990 \\ = 6072486. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (804194 + 13609487) = 14413680 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (804194 + 0 + 6072487 + 0) = 6876680 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (804194 + 6072487 + 0) = 6876680 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 14413680 N , must exceed W = 3504796 N or W1 = 3602414 N
 Path 2-2 = 6876680 N , must exceed W = 3504796 N or W2 = 5914780 N
 Path 3-3 = 6876680 N , must exceed W = 3504796 N or W3 = 5914780 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 N/mm²

The Drop for this Nozzle is : 112.7539 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 422.7539 mm



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 Nozzle Calcs.: N3-Syn Gas inle Nozl: 11 3:22pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: N3-Syn Gas inl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1384.000	mm
Shell Thickness	60.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	10.50	N/mm ²
Radial Load	P	-47250.21	N
Circumferential Shear	Vc	36000.07	N
Longitudinal Shear	Vl	36000.07	N
Circumferential Moment	Mc	50624.98	N-m
Longitudinal Moment	Ml	50624.98	N-m
Torsional Moment	Mt	60750.03	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.503) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (5.048) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 757.2 + 2 \cdot 1.65 \cdot \sqrt{(723.5(60.0 - 3.0))}$$

$$= 1427.348 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1447.000 mm
Nozzle Outside Diameter	(d) =	757.200 mm
Vessel Thickness used	(T) =	57.000 mm
Nozzle Thickness used	(t) =	147.000 mm
T / t	=	0.388
d / t	=	10.000



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Nozzle Calcs.: N3-Syn Gas inle

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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.637$$

Nr/P	=	0.062
Mr/P	=	0.116
M0/P	=	0.036
N0/P	=	0.081
MrD/Mc	=	0.272
NrDL/Mc	=	0.097
M0d/Mc	=	0.082
N0DL/Mc	=	0.082
MrD/M1	=	0.097
NrDL/M1	=	0.060
M0D/M1	=	0.028
N0DL/M1	=	0.097

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1165	1165	1165	1165
Outplane Bending (P)	3130	-3130	3130	-3130
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1992	-1992	1992	1992
Outplane Bending (ML)	-3405	3405	3405	-3405
Normal Pressure Stress	122978	133479	122978	133479
Outplane Stress Summary	121875	132927	132672	130101

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	896	896	896	896
Inplane Bending (P)	10114	-10114	10114	-10114
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1227	-1227	1227	1227
Inplane Bending (ML)	-11934	11934	11934	-11934
Inplane Pressure Stress	61486	61486	61486	61486
Inplane Stress Summary	59334	62975	85658	41560

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	530	530	-530	-530
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1178	1178	1178	1178
Shear Stress Summary	1709	1709	648	648

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	121917	132969	132679	130101

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: N3-Syn Gas inle Nozl: 11 3:22pm Apr 10,2026

(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1165	1165	1165	1165
Outplane Bending (P)	3130	-3130	3130	-3130
Outplane Membrane (Mc)	-1689	-1689	1689	1689
Outplane Bending (Mc)	-10162	10162	10162	-10162
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61486	61486	61486	61486

Outplane Stress Summary	53929	67994	77633	51047
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	896	896	896	896
Inplane Bending (P)	10114	-10114	10114	-10114
Inplane Membrane (Mc)	-1999	-1999	1999	1999
Inplane Bending (Mc)	-33611	33611	33611	-33611
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	122978	133479	122978	133479

Inplane Stress Summary	98379	155873	169600	92649
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-530	-530	530	530
Torsional Shear (Mt)	1178	1178	1178	1178

Shear Stress Summary	648	648	1709	1709
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	98385	155873	169627	92718
------------------------	-------	--------	--------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1165	1165	1165	1165
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1992	-1992	1992	1992
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15064	27668	15064	27668

Hoop Stress Summary	14237	26840	18222	30825
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside



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Axial Membrane (P)	165	165	165	165
Axial Bending (P)	1110	-1110	1110	-1110
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1378	-1378	1378	1378
Axial Bending (ML)	-1234	1234	1234	-1234
Axial Pressure Stress	7528	7528	7528	7528

Axial Stress Summary	6191	6439	11417	6729
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	199	199	-199	-199
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	455	455	455	455

Shear Stress Summary	654	654	255	255
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14285	26861	18229	30825
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1165	1165	1165	1165
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1689	-1689	1689	1689
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15064	27668	15064	27668

Hoop Stress Summary	14540	27144	17919	30522
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	165	165	165	165
Axial Bending (P)	1110	-1110	1110	-1110
Axial Membrane (Mc)	-1378	-1378	1378	1378
Axial Bending (Mc)	-4143	4143	4143	-4143
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7528	7528	7528	7528

Axial Stress Summary	3281	9349	14326	3819
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-199	-199	199	199
Shear due to Torsion	455	455	455	455

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Shear Stress Summary	255	255	654	654
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	14540	27144	18029	30536
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122978	133479	122978	133479	122978	133479	122978	133479
Circ. Pl (SUS)	-827	-827	3157	3157	-1103	-1103	2895	2895
Circ. Q (SUS)	-275	275	6535	-6535	-23497	23497	43725	-43725
Long. Pm (SUS)	61486	61486	61486	61486	61486	61486	61486	61486
Long. Pl (SUS)	-331	-331	2123	2123	-524	-524	2854	2854
Long. Q (SUS)	-1820	1820	22048	-22048	-7032	7032	13292	-13292
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	530	530	-530	-530	-530	-530	530	530
Shear Q (SUS)	1178	1178	1178	1178	1178	1178	1178	1178
Pm (SUS)	122978	133479	122978	133479	122978	133479	122978	133479
Pm+Pl (SUS)	122155	132655	126139	136639	121879	132379	125877	136377
Pm+Pl+Q (Total)	121922	132968	132678	130105	98387	155877	169629	92719

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133479	137989	Passed
Pm+P1 (SUS)	136639	206984	Passed
Pm+P1+Q (TOTAL)	169629	413968	Passed

Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 \cdot S_mh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15064	27668	15064	27668	15064	27668	15064	27668
Circ. Pl (SUS)	-827	-827	3157	3157	-524	-524	2854	2854
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7528	7528	7528	7528	7528	7528	7528	7528
Long. Pl (SUS)	-1213	-1213	1543	1543	-1213	-1213	1543	1543
Long. Q (SUS)	-124	124	2344	-2344	-3033	3033	5253	-5253
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	199	199	-199	-199	-199	-199	199	199
Shear Q (SUS)	455	455	455	455	455	455	455	455



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Pm (SUS)	15064	27668	15064	27668	15064	27668	15064	27668
Pm+Pl (SUS)	14241	26842	18225	30826	14544	27145	17922	30523
Pm+Pl+Q (Total)	14289	26861	18230	30827	14545	27147	18033	30538

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27668	137989	Passed
Pm+Pl (SUS)	30826	206984	Passed
Pm+Pl+Q (TOTAL)	30827	413968	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: Cooling water o

Nozl: 12 3:22pm Apr 10,2026

Input, Nozzle Desc: Cooling water o

From: 70

Pressure for Reinforcement Calculations	P	1.0111	N/mm ²
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	1130.00	mm
Design Length of Section	L	1200.0300	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		14.01	m
-----------------------------------	--	-------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117991.02	kPa
Allowable Stress At Ambient	Sna	117991.02	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137989.50	kPa
Pad Allowable Stress At Ambient	Spa	137989.50	kPa
Diameter of Pad along vessel surface	Dp	697.2000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		120.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

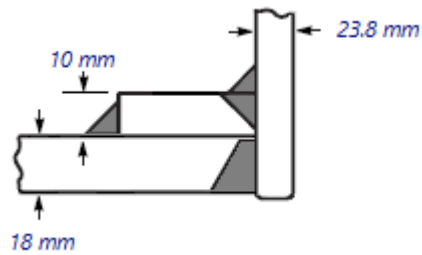


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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026



Reinforcement CALCULATION, Description: Cooling water o

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	16.126 in.
Actual Thickness Used in Calculation	0.937 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.01 \cdot 568.0 / (137990 \cdot 1.0 - 0.6 \cdot 1.01)$
 $= 4.1799 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.01 \cdot 207.8 / (117991 \cdot 1.0 - 0.6 \cdot 1.01)$
 $= 1.7897 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1355 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	831.2008 mm
Parallel to Vessel Wall, opening length	d	415.6004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991.0 / 137989.5)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991.0 / 137989.5)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137989.5 / 137989.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: Cooling water o

Nozl: 12 3:22pm Apr 10,2026

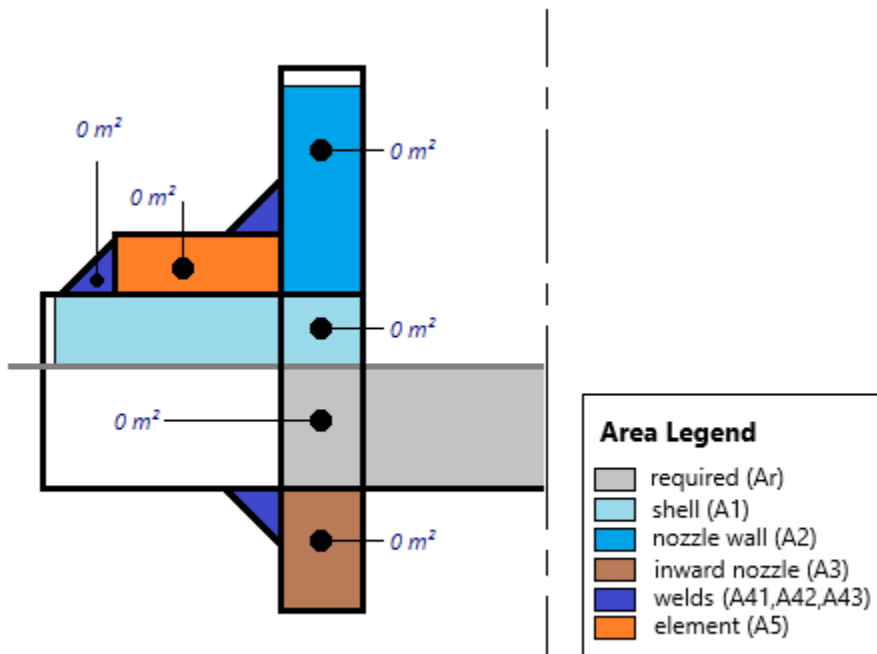


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.002	0.001	NA
Area in Shell	A1	0.004	0.005	NA
Area in Nozzle Wall	A2	0.001	0.001	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.002	0.002	NA
TOTAL AREA AVAILABLE	Atot	0.008	0.008	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (415.6004 \cdot 4.1799 \cdot 1.0 + 2 \cdot 20.7998 \cdot 4.1799 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 0.002 \text{ m}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 415.6(1.0 \cdot 15.0 - 1.0 \cdot 4.18) - 2 \cdot 20.8 \\
 &\quad (1.0 \cdot 15.0 - 1.0 \cdot 4.1799) \cdot (1 - 0.855) \\
 &= 0.004 \text{ m}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.5) \cdot (20.8 - 1.79) \cdot 0.8551
 \end{aligned}$$



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

$$= 0.001 \text{ m}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (697.2 - 457.2) \cdot 10.0 \cdot 1.0$$

$$= 0.002 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.7897 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 7.1799 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.1799 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.4172 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(7.1799, 4.5)]$$

$$= 7.1799 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.7897, 7.1799)$$

$$= 7.1799 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 23.800 = 20.825 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 17091.4, Allowable : 117991.0 kPa Passed

Expansion : 0.0, Allowable : 277886.2 kPa Passed

Occasional : 4809.5, Allowable : 156928.1 kPa Passed

Shear : 9517.4, Allowable : 82593.7 kPa Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.825$, $tr = 1.79$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 13°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.279$, Temp. Reduction = 78°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Govrn. thk, tg = 10.0, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.279$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.279$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 4.18, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.279$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.01/1.96 = 0.516$

Weld Size Calculations, Description: Cooling water o

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$
= $\max(0, (0.0018 - 0.0044 + 2 \cdot 20.7998 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 4.1799))137990)$
= $\max(0, -315208.38) N$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= $(A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(0.0012 + 0.0024 + 0.0002 - 0.0 \cdot 0.86) \cdot 137990$
= 524985.00 N

Weld Load [W2]:

= $(A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
= $(0.0012 + 0.0 + 0.0001 + (0.0005)) \cdot 137990$



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

$$= 253644.31 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (0.0012 + 0.0 + 0.0002 + 0.0024 + (0.0005)) \cdot 137990$$

$$= 598608.69 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 457.2 \cdot 10.0 \cdot 0.49 \cdot 117991$$

$$= 415202. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 697.2 \cdot 10.0 \cdot 0.49 \cdot 137990$$

$$= 740470. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 218.2001) \cdot (23.7998 - 3.0) \cdot 0.7 \cdot 117991$$

$$= 1177604. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 457.2 \cdot 10.0 \cdot 0.74 \cdot 137990$$

$$= 733318. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 457.2 \cdot (8.0 - 0.0) \cdot 0.74 \cdot 137990$$

$$= 586654. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (740470 + 1177604) = 1918074 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (415202 + 733318 + 586654 + 0) = 1735174 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (740470 + 586654 + 0) = 1327125 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1918074 N	, must exceed W = 0 N	or W1 = 524985 N
Path 2-2 = 1735173 N	, must exceed W = 0 N	or W2 = 253644 N
Path 3-3 = 1327124 N	, must exceed W = 0 N	or W3 = 598608 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 N/mm²

Nozzle is O.K. for the External Pressure: 0 N/mm²

The Drop for this Nozzle is : 48.3115 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 316.3115 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 386.62 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 415.60 mm



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: Cooling water :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1130.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	409.600	mm
Nozzle Thickness	23.7998	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 117991.02	kPa
Nozzle Hot S.I. Allowable	SNmh 117991.02	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	697.200	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.01	N/mm ²
Radial Load	P	-33750.00	N
Circumferential Shear	Vc	22499.94	N
Longitudinal Shear	Vl	22499.94	N
Circumferential Moment	Mc	28350.06	N-m
Longitudinal Moment	Ml	28350.06	N-m
Torsional Moment	Mt	40500.07	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{580.5(18.0 - 3.0)}$
 $= 765.136 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1161.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	25.000	mm
Nozzle Thickness used	(t) =	20.800	mm
T / t	=	1.202	
d / t	=	21.981	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.684$$



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Nr/P = 0.054
 Mr/P = 0.097
 MØ/P = 0.035
 NØ/P = 0.120
 MrD/Mc = 0.231
 NrDL/Mc = 0.100
 MØd/Mc = 0.095
 NØDL/Mc = 0.119
 MrD/Ml = 0.106
 NrDL/Ml = 0.051
 MØD/Ml = 0.044
 NØDL/Ml = 0.170

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6467	6467	6467	6467
Outplane Bending (P)	11424	-11424	11424	-11424
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16822	-16822	16822	16822
Outplane Bending (ML)	-26192	26192	26192	-26192
Normal Pressure Stress	22469	23483	22469	23483
Outplane Stress Summary	-2654	27895	83376	9156

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2937	2937	2937	2937
Inplane Bending (P)	31370	-31370	31370	-31370
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5026	-5026	5026	5026
Inplane Bending (ML)	-62968	62968	62968	-62968
Inplane Pressure Stress	11231	11231	11231	11231
Inplane Stress Summary	-22455	40740	113533	-75144

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1247	1247	-1247	-1247
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4929	4929	4929	4929
Shear Stress Summary	6177	6177	3681	3681

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	24220	43222	113974	84617

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6467	6467	6467	6467
Outplane Bending (P)	11424	-11424	11424	-11424
Outplane Membrane (Mc)	-11762	-11762	11762	11762
Outplane Bending (Mc)	-56783	56783	56783	-56783
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11231	11231	11231	11231
Outplane Stress Summary	-39423	51295	97668	-38747

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2937	2937	2937	2937
Inplane Bending (P)	31370	-31370	31370	-31370
Inplane Membrane (Mc)	-9921	-9921	9921	9921
Inplane Bending (Mc)	-137374	137374	137374	-137374
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22469	23483	22469	23483
Inplane Stress Summary	-90519	122503	204073	-132403

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1247	-1247	1247	1247
Torsional Shear (Mt)	4929	4929	4929	4929
Shear Stress Summary	3681	3681	6177	6177

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	90781	122689	204424	132803

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6467	6467	6467	6467
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16822	-16822	16822	16822
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11541	12755	11541	12755
Hoop Stress Summary	1185	2399	34831	36044

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Axial Membrane (P)	1178	1178	1178	1178
Axial Bending (P)	32576	-32576	32576	-32576
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9521	-9521	9521	9521
Axial Bending (ML)	-69166	69166	69166	-69166
Axial Pressure Stress	5770	5770	5770	5770

Axial Stress Summary	-39161	34017	118214	-85272
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1503	1503	-1503	-1503
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5922	5922	5922	5922

Shear Stress Summary	7425	7425	4419	4419
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	42987	35672	118442	121634
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6467	6467	6467	6467
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11762	-11762	11762	11762
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11541	12755	11541	12755

Hoop Stress Summary	6246	7459	29770	30984
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1178	1178	1178	1178
Axial Bending (P)	32576	-32576	32576	-32576
Axial Membrane (Mc)	-9521	-9521	9521	9521
Axial Bending (Mc)	-155438	155438	155438	-155438
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5770	5770	5770	5770

Axial Stress Summary	-125433	120290	204486	-171544
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1503	-1503	1503	1503
Shear due to Torsion	5922	5922	5922	5922

Shear Stress Summary	4419	4419	7425	7425
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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	131969	120462	204797	203066

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{575.5(18.0 - 3.0)}$$

$$= 763.807 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1151.000 mm
Nozzle Outside Diameter	(d) =	697.200 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	140.800 mm
T / t	=	0.107
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 5.306$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.031
MrD/Mc	=	0.240
NrDL/Mc	=	0.095
M0d/Mc	=	0.071
N0DL/Mc	=	0.058
MrD/Ml	=	0.049
NrDL/Ml	=	0.043
M0D/Ml	=	0.015
N0DL/Ml	=	0.029

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4715	4715	4715	4715
Outplane Bending (P)	17898	-17898	17898	-17898
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5170	-5170	5170	5170
Outplane Bending (ML)	-16512	16512	16512	-16512
Normal Pressure Stress	37782	38789	37782	38789
Outplane Stress Summary	38713	36948	82080	14264

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6611	6611	6611	6611
Inplane Bending (P)	63002	-63002	63002	-63002
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Inplane Membrane (ML)	-7852	-7852	7852	7852
Inplane Bending (ML)	-53191	53191	53191	-53191
Inplane Pressure Stress	18891	18891	18891	18891

Inplane Stress Summary	27461	7839	149550	-82838
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1365	1365	-1365	-1365
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3530	3530	3530	3530

Shear Stress Summary	4895	4895	2164	2164
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40540	37747	149619	97193
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4715	4715	4715	4715
Outplane Bending (P)	17898	-17898	17898	-17898
Outplane Membrane (Mc)	-10466	-10466	10466	10466
Outplane Bending (Mc)	-77205	77205	77205	-77205
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18891	18891	18891	18891

Outplane Stress Summary	-46166	72448	129177	-61031
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6611	6611	6611	6611
Inplane Bending (P)	63002	-63002	63002	-63002
Inplane Membrane (Mc)	-17105	-17105	17105	17105
Inplane Bending (Mc)	-260602	260602	260602	-260602
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37782	38789	37782	38789

Inplane Stress Summary	-170310	225894	385104	-261098
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1365	-1365	1365	1365
Torsional Shear (Mt)	3530	3530	3530	3530

Shear Stress Summary	2164	2164	4895	4895
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	170344	225922	385194	261215

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22469	23483	22469	23483	22469	23483	22469	23483
Circ. Pl (SUS)	-10355	-10355	23289	23289	-6984	-6984	12858	12858
Circ. Q (SUS)	-14768	14768	37616	-37616	-106004	106004	168744	-168744
Long. Pm (SUS)	11231	11231	11231	11231	11231	11231	11231	11231
Long. Pl (SUS)	-2089	-2089	7963	7963	-5295	-5295	18229	18229
Long. Q (SUS)	-31598	31598	94338	-94338	-45359	45359	68207	-68207
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1247	1247	-1247	-1247	-1247	-1247	1247	1247
Shear Q (SUS)	4929	4929	4929	4929	4929	4929	4929	4929
Pm (SUS)	22469	23483	22469	23483	22469	23483	22469	23483
Pm+Pl (SUS)	12567	13485	45816	46828	15645	16644	35581	36560
Pm+Pl+Q (Total)	24224	43227	113975	84621	90782	122692	204428	132808

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23483	137989	Passed
Pm+Pl (SUS)	46828	206984	Passed
Pm+Pl+Q (TOTAL)	204428	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37782	38789	37782	38789	37782	38789	37782	38789
Circ. Pl (SUS)	-455	-455	9885	9885	-10494	-10494	23716	23716
Circ. Q (SUS)	1386	-1386	34410	-34410	-197600	197600	323604	-323604
Long. Pm (SUS)	18891	18891	18891	18891	18891	18891	18891	18891
Long. Pl (SUS)	-1241	-1241	14463	14463	-5751	-5751	15181	15181
Long. Q (SUS)	9811	-9811	116193	-116193	-59307	59307	95103	-95103
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1365	1365	-1365	-1365	-1365	-1365	1365	1365
Shear Q (SUS)	3530	3530	3530	3530	3530	3530	3530	3530
Pm (SUS)	37782	38789	37782	38789	37782	38789	37782	38789



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 Nozzle Calcs.: Cooling water o Nozl: 12 3:22pm Apr 10,2026

Pm+Pl (SUS)	37421	38423	47796	48794	27418	28416	61565	62570
Pm+Pl+Q (Total)	40544	37749	149616	97199	170349	225925	385195	261218

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38789	137989	Passed
Pm+Pl (SUS)	62570	206984	Passed
Pm+Pl+Q (TOTAL)	385195	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11541	12755	11541	12755	11541	12755	11541	12755
Circ. Pl (SUS)	-10355	-10355	23289	23289	-5295	-5295	18229	18229
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5770	5770	5770	5770	5770	5770	5770	5770
Long. Pl (SUS)	-8343	-8343	10699	10699	-8343	-8343	10699	10699
Long. Q (SUS)	-36590	36590	101742	-101742	-122862	122862	188014	-188014
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1503	1503	-1503	-1503	-1503	-1503	1503	1503
Shear Q (SUS)	5922	5922	5922	5922	5922	5922	5922	5922
Pm (SUS)	11541	12755	11541	12755	11541	12755	11541	12755
Pm+Pl (SUS)	4813	5810	34952	36158	9317	10473	29937	31138
Pm+Pl+Q (Total)	42994	35673	118444	121638	131977	120461	204797	203072

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12755	117991	Passed
Pm+Pl (SUS)	36158	176986	Passed
Pm+Pl+Q (TOTAL)	204797	353973	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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ASME TS Calc: DESIGN CASE

Case: 1 3:22p Apr 10,2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	10.50 N/mm ²
Shell Maximum Operating Pressure	Psox,max	10.50 N/mm ²
Shell Minimum Operating Pressure	Psox,min	8.30 N/mm ²
Shell Thickness	ts	60.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1130.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel left

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	1.00 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	0.50 N/mm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1130.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	808
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	12000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11750.00 mm
Design Temperature of the Tubes		150.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	157989.59 kPa
Tube Pitch (Center to Center Spacing)	P	30.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	457.685 mm



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ASME TS Calc: DESIGN CASE

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	150.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	137989.50 kPa
Tubesheet Allowable Stress at Ambient	Tt	137989.50 kPa
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1250.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	850.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	44.60 °C
Mean Tube Metal Temp. along Tube length	Ttm	67.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



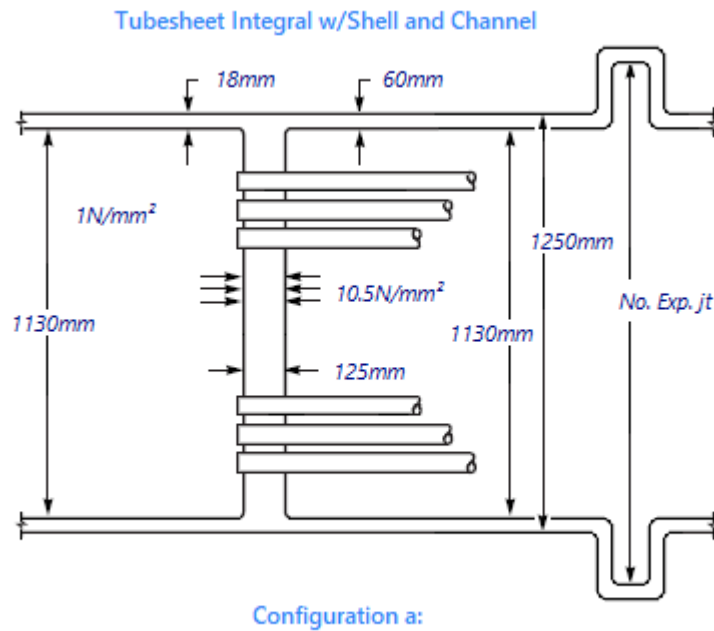
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ASME TS Calc: DESIGN CASE

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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	44.6 °C	0.0000117432 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19507E+06 N/mm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	44.6 °C	0.20137E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	67.0 °C	0.0000118884 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19507E+06 N/mm²
Elastic Mod. at Metal Temp. along Len	67.0 °C	0.20008E+06 N/mm²
Elastic Mod. at Tubsht. Design Temp.	150.0 °C	0.19507E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	150.0 °C	0.19507E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Note:



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ASME TS Calc: DESIGN CASE

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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1.0 \cdot (19.05/2 - 0)) / (92392.97 \cdot 1.0 + 0.4 \cdot 1.0)$$

$$= 0.1026 + 0.0000 = 0.1026 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 150.00 °C
Elastic Modulus for Material 199818.77 N/mm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	11750.00	9.03	50.0000	0.0134948	95780.45
EMAWP = (2.167/(D/T) - 0.0833) * B = 15.0115 N/mm ²						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.5680	19.05	11750.00	12.15	50.0000	0.0074521	95688.52
EMAWP = (4*B)/(3*(D/T)) = (4 * 95688.5234)/(3 * 12.1495) = 10.5019 N/mm ²						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.5680 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm ²
Required Internal Design Pressure	1.00 N/mm ²
Allowable External Pressure at Corroded thickness	15.01 N/mm ²
Required External Design Pressure	10.50 N/mm ²
Required Thickness due to Shell Side pressure	1.5680 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/125.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 2.11 (195074/195074) \cdot (92392/137989) \cdot (0); 19.05 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 457.685 + 19.05$$

$$= 934.420 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (30.0 - 19.05) / 30.0$$

$$= 0.365$$



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ASME TS Calc: DESIGN CASE

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Equivalent outer tube limit radius [ao]:

$$\begin{aligned}
 &= D_o / 2 \\
 &= 934.42 / 2 \\
 &= 467.2100 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))} \\
 &= 30.0 / \sqrt{(1 - 4 \cdot \min(0 \cdot 1000248 ; 4 \cdot 934.42 \cdot 30.0) / (\pi \cdot 934.42^2))} \\
 &= 30.0000 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (30.0 - 19.05) / 30.0 \\
 &= 0.3650
 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned}
 &= a_s / a_o \\
 &= 565.0 / 467.21 \\
 &= 1.2093
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= a_c / a_o \\
 &= 565.0 / 467.21 \\
 &= 1.2093
 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned}
 &= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\
 &= 1 - 808 / ((19.05 - 2 \cdot 2.11) / (2 \cdot 467.21))^2 \\
 &= 0.7965
 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned}
 &= 1 - Nt / (dt / (2 \cdot a_o))^2 \\
 &= 1 - 808 / (19.05 / (2 \cdot 467.21))^2 \\
 &= 0.6642
 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned}
 &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\
 &= \max((0 - 0); 0) \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned}
 &= \pi \cdot t_s (D_s + t_s) E_s / L \\
 &= \pi \cdot 60.0 (1130.0 + 60.0) 195074 / 11750.0 \\
 &= 3723683.0000 \text{ N/mm}
 \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned}
 &= \pi \cdot t_t (D_t - t_t) E_t / L \\
 &= \pi \cdot 2.11 (19.05 - 2.11) 195074 / 11750.0 \\
 &= 1864.1049 \text{ N/mm}
 \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned}
 &= K_s / (Nt \cdot K_t) \\
 &= 3723683 / (808 \cdot 1864.105) \\
 &= 2.47224
 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned}
 &= 1 / (1 + K_s / K_j) \\
 &= 1 / (1 + 3723683 / 0) \\
 &= 1.0000 (= 1 \text{ if No Exp. Jt.})
 \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s) \\
 &= ((12(1 - 0.3^2))^{1/4}) / ((1130.0 + 60.0) 60.0)
 \end{aligned}$$



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= 0.0068 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.007 \cdot 195074 \cdot 60.0^3 / (6(1 - 0.3^2))$$

$$= 52500932.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1130.0 + 60.0) \cdot 52500932 / 125.0^3 \cdot (1 + 125.0 \cdot 0.007 + (125.0 \cdot 0.007)^2 / 2)$$

$$= 424535.1250 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1130.0^2 / (4 \cdot 195074 \cdot 60.0) \cdot (1 - 1130.0 \cdot 0.3 / (2(1130.0 + 60.0)))$$

$$= 0.23390\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1130.0 + 18.0)18.0}$$

$$= 0.0126 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.013 \cdot 199225 \cdot 18.0^3 / (6(1 - 0.3^2))$$

$$= 2691030.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1130.0 + 18.0 / \cos(0)) \cdot 2691030 / 125.0^3 \cdot (1.0 + 125.0 \cdot 0.013 \cdot (1.0 + 1.0) / 2 + (125.0 \cdot 0.013)^2 / 2 \cdot 1.0)$$

$$= 36348.9 \text{ N/mm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1130.0^2 / (4 \cdot 199225 \cdot 18.0) \cdot (1 - 1130.0 \cdot 0.3 / (2(1130.0 + 18.0)))$$

$$= 0.75879\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 4.166667$; $\mu^* = 0.365000$; $E^* = 71604 \text{ N/mm}^2$

$E^*/E = 0.367063$; $\nu^* = 0.320752$

Note: As h/p (4.167) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t_t(dt - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.321^2)808 \cdot 195074 \cdot 2.11(19.05 - 2.11)467.21^2) / (71604 \cdot 11750.0 \cdot 125.0^3))^{1/4}$$

$$= 2.0035$$

Values from Table 4.18.3

$$Z_d = 0.172927 ; Z_v = 0.162518 ; Z_m = 0.676541$$

$$Z_a = 0.803015\text{E+}00 ; Z_w = 0.162518$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:



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$$= A / D_o$$

$$= 1250.0/934.42$$

$$= 1.3377$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 71604 \cdot (424535.12 + 36348.86 + 195074 \cdot \ln(1.34))$$

$$= 4.9104$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.3208) 4.9104$$

$$= 6.4855$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.2093 - 1 - 6.4855 \cdot 0.1625) / (1 + 6.4855 \cdot 0.6765)$$

$$= -0.156784$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.17293 + -0.15678 \cdot 0.16252) / 2 \cdot 2.00353^4$$

$$= 1.1879$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.16252 + -0.15678 \cdot 0.67654) / 2 \cdot 2.00353^4$$

$$= 0.4548$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1625 + (1.2093 - 1) 0.6765) 2.00353^4 / (1 + 6.4855 \cdot 0.6765)$$

$$= 0.9096$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.2093 \cdot 52500932 \cdot 0.0068 \cdot 0.234E-04 (1 + 125.0 \cdot 0.0068)$$

$$= 18693.2441 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$$

$$= 467.21^2 (1.209^2 - 1) \cdot (1.209 - 1) / 4 - 18693.244$$

$$= -13411.4062 \text{ mm}^2$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.209 \cdot 2691030 [0.759E-04 (1.0 + 0.013 \cdot 1.0 \cdot 125.0) - 0 (1.0/125.0 + 1.0 \cdot 0.013/2)]$$

$$= 8058.3179 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c$$

$$= 467.20999^2 [(1.20931^2 + 1) \cdot (1.20931 - 1) / 4 - (1.20931 - 1) / 2] - 8058.31787$$

$$= -2776.4788 \text{ mm}^2$$



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Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 10.501 \{ 0.664 + 2(1 - 0.664) 0.3 + [2/2.472 \cdot (1130.0/934.42)^2] 0.3 - [(1.209^2 - 1)/(1.0 \cdot 2.472)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 2.472)] [(0^2 - (1130.0)^2)/934.42^2] \}$$

$$= 10.8529 \text{ N/mm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.796 + 2(1 - 0.796) 0.3 + 1/(1.0 \cdot 2.472)] 1.0$$

$$= 1.3232 \text{ N/mm}^2$$

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 808 \cdot 1864.105 \cdot 0 / (3.142 \cdot 467.21^2)$$

$$= 0.0000 \text{ N/mm}^2$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.91 \cdot 0 / (2 \cdot \pi \cdot 467.21^2)$$

$$= -0.0000 \text{ N/mm}^2$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(0.91/467.21^2) \cdot (-13411.406 \cdot 10.501 - -2776.479 \cdot 1.0)$$

$$= 0.5752 \text{ N/mm}^2$$

Calculate the pressure [P_ω]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c)$$

$$= (0.91/467.21^2) \cdot (18693.2441 \cdot 0 - 8058.3179 \cdot 0)$$

$$= 0.0000 \text{ N/mm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1)Qz2)) \cdot (P's - P't + P_\gamma + P_\omega + P_w + Prim)$$

$$= 0.1000E+01 \cdot 2.472 / (1 + 1.0 \cdot 2.472 (1.188 + (1.209 - 1) 0.455)) \cdot (10.853 - 1.323 + 0 + 0 + 0 + 0.575)$$

$$= 5.9877 \text{ N/mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega_s \cdot P^* s - \omega_c \cdot P^* c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-13411.406 \cdot 10.501 - -2776.479 \cdot 1.0) - (18693.244 \cdot 0 - 8058.318 \cdot 0)) \cdot 1. + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.48548 \cdot 0.67654)$$

$$= -25621.2676 \text{ N}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.157 + 2 \cdot -25621.268 / (5.988 \cdot 467.21^2)$$

$$= -0.19599$$

Fm Value from Table 4.18.3 = 0.097996



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Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.098 / 0.365) \cdot (2 \cdot 467.21 / (125.0 - 0))^2 \cdot 5.99$$

$$= 134744.0312 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 137989.5 = 206984.23 \text{ kPa}$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0997 / 0.365) \cdot (2 \cdot 467.21 / (98.465 - 0))^2 \cdot 5.61$$

$$= 206983.4375 \text{ kPa}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 98.4654 + 0 + 0$$

$$= 98.4654 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.365 \cdot 125.0 / 467.21) \cdot 110391.594$$

$$= 21560.3965 \text{ kPa}$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (1 / (2 \cdot \mu)) \cdot (a_o / h) \cdot P_e$$

$$= (1 / (2 \cdot 0.365)) \cdot (467.21 / 12.2) \cdot 2.077$$

$$= 108971.0781 \text{ kPa}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 12.2002 + 0 + 0$$

$$= 12.2002 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(98.4654; 12.2002)$$

$$= 98.4654 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.7901, F_{tmax} = 1.1366$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((10.5 \cdot 0.6642 - 1.0 \cdot 0.7965) - (5.988) \cdot 0.79) /$$

$$(0.7965 - 0.6642)$$

$$= 10932.1211 \text{ kPa}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((10.5 \cdot 0.6642 - 1.0 \cdot 0.7965) - (5.988) \cdot 1.137) /$$

$$(0.7965 - 0.6642)$$

$$= -4745.5176 \text{ kPa}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 10932.121 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92392.9688 \text{ kPa}$$



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Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.173 + -0.196 \cdot 0.163)2.004^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4$$

$$= (\sqrt{ 19.05^2 + (19.05 - 2 \cdot 2.11)^2 }) / 4$$

$$= 6.0355 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 195074 / 157989)}$$

$$= 156.1125$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 850.0 / 6.035$$

$$= 140.8340$$

Buckling Allowable Stress [Stb]:

$$= (Sy_t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (157989 / 2.0) \cdot (1 - 140.834 / (2 \cdot 156.113))$$

$$= 43362.957 \text{ kPa (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 10932.12 \cdot 0.0001$$

$$= 1227.55 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.05 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 92392.969$$

$$= 10374.639 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92392.97$$

$$= 10374.6387 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92392.969 / 92392.969$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92392.97$$

$$= 6892.9707 \text{ N}$$

Minimum Length of Weld [ar]:



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$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 10374.64 is >= tube strength 10374.64 N

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= Ss \cdot Esw = 137989.5 \cdot 1.0 = 137989.5 \text{ kPa}$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 467.21^2 / ((1130.0 + 60.0) \cdot 60.0) \cdot [5.99 + (1.209^2 - 1)(10.5 - 1.0)] + 565.0^2 \cdot 1.0 / ((1130.0 + 60.0) \cdot 60.0)$$

$$= 36206.0547 \text{ kPa}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 60.0 mm, the minimum Shell length adjacent to the tubesheet should be 468.692 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 467.21^2 / ((1130.0 + 60.0) \cdot 60.0) \cdot [5.99 + (1.209^2 - 1)(10.5 - 1.0)] + 565.0^2 \cdot 1.0 / ((1130.0 + 60.0) \cdot 60.0)$$

$$= 36206.0547 \text{ kPa}$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot \{ \beta s [\delta s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) \cdot (ao/h)^3 \cdot (1 + h \cdot \beta s / 2) [Pe (Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 52500932 / 60.0^2 \cdot \{ 0.007 [0.234E-04 \cdot 10.501 + 565.0^2 \cdot 0 / (11704440)] + 6(1 - 0.32^2) / (71604.4) \cdot (467.21 / 125.0)^3 \cdot (1 + 125.0 \cdot 0.01 / 2) [6.0 (0.163 + 0.677 \cdot -0.157) + 2/467.21^2 \cdot 0.677 \cdot -25621.268] \}$$

$$= 233887.7500 \text{ kPa}$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot Ss$$

$$:: |36206.05| + |233887.75| \leq 206984.23 \text{ kPa}$$

$$:: 270093.81, \text{ must be } \leq 206984.23 \text{ kPa (See Elastic-Plastic iteration below)}$$

Note:

For a given Channel thickness of 18.0 mm, the minimum Channel length adjacent to the tubesheet should be 256.713 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= ac^2 \cdot Pt / ((Dc + tc) \cdot tc \cdot \cos(\theta))$$

$$= 565.0^2 \cdot 1.0 / ((1130.0 + 18.0) \cdot 18.0 \cdot \cos(0))$$

$$= 15448.362 \text{ kPa}$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot \{ (\delta c \cdot Cm \Delta + \beta c - \zeta c \cdot Cm \Theta) \cdot Pt - 6(1 - \nu^2) / (E \cdot h) \cdot (ao/h)^3 \cdot (Cm \Theta + h \cdot \beta c / 2 \cdot Cm \Delta) \cdot [Pe (Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 2691030 / 18.0^2 \cdot \{ (0.759E-04 \cdot 1.0 \cdot 0.013 - 0 \cdot 1.0) \cdot 1.0 \}$$



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ASME TS Calc: DESIGN CASE

Case: 1 3:22p Apr 10,2026

$$\begin{aligned}
 & -6((1 - 0.321^2)/71604) \cdot \\
 & (467.21/125.0)^3 \cdot (1.0 + 125.0 \cdot 0.013/2 \cdot 1.0) \cdot [6.0(0.16252 + 0.67654 \\
 & \cdot -0.157) + 2/467.21^2 \cdot 0.67654 \cdot -25621.268] \\
 & = -14925.7656 \text{ kPa}
 \end{aligned}$$

Channel Stress Summation vs. Allowable

$$:: |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$$

$$:: |15448.36| + |-14925.77| \leq 206984.23 \text{ kPa}$$

$$:: 30374.13 \text{ must be } \leq 206984.23 \text{ kPa}$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	134744.0	<= 206984.2 kPa	Pass
Tubesheet Shear Stress	30656.1	<= 110391.6 kPa	Pass
Maximum Tube Stress	10932.1	<= 92393.0 kPa	Pass
Minimum Tube Stress (Buckling)	-4745.5	<= -43363.0 kPa	Pass
Maximum Force on any one Tube	1227.5	<= 10374.6 N	Pass
Axial Membrane Stress in Shell	36206.1	<= 137989.5 kPa	Pass
Shell Stress (jt. inter.)	270093.8	<= 206984.2 kPa	See below
Channel Stress (jt. inter.)	30374.1	<= 206984.2 kPa	Pass

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

$$= \min(\text{Sys}, \text{SPS}, s/2)$$

$$= \min(231994, 463989/2) = 231994.89 \text{ kPa}$$

Compute Allowable [S*c]:

$$= \min(\text{Syc}, \text{SPS}, c/2)$$

$$= \min(242594, 485189/2) = 242594.55 \text{ kPa}$$

Compute Factor [facts]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_{c,b}) / \text{Sstars}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(233887.75) / 231994.89], 1) = 0.997$$

Compute Factor [factc]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_{c,b}) / \text{Sstarc}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(-14925.77) / 242594.55], 1) = 1.0$$

Compute Modified Modulus [Es]:

$$= E_s \cdot \text{facts}$$

$$= 195074 \cdot 0.9967 = 194437 \text{ N/mm}^2$$

Tubesheet Bending Stress - final thickness [σf]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0996 / 0.365) \cdot (2 \cdot 467.21 / (98.433 - 0))^2 \cdot 5.61$$

$$= 206983.3594 \text{ kPa}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + \text{Cats} + \text{Catc}$$

$$= 98.4325 + 0 + 0$$

$$= 98.4325 \text{ mm}$$

For the Elastic-Plastic iteration, Shell Allowable = SPS,s

$$\text{ShellAllow} = 463989.7812 \text{ kPa}$$



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ASME TS Calc: DESIGN CASE

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ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Plastic iteration, the channel allowable stress = SPS,c
Challow = 485189.0938 kPa

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	134662.5	<= 206984.2 kPa	Pass
Tubesheet Shear Stress	30656.1	<= 110391.6 kPa	Pass
Maximum Tube Stress	10932.1	<= 92393.0 kPa	Pass
Minimum Tube Stress (Buckling)	-4745.5	<= -43363.0 kPa	Pass
Maximum Force on any one Tube	1227.5	<= 10374.6 N	Pass
Axial Membrane Stress in Shell	36206.1	<= 137989.5 kPa	Pass
Shell Stress (jt. inter.)	270093.8	<= 463989.8 kPa	Pass
Channel Stress (jt. inter.)	30374.1	<= 485189.1 kPa	Pass

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	12.143	...	18939	206984	4049	110392	Ps+Pt-Th	D1	Ok
D2uc	104.869	...	148323	206984	34705	110392	Ps+Pt-Th	D2	Ok
D3uc	98.465	...	134663	206984	30656	110392	Ps+Pt-Th	D3	Ok
O1uc	30.247	...	81971	437971	17069	110392	Ps+Pt+Th	O1	Ok
O2uc	86.992	...	245985	437971	55824	110392	Ps+Pt+Th	O2	Ok
O3uc	83.218	...	227045	437971	51774	110392	Ps+Pt+Th	O3	Ok
O4uc	37.847	...	100911	437971	21118	110392	Ps+Pt+Th	O4	Ok
D1c	17.984	...	20243	206984	4127	110392	Ps+Pt-Th-c	D1	Ok
D2c	107.650	...	155520	206984	35285	110392	Ps+Pt-Th-c	D2	Ok
D3c	101.496	...	138380	206984	31158	110392	Ps+Pt-Th-c	D3	Ok
O1c	35.137	...	85244	437971	17154	110392	Ps+Pt+Th-c	O1	Ok
O2c	90.007	...	260000	437971	56567	110392	Ps+Pt+Th-c	O2	Ok
O3c	86.347	...	239757	437971	52440	110392	Ps+Pt+Th-c	O3	Ok
O4c	42.363	...	105487	437971	21282	110392	Ps+Pt+Th-c	O4	Ok
Max:	107.6500	... mm	0.751	0.512	(Str. Ratio)				

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+/-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:



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ASME TS Calc: DESIGN CASE

Case: 1 3:22p Apr 10,2026

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	639	137989	...	...	:	...	...	...	...	:	Ok	
D2uc	35567	137989	...	...	:	...	...	...	...	:	Ok	
D3uc	36206	137989	...	...	:	...	...	...	...	:	Ok	
O1uc	13249	463989	...	...	:	...	...	...	...	:	Ok	
O2uc	48176	463989	...	...	:	...	...	...	...	:	Ok	
O3uc	48816	463989	...	...	:	...	...	...	...	:	Ok	
O4uc	12610	463989	...	...	:	...	...	...	...	:	Ok	
D1c	747	137989	...	...	:	...	...	...	...	:	Ok	
D2c	37171	137989	...	...	:	...	...	...	...	:	Ok	
D3c	37918	137989	...	...	:	...	...	...	...	:	Ok	
O1c	13449	463989	...	...	:	...	...	...	...	:	Ok	
O2c	49873	463989	...	...	:	...	...	...	...	:	Ok	
O3c	50620	463989	...	...	:	...	...	...	...	:	Ok	
O4c	12702	463989	...	...	:	...	...	...	...	:	Ok	
Max RATIO	0.275		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	1543	92392	-1543	-43362	173	10374	20928	206984	77780	206984	Ok
D2uc	12475	92392	-5780	-43362	1401	10374	289743	463989	77257	485189	Ok
D3uc	10932	92392	-4746	-43362	1228	10374	270094	463989	30374	485189	Ok
O1uc	35999	184785	-35999	-43362	4042	20749	106928	463989	34655	485189	Ok
O2uc	42812	184785	-42812	-43362	4807	20749	416321	463989	158795	485189	Ok
O3uc	41778	184785	-41778	-43362	4691	20749	396672	463989	111912	485189	Ok
O4uc	37033	184785	-37033	-43362	4158	20749	126578	463989	81538	485189	Ok
D1c	1922	92392	-1922	-43362	216	10374	24102	206984	92551	206984	Ok
D2c	15548	92392	-5218	-43362	1746	10374	318026	463989	86463	485189	Ok
D3c	13626	92392	-4197	-43362	1530	10374	295417	463989	31285	485189	Ok
O1c	35421	184785	-35421	-43362	3977	20749	114939	463989	30634	485189	Ok
O2c	41661	184785	-41661	-43362	4678	20749	455574	463989	172275	485189	Ok
O3c	40639	184785	-40639	-43362	4563	20749	432965	463989	117097	485189	Ok
O4c	36443	184785	-36443	-43362	4092	20749	137548	463989	85812	485189	Ok
Max RATIO	0.232		0.987		0.232		0.982		0.447		

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	107.650	125.000	Ok
Tube thickness :	1.568	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (60.000) adj. to tubesheet: 468.692 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 256.713 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +3.171 mm



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ASME TS Calc: DESIGN CASE

Case: 1 3:22p Apr 10,2026

Due to Pressure Shell Compresses by : -0.749 mm
Due to Pressure + Thermal Shell Extends by : +3.191 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	10.116	0.996	13.726	0.995
Tubesheet Shear Stress	26.748	1.000	26.518	1.000
Tube Tensile Stress	48.070	1.000	62.400	1.000
Tube Compressive Stress	11.634	1.000	11.499	1.000
Tube-Tubesheet Joint load	48.070	1.000	62.400	1.000
Shell Stress (Axial, Junction)	7.287	0.137	10.778	1.000
Tubesheet-Channel Junction	5.243	1.000	10.778	0.360
Tube Pressure Stress	22.458	1.000	15.011	1.000
Minimum MAWP	5.243		10.778	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	10.860	1.000	14.779	0.999
Tubesheet Shear Stress	27.267	1.000	33.403	1.000
Tube Tensile Stress	59.833	1.000	77.727	1.000
Tube Compressive Stress	30.755	1.000	86.301	1.000
Tube-Tubesheet Joint load	59.833	1.000	77.727	1.000
Shell Stress (Axial, Junction)	10.860	1.000	14.779	0.999
Tubesheet-Channel Junction	10.860	1.618	14.779	0.999
Tube Pressure Stress	22.458	1.000	15.020	1.000
Minimum MAPnc	10.860		14.779	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0201	0.0201	...	...	...	...
2	-0.7489	-0.7492	...	...	...	...
3	-0.7287	-0.7291	...	...	...	...
4	...	...	...	...	...	...
5	0.0201	0.0201	3.1691	3.1707	...	...
6	-0.7489	-0.7492	3.1691	3.1707	...	...
7	-0.7287	-0.7291	3.1691	3.1707	...	...
8	...	...	3.1691	3.1707	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(125.0/4, \min(125.0, 60.0))$$

$$= 60.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(ps/\text{Shellside MAPnc}, 1)$$

$$= \min(10.5/14.78, 1)$$

$$= 0.710$$



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ASME TS Calc: DESIGN CASE

Case: 1 3:22p Apr 10,2026

Thickness Ratio = 0.71, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(125.0/4, min(125.0, 18.0))
= 31.250 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(1.0/10.86, 1)
= 0.092

Thickness Ratio = 0.092, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

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ASME TS Calc: START-UP

Case: 2 3:22p Apr 10,2026

Input Echo, Tubesheet Number 2, Description:

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	0.40 N/mm ²
Shell Maximum Operating Pressure	Psox,max	0.40 N/mm ²
Shell Minimum Operating Pressure	Psox,min	8.30 N/mm ²
Shell Thickness	ts	60.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1130.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	38.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel left

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	0.10 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	0.10 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	0.50 N/mm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1130.000 mm
Channel Design Temperature	TEMPC	38.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	808
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	12000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11750.00 mm
Design Temperature of the Tubes		38.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	178986.38 kPa
Tube Pitch (Center to Center Spacing)	P	30.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	457.685 mm



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ASME TS Calc: START-UP

Case: 2 3:22p Apr 10,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	38.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	137989.50 kPa
Tubesheet Allowable Stress at Ambient	Tt	137989.50 kPa
Thickness of Tubesheet	h	125.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1250.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
---------------------------	----	----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	850.000 mm
Tube End condition corresponding to Span (l)	k	1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	44.60 °C
Mean Tube Metal Temp. along Tube length	Ttm	67.00 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----



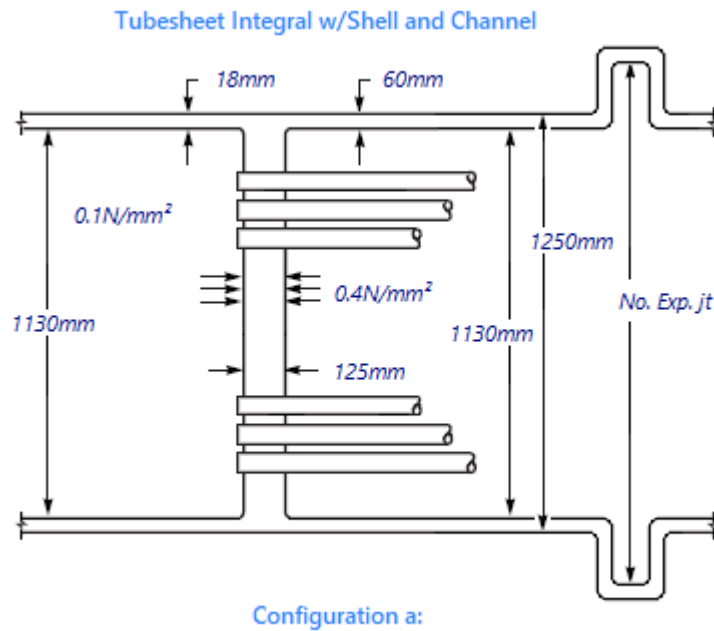
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ASME TS Calc: START-UP

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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	44.6 °C	0.0000117432 /°C
Elastic Mod. at Design Temperature	38.0 °C	0.20175E+06 N/mm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	44.6 °C	0.20137E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	38.0 °C	0.20175E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	67.0 °C	0.0000118884 /°C
Elastic Mod. at Design Temperature	38.0 °C	0.20175E+06 N/mm²
Elastic Mod. at Metal Temp. along Len	67.0 °C	0.20008E+06 N/mm²
Elastic Mod. at Tubsht. Design Temp.	38.0 °C	0.20175E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	38.0 °C	0.20037E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20133E+06 N/mm²

Note:



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ASME TS Calc: START-UP

Case: 2 3:22p Apr 10,2026

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(0.1 \cdot (19.05/2 - 0)) / (92392.97 \cdot 1.0 + 0.4 \cdot 0.1)$

= $0.0104 + 0.0000 = 0.0104$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 38.00 °C
Elastic Modulus for Material 199955.00 N/mm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
2.1100	19.05	11750.00	9.03	50.0000	0.0134948	95834.94

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15.0201$ N/mm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
0.2662	19.05	11750.00	71.57	50.0000	0.0002148	21470.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 21470.9375) / (3 \cdot 71.5664) = 0.4$ N/mm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2662 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm²
Required Internal Design Pressure	0.10 N/mm²
Allowable External Pressure at Corroded thickness	15.02 N/mm²
Required External Design Pressure	0.40 N/mm²
Required Thickness due to Shell Side pressure	0.2662 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	3.810	...	1914	206984	410	110392	Ps+Pt-Th	D1 Ok
D2uc	3.810	...	5697	206984	1321	110392	Ps+Pt-Th	D2 Ok
D3uc	3.810	...	4199	206984	912	110392	Ps+Pt-Th	D3 Ok
O1uc	33.618	...	102176	495962	21417	110392	Ps+Pt+Th	O1 Ok
O2uc	36.578	...	109605	495962	23148	110392	Ps+Pt+Th	O2 Ok
O3uc	35.905	...	107691	495962	22739	110392	Ps+Pt+Th	O3 Ok
O4uc	34.298	...	104090	495962	21827	110392	Ps+Pt+Th	O4 Ok
D1c	9.810	...	2045	206984	418	110392	Ps+Pt-Th-c	D1 Ok
D2c	9.810	...	5900	206984	1344	110392	Ps+Pt-Th-c	D2 Ok
D3c	9.810	...	4368	206984	926	110392	Ps+Pt-Th-c	D3 Ok
O1c	38.371	...	106739	495962	21577	110392	Ps+Pt+Th-c	O1 Ok
O2c	41.183	...	114656	495962	23339	110392	Ps+Pt+Th-c	O2 Ok
O3c	40.546	...	112611	495962	22921	110392	Ps+Pt+Th-c	O3 Ok
O4c	39.016	...	108783	495962	21995	110392	Ps+Pt+Th-c	O4 Ok
Max:	41.1832	...	mm	0.231		0.211	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+/-)Th]:



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ASME TS Calc: START-UP

Case: 2 3:22p Apr 10,2026

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	65	137989	...	...	:	...	...	...	...	:	Ok	
D2uc	1355	137989	...	...	:	...	...	...	...	:	Ok	
D3uc	1419	137989	...	...	:	...	...	...	...	:	Ok	
O1uc	13098	523984	...	...	:	...	...	...	...	:	Ok	
O2uc	14387	523984	...	...	:	...	...	...	...	:	Ok	
O3uc	14452	523984	...	...	:	...	...	...	...	:	Ok	
O4uc	13033	523984	...	...	:	...	...	...	...	:	Ok	
D1c	76	137989	...	...	:	...	...	...	...	:	Ok	
D2c	1416	137989	...	...	:	...	...	...	...	:	Ok	
D3c	1491	137989	...	...	:	...	...	...	...	:	Ok	
O1c	13204	523984	...	...	:	...	...	...	...	:	Ok	
O2c	14543	523984	...	...	:	...	...	...	...	:	Ok	
O3c	14619	523984	...	...	:	...	...	...	...	:	Ok	
O4c	13128	523984	...	...	:	...	...	...	...	:	Ok	
Max RATIO	0.028		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	157	92392	-157	-47243	18	10374	2125	206984	7852	206984	Ok
D2uc	478	92392	-220	-47243	54	10374	11045	206984	2887	206984	Ok
D3uc	321	92392	-116	-47243	36	10374	9050	206984	4964	206984	Ok
O1uc	38196	184785	-38196	-47243	4289	20749	129162	523984	78028	523984	Ok
O2uc	38521	184785	-38521	-47243	4325	20749	142202	523984	85637	523984	Ok
O3uc	38416	184785	-38416	-47243	4314	20749	140207	523984	80916	523984	Ok
O4uc	38301	184785	-38301	-47243	4301	20749	131157	523984	82750	523984	Ok
D1c	196	92392	-196	-47243	22	10374	2446	206984	9345	206984	Ok
D2c	595	92392	-199	-47243	67	10374	12122	206984	3230	206984	Ok
D3c	400	92392	-95	-47243	45	10374	9828	206984	6114	206984	Ok
O1c	37587	184785	-37587	-47243	4221	20749	140192	523984	81503	523984	Ok
O2c	37890	184785	-37890	-47243	4255	20749	154609	523984	90292	523984	Ok
O3c	37786	184785	-37786	-47243	4243	20749	152315	523984	84733	523984	Ok
O4c	37691	184785	-37691	-47243	4232	20749	142487	523984	87061	523984	Ok
Max RATIO	0.208		0.815			0.208		0.295		0.172	

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	41.183	125.000	Ok
Tube thickness :	0.266	2.110	Ok



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ASME TS Calc: START-UP

Case: 2 3:22p Apr 10,2026

Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (60.000) adj. to tubesheet: 468.692 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 256.713 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +3.171 mm
 Due to Pressure Shell Compresses by : -0.028 mm
 Due to Pressure + Thermal Shell Extends by : +3.173 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	10.197	0.999	13.863	0.996
Tubesheet Shear Stress	26.765	1.000	26.320	1.000
Tube Tensile Stress	47.814	1.000	62.082	1.000
Tube Compressive Stress	13.237	1.000	16.234	1.000
Tube-Tubesheet Joint load	47.814	1.000	62.083	1.000
Shell Stress (Axial, Junction)	6.761	0.312	12.589	1.000
Tubesheet-Channel Junction	5.681	1.000	12.589	0.360
Tube Pressure Stress	22.458	1.000	15.020	1.000
Minimum MAWP	5.681		12.589	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	10.890	1.000	14.813	0.999
Tubesheet Shear Stress	27.280	1.000	33.417	1.000
Tube Tensile Stress	59.507	1.000	77.307	1.000
Tube Compressive Stress	30.587	1.000	86.214	1.000
Tube-Tubesheet Joint load	59.507	1.000	77.307	1.000
Shell Stress (Axial, Junction)	10.890	1.000	14.813	0.999
Tubesheet-Channel Junction	10.890	1.623	14.813	0.999
Tube Pressure Stress	22.458	1.000	15.020	1.000
Minimum MAPnc	10.890		14.813	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0020	0.0020	...	...	...	...
2	-0.0276	-0.0276	...	...	...	...
3	-0.0256	-0.0256	...	...	...	...
4	...	...	...	...	...	...
5	0.0020	0.0020	3.1691	3.1707	...	...
6	-0.0276	-0.0276	3.1691	3.1707	...	...
7	-0.0256	-0.0256	3.1691	3.1707	...	...
8	...	...	3.1691	3.1707	...	...